Aiyagari: Proof Ledger

55 GREEN / 0 REVIEW_READY / 2 UNFORMALIZED

September 11, 2026 -- Version 1.0

Economic status: P01, P02, P03, H01, H02, H03, H04, H05, H06, H07, H08, H09, H10, H11, H12, H13, H14, D01, D02, D03, S01, S02, S03, S04, S05, S06, A01, A02, A03, A04, A05, N01, N02, N03, N04, N05, N06, N07, B01, B02, B03, F01, F02, G01, G02, G03, G04, G05, G06, G07, G08, NP01, NP02, NP03, E01 are **GREEN**; E02, E03 are **UNFORMALIZED**. M00 bootstrap acceptance remains infrastructure only. Exact acceptance records are in `reviews/`. Proposed proof plans remain proposed until checked.

Current existence boundary: G02 and G03 are accepted GREEN existence results. They construct, respectively, a stationary equilibrium for each supplied finite cap and a natural-limit equilibrium at an endogenous rate strictly between zero and the impatience rate. Both retain the supplied beta, utility, and complete income law, use endogenous household prices, and establish existence only. Uniqueness remains unformalized. The universal every-equilibrium restriction G04 is accepted GREEN and is independent of which existence construction supplies an equilibrium. G05 is the accepted mean-income certainty benchmark; G06 and G07 are accepted GREEN comparisons. G08 is the universal goods-clearing identity. A05 is the accepted partial-equilibrium comparison and makes no equilibrium claim.

The completed ledger must replace each plan pointer with the actual readable proof, exact elaborated Lean signature, all economic hypotheses, axiom output and review evidence.

Milestone 00 — compiled API infrastructure

Status: GREEN / accepted 2026-09-11. Scope: bootstrap/environment/source/API infrastructure only, including build/audit infrastructure and representation preflight. See `reviews/00_acceptance.md` for the external decision and four reviewer qualifications. No economic adequacy is inferred from the probes. No economic contract is assigned to this milestone and none is promoted. Source bytes are verified; mathematical page locators below remain the approved design locators. The clean Lean 4.32.0 / pinned Mathlib build and audit evidence are in `reports/00_environment.md` and `reports/logs/00_axioms.log`.

Actual economic assumptions: none. The generic Banach result assumes `ContractingWith K T`; the kernel results assume the specified Markov instances and measurability; weak test convergence assumes weak convergence; family compactness assumes uniform tightness. These mathematical hypotheses are not household primitives or proved economic conclusions.

Readable proofs. The bounded continuous function instance gives completeness, and evaluation distance is at most the supremum distance. NNReal inherits its complete, separable metric and Borel structure. Banach supplies a fixed point and its uniqueness for a contraction on the bounded-value space. The constant zero operator has Lipschitz constant zero, strictly below one; uniqueness identifies its fixed point as zero. This is a concrete infrastructure witness, not the primitive economic consistency witness P03.

A bounded continuous real function is measurable and norm bounded, hence integrable against any finite probability measure. A measurable pushforward is again a probability law. The map-integrability

equivalence proves integrability of the composed test, and the change-of-variables theorem gives the integral identity. The pushforward probe exports both integrability statements. Markov composition preserves probability mass; composing with the identity changes nothing. A deterministic kernel transports a measure to its pushforward. These are generic kernel facts, with no stationarity claim.

Weak convergence implies convergence of every bounded continuous test by its installed characterization. On the compact interval [0,1], the probability-law space is compact. Every single law on NNReal is tight by complete second-countable metrizability. Prokhorov gives compact closure for a uniformly tight family. It does not provide an unbounded-moment limit. Finally, a constant-infinity ENNReal function demonstrates that the marginal representation can retain infinity at zero; it asserts no derivative theorem.

Exact elaborated declaration inventory and axiom output. The following is the actual successful `lake env lean Audit.lean` output. Lean suppresses proof arguments with an ellipsis; the explicit source signatures and proofs are in `Probes/Core.lean`. All declarations pass `assert_no_sorry`; only the standard foundational axioms occur. The abbreviations are State = NNReal, ValueSpace = bounded continuous State-to-Real functions, and CompactInterval = [0,1] as a subtype.

```
Aiyagari1994.Probes.State : Type
'Aiyagari1994.Probes.State' depends on axioms: [propext, Classical.choice, Quot.sound]
Aiyagari1994.Probes.ValueSpace : Type
'Aiyagari1994.Probes.ValueSpace' depends on axioms: [propext, Classical.choice, Quot.sound]
Aiyagari1994.Probes.CompactInterval : Set ℝ
'Aiyagari1994.Probes.CompactInterval' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.value_complete : CompleteSpace Aiyagari1994.Probes.ValueSpace
'Aiyagari1994.Probes.value_complete' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.sup_distance (f g : Aiyagari1994.Probes.ValueSpace) (z :
↪  Aiyagari1994.Probes.State) :
  dist (f z) (g z) ≤ dist f g
'Aiyagari1994.Probes.sup_distance' depends on axioms: [propext, Classical.choice, Quot.sound]
Aiyagari1994.Probes.state_complete : CompleteSpace Aiyagari1994.Probes.State
'Aiyagari1994.Probes.state_complete' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.state_separable : TopologicalSpace.SeparableSpace
↪  Aiyagari1994.Probes.State
'Aiyagari1994.Probes.state_separable' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.state_borel : BorelSpace Aiyagari1994.Probes.State
'Aiyagari1994.Probes.state_borel' depends on axioms: [propext, Classical.choice, Quot.sound]
Aiyagari1994.Probes.state_measurable_id : Measurable id
'Aiyagari1994.Probes.state_measurable_id' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.banach_on_values (T : Aiyagari1994.Probes.ValueSpace →
↪  Aiyagari1994.Probes.ValueSpace) (K : NNReal)
  (hT : ContractingWith K T) : ∃! v : Aiyagari1994.Probes.ValueSpace, T v = v
'Aiyagari1994.Probes.banach_on_values' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.zero_contraction : ContractingWith 0 fun (x :
↪  Aiyagari1994.Probes.ValueSpace) => 0
'Aiyagari1994.Probes.zero_contraction' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.zero_fixed_point :
  ContractingWith.fixedPoint (fun (x : Aiyagari1994.Probes.ValueSpace) => 0)
  ↪  Aiyagari1994.Probes.zero_contraction = 0
```

```
'Aiyagari1994.Probes.zero_fixed_point' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.bounded_test_integrable (μ : MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State)
  (f : Aiyagari1994.Probes.ValueSpace) : MeasureTheory.Integrable ⇑f ↑μ
'Aiyagari1994.Probes.bounded_test_integrable' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.pushforward_test (μ : MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State)
  (g : Aiyagari1994.Probes.State → Aiyagari1994.Probes.State) (hg : Measurable g) (f :
  ↪  Aiyagari1994.Probes.ValueSpace) :
  MeasureTheory.Integrable ⇑f ↑(μ.map ⋯) ∧
    MeasureTheory.Integrable (fun (x : Aiyagari1994.Probes.State) => f (g x)) ↑μ ∧
      ∫ (y : Aiyagari1994.Probes.State), f y ∂↑(μ.map ⋯) = ∫ (x :
      ↪  Aiyagari1994.Probes.State), f (g x) ∂↑μ
'Aiyagari1994.Probes.pushforward_test' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.kernel_composition
  (κ η : ProbabilityTheory.Kernel Aiyagari1994.Probes.State Aiyagari1994.Probes.State)
  [ProbabilityTheory.IsMarkovKernel κ] [ProbabilityTheory.IsMarkovKernel η] :
  ProbabilityTheory.IsMarkovKernel (η.comp κ)
'Aiyagari1994.Probes.kernel_composition' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.kernel_identity (κ : ProbabilityTheory.Kernel Aiyagari1994.Probes.State
↪  Aiyagari1994.Probes.State) :
  κ.comp ProbabilityTheory.Kernel.id = κ
'Aiyagari1994.Probes.kernel_identity' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.kernel_transport (μ : MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State)
  (g : Aiyagari1994.Probes.State → Aiyagari1994.Probes.State) (hg : Measurable g) :
  (↑μ).bind ⇑(ProbabilityTheory.Kernel.deterministic g hg) = ↑(μ.map ⋯)
'Aiyagari1994.Probes.kernel_transport' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.deterministic_markov (g : Aiyagari1994.Probes.State →
↪  Aiyagari1994.Probes.State)
  (hg : Measurable g) : ProbabilityTheory.IsMarkovKernel
  ↪  (ProbabilityTheory.Kernel.deterministic g hg)
'Aiyagari1994.Probes.deterministic_markov' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.transport_mass (κ : ProbabilityTheory.Kernel Aiyagari1994.Probes.State
↪  Aiyagari1994.Probes.State)
  [ProbabilityTheory.IsMarkovKernel κ] (μ : MeasureTheory.ProbabilityMeasure
  ↪  Aiyagari1994.Probes.State) :
  ((↑μ).bind ⇑κ) Set.univ = 1
'Aiyagari1994.Probes.transport_mass' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.weak_test_convergence (μs : ℕ → MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State)
  (μ : MeasureTheory.ProbabilityMeasure Aiyagari1994.Probes.State) (h : Filter.Tendsto μs
  ↪  Filter.atTop (nhds μ))
  (f : Aiyagari1994.Probes.ValueSpace) :
  Filter.Tendsto (fun (n : ℕ) => ∫ (x : Aiyagari1994.Probes.State), f x ∂↑(μs n))
  ↪  Filter.atTop
    (nhds (∫ (x : Aiyagari1994.Probes.State), f x ∂↑μ))
```

```
'Aiyagari1994.Probes.weak_test_convergence' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.compact_interval_laws :
  CompactSpace (MeasureTheory.ProbabilityMeasure ↑Aiyagari1994.Probes.CompactInterval)
'Aiyagari1994.Probes.compact_interval_laws' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.individual_law_tight (μ : MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State) :
  MeasureTheory.IsTightMeasureSet {↑μ}
'Aiyagari1994.Probes.individual_law_tight' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.tight_family_compact (S : Set (MeasureTheory.ProbabilityMeasure
↪  Aiyagari1994.Probes.State))
  (hS : MeasureTheory.IsTightMeasureSet {x : MeasureTheory.Measure Aiyagari1994.Probes.State
  ↪  | ∃ μ ∈ S, ↑μ = x}) :
  IsCompact (closure S)
'Aiyagari1994.Probes.tight_family_compact' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
Aiyagari1994.Probes.extended_zero_value : ∃ (q : Aiyagari1994.Probes.State → ENNReal), q 0 =
↪  ⊤
'Aiyagari1994.Probes.extended_zero_value' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

M00 review boundary: the bootstrap acceptance certifies only its stated infrastructure scope. M01 was separately assigned and its results follow below; generic API success does not certify the later economic proofs.

P01 — Shifted budget normalization

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Milestone:** 01.

Target: `Aiyagari1994.shifted_budget_iff` in `Aiyagari1994/Budget/Normalization.lean`.

Actual statement and assumptions. For arbitrary real net rate, wage, current labor, debt shift, current net assets, next net assets and consumption, the original budget equality together with the next borrowing inequality is equivalent to the shifted budget equality together with nonnegative next shifted assets. There are no economic premises: this algebraic result is stronger than its restriction to admissible prices. Gross return is structurally $1 + r$; it is never independent data alongside r in the original-price structure.

Readable proof. Expand $(1 + r)(a + \phi)$ and cancel $r\phi$. Relative to the original equality, both sides acquire exactly ϕ. Adding or subtracting ϕ also proves $-\phi \le a_{next}$ iff $0 \le a_{next} + \phi$. The main theorem joins these two genuine equivalences. Consumption is the same real number in both systems, so a separate nonnegativity restriction on consumption is preserved without an additional premise in the algebraic identity.

The audited supporting declarations keep the components separate:

- `shifted_budget_equality_iff`: the budget equality component.
- `shifted_borrowing_iff`: the borrowing inequality component.
- `next_resource_identity`: next labor and shifted savings in the next-period resource equation.
- `original_normalized_coordinates`: the constructed map has $R = 1 + r$ and $k = -r\phi$.

Source correspondence. A94 equations (1b), (3a)-(4b), printed pp. 665-666 / PDF pp. 8-9, visually inspected in M01. No No-Ponzi equivalence or policy theorem is claimed.

Exact elaborated signature and transitive axioms:

```
Aiyagari1994.shifted_budget_iff (r w l phi a a_next c : ℝ) :
  c + a_next = (1 + r) * a + w * l ∧ -phi ≤ a_next ↔
    c + (a_next + phi) = (1 + r) * (a + phi) + w * l - r * phi ∧ 0 ≤ a_next + phi
'Aiyagari1994.shifted_budget_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Integrability: none; this is pathwise algebra. **Dependencies:** primitive real arithmetic only. **Adequacy caveat:** externally accepted in `reviews/01_acceptance.md`; no No-Ponzi claim is included.

P02 — Effective borrowing limits

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Milestone:** 01.

Target: `Aiyagari1994.effectiveLimit_admissible_continuous`.

Module: `Aiyagari1994/Budget/EffectiveLimit.lean`.

Actual assumptions. The cap b is a fixed nonnegative real and the labor floor l_{min} is strictly positive. Scalar admissibility quantifies over positive wages and labor at least the floor. Finite-cap continuity is joint in (w, r) on $w > 0, r > -1$, for each fixed b, l_{min}. Its nonnegativity and effective-income inequalities are actually proved for every real rate. The original-price constructor additionally requires $r > -1$ to ensure positive gross return. The natural branch separately requires $r > 0$ and has no finite-cap premise.

Readable proof. At $r > 0$, both entries of $\min(b, wl_{min}/r)$ are nonnegative, and $r\phi \le wl_{min} \le wl$. At $r \le 0$, $\phi = b \ge 0$ and $rb \le 0 < wl_{min} \le wl$. Thus effective income is nonnegative in each branch.

For continuity, the proof establishes the exact auxiliary identity

$$\phi(b, l_{min}, w, r) = \frac{b(wl_{min})}{\max(wl_{min}, br)}.$$

When $br \le wl_{min}$ the expression equals b; otherwise the positive-rate minimum equals wl_{min}/r. The denominator is at least $wl_{min} > 0$. Continuity of multiplication, maximum and division therefore proves joint continuity on the stated price domain. The case $b = 0$ never requires division by b.

The explicit `effectiveLimit_zero_neighborhood` theorem also proves the source-sensitive boundary step: the continuous expression $w'l_{min} - br'$ is positive near $(w, 0)$. Consequently, if $r' > 0$, then $b \le w'l_{min}/r'$; the minimum equals b. For nearby nonpositive rates the definition is already b. This proves local constancy at zero, not a reliance on a convention for division by zero.

For the separate natural limit $\phi_{nat} = wl_{min}/r$, cancellation with $r > 0$ gives $-r\phi_{nat} = -wl_{min}$. Effective income is exactly $w(l - l_{min}) \ge 0$. Raw natural-limit continuity is proved only on positive rates. No continuation of the raw debt limit through zero is claimed.

`finiteCapPrices` and `naturalCapPrices` construct the same `OriginalPrices` type with these proved admissibility facts. `OriginalPrices.normalized` constructs normalized prices; `NormalizedPrices.nextResources` returns NNReal using nonnegative shifted savings and nonnegative effective income. The audited helper `generated_resources_admissible` gives the explicit generated-state inequality. No invariant or absorbing bound is used.

Source correspondence. A94 equations (2a)-(2b), printed p. 666 / PDF p. 9, visually inspected in M01. The explicit continuity argument and the rational/max helper are implementation mathematics consistent with architecture section 2.

Exact elaborated signature and transitive axioms:

```
Aiyagari1994.effectiveLimit_admissible_continuous (b lo : ℝ) (hb : 0 ≤ b) (hl : 0 < lo) :
  (∀ (w r : ℝ), 0 < w → 0 ≤ Aiyagari1994.effectiveLimit b lo w r) ∧
    (∀ (w r l : ℝ), 0 < w → lo ≤ l → 0 ≤ w * l - r * Aiyagari1994.effectiveLimit b lo w r)
    ↪  ∧
```

```
      ContinuousOn (fun (p : ℝ × ℝ) => Aiyagari1994.effectiveLimit b lo p.1 p.2) {p : ℝ × ℝ
      ↪  | 0 < p.1 ∧ -1 < p.2} ∧
        (∀ (w : ℝ),
           0 < w →
             ∀ᶠ (p : ℝ × ℝ) in nhds (w, 0),
               (0 < p.2 → b ≤ p.1 * lo / p.2) ∧ Aiyagari1994.effectiveLimit b lo p.1 p.2 =
               ↪  b) ∧
          (∀ (w r : ℝ),
             0 < w →
               0 < r →
                 0 ≤ Aiyagari1994.naturalLimit lo w r ∧
                   -r * Aiyagari1994.naturalLimit lo w r = -w * lo ∧
                     ∀ (l : ℝ),
                       lo ≤ l →
                         w * l - r * Aiyagari1994.naturalLimit lo w r = w * (l - lo) ∧
                           0 ≤ w * l - r * Aiyagari1994.naturalLimit lo w r) ∧
            ContinuousOn (fun (p : ℝ × ℝ) => Aiyagari1994.naturalLimit lo p.1 p.2) {p : ℝ ×
            ↪  ℝ | 0 < p.1 ∧ 0 < p.2}
'Aiyagari1994.effectiveLimit_admissible_continuous' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
```

Integrability: none; scalar budget/admissibility results. **Dependencies:** primitive real order, algebra and topology; P01 supplies the normalized-price context. **Adequacy caveat:** the quantified finite and natural clauses are independent branches. The statement does not require a household simultaneously to have both borrowing rules. External M01 review accepted this scope on 2026-09-11.

P03 — Exact primitive consistency witness

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Milestone:** 01.

Target: `Aiyagari1994.corePrimitives_nonempty` in `Aiyagari1994/Primitives/Examples.lean`.

Actual assumptions. None are assumed by the existential theorem. It constructs an element of the same `HouseholdPrimitives` type used for arbitrary compact income laws, together with `CoreRegularity`. The exact output data are $\beta = 1/2$, $R = 1$, $w = 1$, intercept zero, original rate zero, cap zero, debt limit zero, utility $U(c) = c/(1+c)$, and the law placing mass $1/2$ at each labor endpoint $1/2$ and $3/2$. Assets/resources remain NNReal. The finite shock distribution is only a witness, not a restriction on the model family.

Readable utility proof. The utility is a real-valued function by its declared type. On $c \geq 0$, its denominator is positive, quotient continuity applies, and $0 \leq U(c) < 1$, so it is bounded above and below. Quotient differentiation proves

$$U'(c) = \frac{1}{(1+c)^2}, \qquad U''(c) = -\frac{2}{(1+c)^3}.$$

The first derivative formula is proved for $c \geq 0$ and the second for $c > 0$. To differentiate the derivative, the proof first establishes the quotient's derivative formula on a neighborhood of each positive c, then transfers the derivative through that local equality. Positive first derivative and negative second derivative on the interior, together with continuity at zero, give strict increase and strict concavity on $[0, \infty)$. Quotient smoothness supplies C1/C2 (indeed every finite order) on $(0, \infty)$. Algebra gives

$$-cU''(c)/U'(c) = 2c/(1+c) \leq 2.$$

The curvature record uses threshold one and finite bound two. There is no assumption of global continuity or concavity across the singularity at $c = -1$, and the general primitive records require neither finite nor infinite marginal utility at zero.

Readable income proof and integrability. `witnessMeasure` is the sum of two half-weighted Dirac measures on the full subtype $[1/2, 3/2]$. Its total mass is one. Each endpoint singleton has mass exactly one half; the complement of the two endpoints has zero mass. Every relative endpoint neighborhood contains the corresponding singleton and therefore has positive mass. These facts also prove the exact topological support and nondegeneracy.

`labor_integrable` proves labor integrability for every `IncomeData` law from continuity on the compact labor subtype and finiteness of the probability law. The direct mean computation separately checks integrability for each weighted Dirac term before using integral additivity, giving $(1/2)(1/2)+(1/2)(3/2) = 1$.

`historyLaw` is the finite product of copies of the general labor law. For every finite horizon it is a probability measure, its coordinate pushforwards are the original law, its coordinates are independent, and the `piFinSuccAbove` measurable equivalence gives the current-draw/tail product decomposition. The same proved `history_probability`, `history_marginal`, `history_independent` and `history_step` declarations apply directly to `witnessIncome`; no IID conclusion is stored as an assumed primitive field. Lifetime Bellman verification and infinite-path construction remain later work.

Finally, `witnessModel` combines the utility, income and the normalized map of the finite-cap original prices. `witness_core_regular` supplies every required regularity proof. The main theorem packages that actual inhabitant and checks the exact parameter identities.

Source correspondence. New exact consistency witness specified by architecture section 2 and the accepted M01 instructions; not claimed to appear in the papers. A93 is the manifest source key for the modeled primitive context. The budget/income conventions match A94 printed pp. 665-666 / PDF pp. 8-9, inspected visually.

Exact elaborated signature and transitive axioms:

```
Aiyagari1994.corePrimitives_nonempty :
  ∃ (m : Aiyagari1994.HouseholdPrimitives),
    Aiyagari1994.CoreRegularity m ∧
      m.beta = 1 / 2 ∧
        m.prices.grossReturn = 1 ∧
          m.prices.wage = 1 ∧
            m.prices.intercept = 0 ∧
              m.utility = Aiyagari1994.witnessUtility ∧
                m.income = Aiyagari1994.witnessIncome ∧
                  ∃ (p : Aiyagari1994.OriginalPrices m.income),
                    p.normalized = m.prices ∧
                      p.netRate = 0 ∧
                        p.wage = 1 ∧ p.debtLimit = 0 ∧ p.debtLimit =
                        ↪  Aiyagari1994.effectiveLimit 0 m.income.lower 1 0
'Aiyagari1994.corePrimitives_nonempty' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

Primitive audit. All twelve structures are printed in the direct audit log. Their fields are utility data and authorized regularity, support/probability data, optional mean normalization, prices/admissibility, and discount data. No value function, policy, Bellman theorem, Euler/envelope equation, Feller/mixing condition, invariant distribution, asset bound, asset supply or equilibrium conclusion is a field. `HouseholdPrimitives` imposes $R > 0$ and never $\beta R < 1$.

Adequacy caveat: consistency of primitives does not prove optimality, stationarity, equilibrium or any later contract. P03 is GREEN by external acceptance on 2026-09-11, within this primitive-consistency scope. Helpers and exact source files are inventoried in `reports/01_declarations.json`.

H01 — Bellman self-map and contraction

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Review gate:** M02A. See `reviews/02a_acceptance.md`.

Target: `Aiyagari1994.bellman_selfmap_contracting` in `Aiyagari1994/Household/Bellman.lean`.

Actual assumptions. Only `HouseholdPrimitives`. The construction uses $0 < \beta < 1$, utility continuity and boundedness on nonnegative consumption, the finite probability law on the compact labor subtype, and nonnegative affine generated resources. Strict utility increase and concavity are available in BASIC but not needed for H01. No return-discount restriction is imposed: every admissible $R > 0$ is covered.

Definition and integrability. For $a \in [0, z]$,

$$Q_v(z, a) = U(z - a) + \beta \int v(Ra + e(l))\, d\nu(l).$$

The transition and integrand facts are proved separately:

- `nextResources_continuous`: joint transition continuity.
- `continuation_integrand_continuous`: joint integrand continuity.
- `continuation_integrand_bound`: the bound by $\|v\|_\infty$.

`continuation_integrable` proves integrability for every action by continuity on the compact labor subtype and finiteness of its probability measure. `continuation_continuous` applies the installed compact parametric integral theorem on the whole labor subtype. Integral subtraction in the contraction proof uses the two explicit integrability results.

The objective has a continuous extension using NNReal subtraction outside the feasible set. `bellmanObjective_feasible` identifies it with the displayed real-subtraction formula whenever $a \le z$. Thus no negative-consumption utility value enters the maximization. Joint objective continuity is proved by composing primitive utility continuity with nonnegative consumption and adding the continuous continuation term.

Maximum and boundedness proof. The actual action is shifted savings a. For maximum-value continuity only, `shareObjective` evaluates $a = tz$ on the fixed compact subtype $t \in [0, 1]$. `share_feasible` and `feasible_share` prove that this parametrization covers exactly $[0, z]$; at zero, all shares give the same singleton action. `compactMax_continuous` wraps the installed compact supremum theorem; `compactMax_attained`, `le_compactMax` and `compactMax_le` give attainment and bounds. These are reusable generic theorems in `Analysis/ParametricMax.lean`, with their mathematical hypotheses discharged by `shareObjective_continuous`.

For a primitive bound $|U(c)| \le C$ and bounded continuous v,

$$|Q_v(z, a)| \le C + \beta \|v\|_\infty.$$

`bellmanValue_bound` transfers the bound through an attained optimizer. The distance between any two Bellman values is consequently bounded by twice this constant, supplying the bounded-continuous constructor. There is no bound on z, assets or the state space.

Contraction proof. The probability mass is one, so `continuation_sub_bound` gives $|E[v(Ra + e)] - E[w(Ra + e)]| \le \|v - w\|_\infty$. Current utility cancels at the same action, giving `objective_sub_bound`.

Take an optimizer for v at z. Its objective under w is no greater than the maximum for w, hence $Tv(z) - Tw(z) \leq \beta\|v - w\|_\infty$. Reverse the candidates and use symmetry of the norm to obtain the absolute-value bound. `bellman_pointwise_contraction` and `bellman_norm_contraction` turn this into the supremum-norm bound, then the installed `ContractingWith` type. No subtraction is moved through a maximum.

Exact elaborated anchor and canonical self-map:

```
Aiyagari1994.bellman_selfmap_contracting (m : Aiyagari1994.HouseholdPrimitives) :
  ContractingWith ⟨m.beta, ⋯⟩ (Aiyagari1994.bellmanOperator m)
'Aiyagari1994.bellman_selfmap_contracting' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

```
Aiyagari1994.bellmanOperator (m : Aiyagari1994.HouseholdPrimitives) (v :
↪  Aiyagari1994.ValueSpace) :
  Aiyagari1994.ValueSpace
'Aiyagari1994.bellmanOperator' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Source correspondence: A94 equations (4)-(7), printed pp. 666-667 / PDF pp. 9-10, visually inspected. A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39, is the manifest context and was also inspected. Its differentiability and envelope claims are outside M02A; the BASIC Bellman construction is explicitly reconstructed under architecture section 3.

Dependencies and qualification. P01 supplies the established shifted-budget interpretation; the normalized-price type is unchanged. The kernel-checked result is a Bellman self-map, not lifetime verification or an equilibrium result. All H01 exports pass direct no-sorry and transitive-axiom audits; only the accepted foundational axioms occur.

H02 — Canonical bounded continuous value function

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Review gate:** M02A. See `reviews/02a_acceptance.md`.

Target: `Aiyagari1994.valueFunction_unique_fixedPoint`.

Module: `Aiyagari1994/Household/Value.lean`.

Actual assumptions and dependencies. Only `HouseholdPrimitives`, using H01. The bounded-continuous real function space on all NNReal is complete. No value function is supplied as a premise, and no extra economic assumption is introduced.

Fixed-point and convergence proof. Define `valueFunction m` once using the installed `ContractingWith.fixedPoint` applied to `bellmanOperator m` and `bellman_selfmap_contracting m`. The installed fixed-point and uniqueness theorems prove `valueFunction_fixedPoint` and `valueFunction_unique` among all bounded continuous candidates. `valueIteration_tendsto` gives convergence in the supremum metric from every initial bounded continuous function. `valueIteration_uniform` uses the installed equivalence with uniform convergence, not merely a pointwise assertion. Later modules use exactly this canonical definition.

Exact range bounds. `utilityRange` is the image of all nonnegative real consumption under U. `utilityRange_nonempty` witnesses consumption zero. `utilityRange_bounded` derives both real bounds from the primitive absolute bound. Define

$$U_{inf} = \mathrm{sInf}(\mathrm{range}\, U), \qquad U_{sup} = \mathrm{sSup}(\mathrm{range}\, U).$$

The functions here have NNReal consumption arguments coerced to real. Thus these are exactly the finite real infimum and supremum over $c \geq 0$. `utility_bounds` supplies the order inequalities. Likewise `valueRange_bounded` bounds the range of any bounded continuous candidate by its norm.

Let $L = \inf V$ and $H = \sup V$. Explicit integrability and integral monotonicity imply $L \leq E[V(Ra+e)] \leq H$ (`continuation_range_bounds`). The feasible action zero gives $V(z) \geq U_{inf} + \beta L$; an attained optimizer gives $V(z) \leq U_{sup} + \beta H$. Taking infimum and supremum, respectively, yields

$$U_{inf} + \beta L \leq L, \qquad H \leq U_{sup} + \beta H.$$

Since $1 - \beta > 0$, division and $L \leq V(z) \leq H$ give exactly

$$\frac{U_{inf}}{1-\beta} \leq V(z) \leq \frac{U_{sup}}{1-\beta}.$$

No attainment of the infimum or supremum of V on the unbounded state space is assumed. This argument uses conditional-completeness lemmas with proved nonemptiness and boundedness; it does not replace the bounds by an arbitrary symmetric constant.

Exact elaborated anchor and canonical definition type:

```
Aiyagari1994.valueFunction_unique_fixedPoint (m : Aiyagari1994.HouseholdPrimitives) :
  Aiyagari1994.bellmanOperator m (Aiyagari1994.valueFunction m) = Aiyagari1994.valueFunction
  ↪  m ∧
    (∀ (v : Aiyagari1994.ValueSpace), Aiyagari1994.bellmanOperator m v = v → v =
    ↪  Aiyagari1994.valueFunction m) ∧
      (∀ (v : Aiyagari1994.ValueSpace),
          TendstoUniformly (fun n z => ((Aiyagari1994.bellmanOperator m)^[n] v) z)
          ↪  (⇑(Aiyagari1994.valueFunction m))
            Filter.atTop) ∧
        ∀ (z : Aiyagari1994.Resources),
          Aiyagari1994.utilityInf m / (1 - m.beta) ≤ (Aiyagari1994.valueFunction m) z ∧
            (Aiyagari1994.valueFunction m) z ≤ Aiyagari1994.utilitySup m / (1 - m.beta)
'Aiyagari1994.valueFunction_unique_fixedPoint' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]

Aiyagari1994.valueFunction (m : Aiyagari1994.HouseholdPrimitives) : Aiyagari1994.ValueSpace
'Aiyagari1994.valueFunction' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Source correspondence: A94 equations (4)-(7), printed pp. 666-667 / PDF pp. 9-10, visually inspected. A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39, is the manifest context and was also inspected. Its differentiability and envelope claims are outside M02A; the BASIC Bellman construction is explicitly reconstructed under architecture section 3.

Qualification. The canonical function is the unique bounded continuous Bellman fixed point. Its interpretation as lifetime-optimal expected utility remains H05. Audit results contain only `propext`, `Classical.choice`, and `Quot.sound`.

H03 — Value concavity and strict increase

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Review gate:** M02A. See `reviews/02a_acceptance.md`.

Target: `Aiyagari1994.valueFunction_concave_strictMono`.

Module: `Aiyagari1994/Household/Value.lean`.

Actual assumptions and dependencies. Only BASIC, through the canonical H02 value function, primitive strict concavity and strict monotonicity of utility, and the affine nonnegative transition. No differentiability, IID histories, income nondegeneracy or impatience condition is used.

Ordinary concavity. `NNRealConcave` quantifies over all states x, y and all nonnegative real weights a, b summing to one, represented as NNReal. It asserts $af(x) + bf(y) \leq f(ax + by)$. `NNRealConcave.real_combination` explicitly yields the usual formula with any real $\theta \in [0, 1]$. This avoids imposing a real vector-space instance on NNReal without weakening real convex-combination concavity.

Preservation proof. Given feasible savings $s \leq x$ and $t \leq y$, $as + bt \leq ax + by$. Consumption at the combined pair is $a(x - s) + b(y - t)$. Primitive strict concavity supplies ordinary utility concavity. `transition_convex_combination` proves the affine identity for next resources, including the constant income term by $a + b = 1$. Concavity of v therefore gives a pointwise continuation inequality. `continuation_concave` integrates it with explicit integrability of each summand and uses integral addition and scalar multiplication. Multiplication by positive beta and addition of the utility inequality prove `objective_convex_combination`.

Take attained optimizers at the two original states. Their convex combination is feasible at the combined state and cannot exceed that state's maximum. This proves `bellman_preserves_concavity`.

Iteration and limit. Zero is concave and weakly increasing (`nnrealConcave_zero`, `monotone_const`). Induction gives `valueIteration_concave` and `valueIteration_monotone` for zero-start iterates. The uniform convergence established in H02 supplies the convergent evaluations at each of the three states in the concavity inequality, and at both states in the monotonicity inequality. Passing inequalities through these proved limits establishes `valueFunction_concave` and `valueFunction_monotone`. No unproved limit principle is used.

Strict increase. For $x < y$, take an optimizer at x and keep the same saving action at y. It remains feasible. Current consumption strictly increases; the continuation term is exactly unchanged. Strict utility monotonicity makes the new feasible objective strictly exceed the old maximum. Therefore Tv is strictly increasing for every candidate v (`bellman_strictMono`), a slightly stronger intermediate result than monotonicity preservation. Applying the fixed-point identity proves `valueFunction_strictMono`. No envelope theorem or derivative is involved.

Exact elaborated anchor and real concavity formula:

```
Aiyagari1994.valueFunction_concave_strictMono (m : Aiyagari1994.HouseholdPrimitives) :
  Aiyagari1994.NNRealConcave ⇑(Aiyagari1994.valueFunction m) ∧ StrictMono
  ↪ ⇑(Aiyagari1994.valueFunction m)
'Aiyagari1994.valueFunction_concave_strictMono' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

```
Aiyagari1994.NNRealConcave.real_combination {f : Aiyagari1994.Resources → ℝ} (hf :
↪ Aiyagari1994.NNRealConcave f)
  (x y : Aiyagari1994.Resources) (theta : ℝ) (h0 : 0 ≤ theta) (h1 : theta ≤ 1) :
  theta * f x + (1 - theta) * f y ≤ f ⟨theta * ↑x + (1 - theta) * ↑y, ⋯⟩
'Aiyagari1994.NNRealConcave.real_combination' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Source correspondence: A94 equations (4)-(7), printed pp. 666-667 / PDF pp. 9-10, visually inspected. A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39, is the manifest context and was also inspected. Its differentiability and envelope claims are outside M02A; the BASIC Bellman construction is explicitly reconstructed under architecture section 3.

Qualification. This is concavity and strict increase of the canonical value function only. Policy monotonicity, differentiability and drift remain later contracts. The no-sorry and transitive-axiom audits pass with the accepted foundational axioms.

H04 — Unique continuous shifted-asset policy

Status: GREEN, externally accepted 2026-09-11. **Scope:** core. **Review gate:** M02A. See `reviews/02a_acceptance.md`.

Target: `Aiyagari1994.assetPolicy_unique_continuous`.

Module: `Aiyagari1994/Household/Policy.lean`.

Actual assumptions and dependencies. Only BASIC and H03's canonical concave value function. In particular, there is no supplied optimizer/policy, derivative, Euler condition, positive-consumption premise, finite labor law, bounded asset space or invariant law.

Existence and uniqueness. `AssetOptimal m z a` means $a \le z$ and $Q_V(z,a) = V(z)$; a is NNReal. H01 attainment and the H02 fixed-point equation prove `assetOptimal_exists`. For two distinct feasible actions, consumption $z - a$ is distinct. `objective_strictConcave_assets` proves strict concavity for every positive pair of real convex weights by adding strict current utility concavity and weak expected continuation concavity. Combined actions remain feasible.

At zero all feasible actions equal zero. Otherwise two distinct maximizing actions would give a midpoint with objective strictly above their common maximum, a contradiction. This proves `assetOptimal_unique` and then `assetOptimal_existsUnique`. Classical choice from this proved unique existence defines the canonical `assetPolicy`. It is never a primitive field. `assetPolicy_optimal` and `assetPolicy_le_state` give maximization and feasibility. `assetOptimal_iff_maximizes` proves equivalence with maximization over every feasible shifted asset; `assetPolicy_existsUnique_maximizer` exposes this as genuine unique existence for use by H05.

Reusable continuity lemma: `compact_unique_argmax_continuous`.

Analysis module: `Analysis/ParametricMax.lean`.

Continuity proof. The reusable lemma uses a closed-graph proof. For a closed choice set S, intersect the graph $f(x,k) = \max_j f(x,j)$ with $X \times S$. It is closed by joint objective and maximum-value continuity. Projection to X is closed because the choice space is compact. Uniqueness identifies this projection with the inverse image of S under the selected optimizer; hence the optimizer is continuous.

Apply this theorem to the share optimizer only on the open subtype of strictly positive resources. Define `positiveAssetShare` as $A(z)/z$; feasibility puts it in $[0,1]$. `positiveAssetShare_mul` recovers the actual asset choice. At positive z, multiplication by z is injective, so uniqueness of assets implies uniqueness of the share. Joint share-objective continuity supplies the remaining generic hypothesis. Multiplying the continuous share by z gives `assetPolicy_continuous_positive`.

At zero, `assetPolicy_zero` and $0 \le A(z) \le z$ imply convergence to zero by squeezing. Joining this boundary argument with continuity on the open positive domain proves `assetPolicy_continuous`. No unique share at zero is asserted. Continuity of V alone is never used as a policy-continuity argument.

Consumption and forward interface. Define `consumptionPolicy m z = z - assetPolicy m z` in NNReal. The proven asset bound gives its exact real-subtraction identity, nonnegativity, and $c(z)+A(z) = z$ (`consumptionPolicy_coe`, `consumptionPolicy_budget`). It is continuous by subtraction. The main

theorem exposes canonical uniqueness, feasibility, domination of every feasible objective, budget and continuity.

Exact elaborated anchor and maximization interface:

```
Aiyagari1994.assetPolicy_unique_continuous (m : Aiyagari1994.HouseholdPrimitives) :
  (∀ (z : Aiyagari1994.Resources),
      Aiyagari1994.AssetOptimal m z (Aiyagari1994.assetPolicy m z) ∧
        ∀ (a : Aiyagari1994.Resources), Aiyagari1994.AssetOptimal m z a → a =
        ↪  Aiyagari1994.assetPolicy m z) ∧
    Continuous (Aiyagari1994.assetPolicy m) ∧
      (∀ (z a : Aiyagari1994.Resources),
          a ≤ z →
            Aiyagari1994.bellmanObjective m (Aiyagari1994.valueFunction m) z a ≤
              Aiyagari1994.bellmanObjective m (Aiyagari1994.valueFunction m) z
              ↪  (Aiyagari1994.assetPolicy m z)) ∧
        (∀ (z : Aiyagari1994.Resources), 0 ≤ Aiyagari1994.assetPolicy m z ∧
        ↪  Aiyagari1994.assetPolicy m z ≤ z) ∧
          (∀ (z : Aiyagari1994.Resources),
              ↑(Aiyagari1994.consumptionPolicy m z) = ↑z - ↑(Aiyagari1994.assetPolicy m z) ∧
                Aiyagari1994.consumptionPolicy m z + Aiyagari1994.assetPolicy m z = z) ∧
            Continuous (Aiyagari1994.consumptionPolicy m)
'Aiyagari1994.assetPolicy_unique_continuous' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]

Aiyagari1994.assetPolicy_existsUnique_maximizer (m : Aiyagari1994.HouseholdPrimitives) (z :
↪  Aiyagari1994.Resources) :
  ∃! a,
    a ≤ z ∧
      ∀ b ≤ z,
        Aiyagari1994.bellmanObjective m (Aiyagari1994.valueFunction m) z b ≤
          Aiyagari1994.bellmanObjective m (Aiyagari1994.valueFunction m) z a
'Aiyagari1994.assetPolicy_existsUnique_maximizer' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]

Aiyagari1994.assetPolicy (m : Aiyagari1994.HouseholdPrimitives) (z : Aiyagari1994.Resources)
↪  : Aiyagari1994.Resources
'Aiyagari1994.assetPolicy' depends on axioms: [propext, Classical.choice, Quot.sound]

Aiyagari1994.consumptionPolicy (m : Aiyagari1994.HouseholdPrimitives) (z :
↪  Aiyagari1994.Resources) :
  Aiyagari1994.Resources
'Aiyagari1994.consumptionPolicy' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Source correspondence: A94 equations (4)-(7), printed pp. 666-667 / PDF pp. 9-10, visually inspected. A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39, is the manifest context and was also inspected. Its differentiability and envelope claims are outside M02A; the BASIC Bellman construction is explicitly reconstructed under architecture section 3.

Qualification. The asset action is shifted next-period assets, not a share or net assets. These results do not claim policy monotonicity, strict positivity of consumption, or H05 lifetime optimality. Only accepted foundational axioms occur. H05 was UNFORMALIZED at the accepted M02A boundary; the separate M02B entry below records its implementation and external acceptance.

H05 — Canonical Policy lifetime optimal

Status: GREEN, externally accepted on 2026-09-11. **Scope:** core. **Milestone:** M02B.

Target declaration: `Aiyagari1994.canonicalPolicy_lifetime_optimal`.

Module: `Aiyagari1994/Household/Verification.lean`.

Support module: `Aiyagari1994/Household/History.lean`.

Mathematical statement. For every primitive household and initial resource $z_0 \geq 0$, every measurable nonanticipative feasible plan has an absolutely convergent expected discounted utility series. Its sum is at most $V(z_0)$. The recursively constructed canonical plan attains $V(z_0)$ and hence weakly dominates every such plan.

Actual assumptions and scope. BASIC is the existing `HouseholdPrimitives`: $0 < \beta < 1$, utility continuous, bounded, strictly increasing and strictly concave on nonnegative consumption, the existing compact labor support and probability law, and admissible normalized prices. H02 and H04 supply the constructed bounded continuous value and continuous feasible optimal asset policy. IID is discharged by the existing finite product `historyLaw`; there is no new independence field. No CoreRegularity, smoothness, Inada, $\beta R < 1$, nondegeneracy, mean-one labor, bounded assets or stationary law is assumed. The entire NNReal resource space and a general compactly supported labor distribution remain covered. Dependencies P01, H02 and H04 are GREEN. Primitive consistency remains witnessed by P03.

History and comparison class. `History m t` is `Fin t` to labor. Coordinate zero is the newest shock. `historySplit` is the installed measurable equivalence `piFinSuccAbove` at zero; its two projections define `newestShock` and `previousHistory`, and its inverse defines `extendHistory`. Thus at date t a plan can use all t observed shocks, with no dependence on future shocks. The empty date-zero history is unique.

A `FeasiblePlan m z0` contains exactly an action family, its measurability proof and pointwise feasibility. The family has type `(t : Nat) -> History m t -> Resources`. It may vary with time and every observed coordinate; it need not factor through current resources and has no stationarity or optimality field. These are measurable functions of the given shock histories. No separate private randomization variable is added to the approved information space; the comparison class is not restricted to deterministic time sequences or current-state feedback. Pointwise feasibility at every history is the explicit M02B interface, replacing the older design's almost-everywhere wording.

Derived resources and consumption. The initial state is z_0. At date $t+1$, resources are `nextResources` of the date-t action on the previous history and the newest shock. In particular the transition does not use the date-$t+1$ choice. These states are derived from the action family, not stored independently. Measurability follows by composition of measurable history projections, measurable actions and the continuous resource transition. Consumption is NNReal subtraction. Feasibility gives the exact real subtraction identity and $c_t + a_t = z_t$, so utility always receives a nonnegative consumption argument. Measurability of consumption and flow utility follows by subtraction and continuity of utility on NNReal.

Integrability and the criterion. `utilityBound` selects a finite bound from the proved primitive boundedness property; `utilityBound_spec` and `utilityBound_nonneg` certify it. Uniformly over plans, dates and histories, $|U(c_t)| \leq C$. Since each finite history law is a probability measure, `flowUtility_integrable` proves integrability and `expectedFlow_bound` proves $|EU(c_t)| \leq C$. The terminal value composed with measurable resources is likewise integrable, with expected absolute value bounded by $\|V\|_\infty$. The definition `lifetimeUtility` is the tsum of $\beta^t EU(c_t)$. Comparison with $C\beta^t$ proves absolute summability and convergence of the finite sums to this tsum, for every feasible plan. No unproved limiting expectation defines utility.

The stochastic bridge. `history_integral_step` takes an explicit integrability premise on the next-history function. The already constructed `history_step` proves that the history split preserves measure from the next finite product to labor law times previous-history law. Its inverse changes variables in the integral and transfers integrability. Mathlib's `integral_prod_symm` then yields previous-history outer integration and newest-shock inner integration. Applying this result to the next value function identifies the integrated continuation with the next expected terminal value. IID is proved through this product construction, rather than postulated as an expectation identity.

One-step and finite-horizon verification. At every history, feasibility places the plan's actual shifted-asset action in $[0, z_t]$. H04 maximization and the Bellman equality give

$$U(c_t) + \beta E_l V(Ra_t + e(l)) \leq V(z_t).$$

All three terms have explicit integrability theorems. Integrating this inequality and using the history bridge gives

$$EU(c_t) + \beta T_{t+1} \leq T_t,$$

where T_t is the plan's expected terminal value. At horizon zero the history law integrates the constant initial value to $V(z_0)$. Induction, multiplying the one-step inequality by nonnegative β^N, and the finite-sum successor identity give

$$\sum_{t<N} \beta^t EU(c_t) + \beta^N T_N \leq V(z_0).$$

This is the proved `finiteHorizon_verification`, with no telescoping assumption.

Canonical construction and equality. `canonicalResources` is defined recursively from the accepted asset policy and resource transition. Continuity of the asset policy and measurability of the history operations prove measurable canonical actions. The resource family derived from these actions agrees with the recursive family. The accepted bound $A(z) \leq z$ certifies feasibility, giving `canonicalPlan`. Every canonical action attains the Bellman maximum. Integrating the pointwise equality and repeating the same induction proves `canonical_finiteHorizon_verification` as an equality for every horizon.

Infinite-horizon limit. For every comparison plan,

$$|\beta^N T_N| \leq \beta^N \|V\|_\infty \longrightarrow 0.$$

The proof uses only bounded V and $0 < \beta < 1$, regardless of the growth of resources or assets. The finite discounted sums converge by absolute summability. Closedness of the real order gives lifetime utility at most $V(z_0)$; uniqueness of limits applied to the canonical finite-horizon identity gives canonical lifetime utility exactly $V(z_0)$. Substitution proves dominance.

Original-budget bridge. `original_shifted_trajectory_budget_iff` applies P01 at every date of arbitrary real asset and consumption trajectories. Writing shifted holdings as $\widehat{a}_t = a_t + \phi$, the conjunction of the original budget and $a_{t+1} \geq -\phi$ is equivalent to the shifted budget and $\widehat{a}_{t+1} \geq 0$. For a concrete feasible history plan whose prices are `OriginalPrices.normalized`, `plan_original_budget_step` proves the original budget on every next history, identifying net assets with shifted actions minus ϕ. The separate initial-date theorem uses the exact initial coordinate relation. Consumption is unchanged by the coordinate shift. These are conditional coordinate identities, not new assumptions on H05's general normalized theorem.

Source correspondence. A93 Appendix Proposition 2, printed pp. 37–38 / PDF pp. 38–39; A94 equations (5)–(7), printed pp. 666–667 / PDF pp. 9–10. These approved source pages were visually checked in the household source review. The finite-history product bridge, integrability and explicit verification details are reconstructed proofs, not claims that the papers contain Lean-ready statements. P01 supplies the original coordinate algebra.

Exact elaborated main theorem and transitive axioms.

```
Aiyagari1994.canonicalPolicy_lifetime_optimal (m : Aiyagari1994.HouseholdPrimitives) (z0 :
↪  Aiyagari1994.Resources) :
  (∀ (p : Aiyagari1994.FeasiblePlan m z0), Summable fun (t : ℕ) => |m.beta ^ t *
  ↪  Aiyagari1994.expectedFlow p t|) ∧
    (∀ (p : Aiyagari1994.FeasiblePlan m z0), Aiyagari1994.lifetimeUtility p ≤
    ↪  (Aiyagari1994.valueFunction m) z0) ∧
      Aiyagari1994.lifetimeUtility (Aiyagari1994.canonicalPlan m z0) =
      ↪  (Aiyagari1994.valueFunction m) z0 ∧
        ∀ (p : Aiyagari1994.FeasiblePlan m z0),
          Aiyagari1994.lifetimeUtility p ≤ Aiyagari1994.lifetimeUtility
          ↪  (Aiyagari1994.canonicalPlan m z0)
'Aiyagari1994.canonicalPolicy_lifetime_optimal' depends on axioms: [propext,
↪  Classical.choice, Quot.sound]
```

Audit and review evidence. Fresh builds of both M02B modules, the full project build, direct `Audit.lean`, exact signature probe, contract checker and prohibited-pattern scan pass. All 65 new exported declarations, including the feasible-plan projections, pass `assert_no_sorry` and transitive axiom printing. Across the entire audit there are 254 checked declarations; only `propext`, `Classical.choice` and `Quot.sound` occur. Exact required interfaces and structure fields appear in `reports/02b_signatures.md`; raw evidence is in `reports/logs/02b/`. H05 is GREEN by external review in `reviews/02b_acceptance.md`. Lifetime utility is a series of expectations under consistent finite-history laws; no literal infinite-product path random variable has been constructed. At the accepted M02B boundary, H06 and all later contracts were UNFORMALIZED; no M03 work was included. The M03A entries below record the separate H07/H08 review submission.

H06 — Policy jointly continuous

Status: GREEN. Independent Astra acceptance: `reviews/m04a_acceptance.md`. **Scope:** core. **Review gate:** M04A.

Target declaration: `Aiyagari1994.policy_jointly_continuous`.
Module: `Aiyagari1994/Household/ParameterContinuity.lean`.

Mathematical statement and parameter domain. Fix a `HouseholdPrimitives` object `m`, hence its utility, utility proof, compact iid labor law, income support and discount factor. `AdmissibleNormalizedPrices m.income` is the subtype of real triples (R, w, k) satisfying $R > 0$, $w > 0$, and $w\ell + k \geq 0$ on the fixed labor support. `m.withPrices q` changes only this normalized price triple. The theorem proves that

$$(q, z) \longmapsto V_{m[q]}(z), \qquad (q, z) \longmapsto A_{m[q]}(z)$$

are continuous on the full product of admissible prices and nonnegative resources. There is no condition on βR; critical and supercritical returns are included.

Actual assumptions and dependencies. BASIC only. H02 supplies the canonical bounded continuous Bellman fixed point and contraction iteration; H04 supplies the canonical unique shifted-asset maximizer and $0 \leq A(z) \leq z$. Utility, the labor probability law, and beta are fixed definitionally by `withPrices`. No smoothness, Inada condition, CoreRegularity, income nondegeneracy, mean normalization, consumption positivity, invariant law, bounded state space, or impatience condition $\beta R < 1$ is assumed.

Value proof. Starting from the zero bounded value, each finite Bellman iterate is jointly continuous in (q, z). The transition $(q, a, \ell) \mapsto Ra + w\ell + k$ is continuous, compact-support parametric integration preserves continuity, and maximization uses the fixed compact share set $[0, 1]$. If $|U| \leq C$, contraction gives the price-uniform tail estimate

$$\left|T_q^n 0(z) - V_q(z)\right| \leq \beta^n \frac{C}{1-\beta}.$$

Thus the jointly continuous finite iterates converge uniformly over all admissible (q, z) to the parameterized value function. This does not assert or use global sup-norm continuity of $q \mapsto T_q$ on the unbounded state space.

Policy proof and zero boundary. On $z > 0$, express the policy as the unique maximizing share times resources. The objective is jointly continuous because the newly proved value function is jointly continuous and the labor integral is over the fixed compact probability space. The compact unique-argmax theorem makes the share continuous, hence $A_q(z)$ is jointly continuous on positive resources. At $z = 0$, uniqueness of shares is neither claimed nor needed: H04 gives $0 \leq A_q(z) \leq z$ uniformly in prices, so the squeeze theorem proves joint continuity at every $(q, 0)$.

Exact elaborated contract signature and transitive axioms.

```
Aiyagari1994.policy_jointly_continuous (m : Aiyagari1994.HouseholdPrimitives) :
  (Continuous fun x =>
      (Aiyagari1994.valueFunction (Aiyagari1994.HouseholdPrimitives.withPrices m x.1)) x.2) ∧
    Continuous fun x =>
      Aiyagari1994.assetPolicy (Aiyagari1994.HouseholdPrimitives.withPrices m x.1) x.2
'Aiyagari1994.policy_jointly_continuous' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

Every new public declaration is covered by `#check`, `assert_no_sorry`, and `#print axioms` in `Audit.lean` and `Probes/M04ASignatures.lean`. Only `propext`, `Classical.choice`, and `Quot.sound` occur.

Source correspondence and qualifications. Approved locator: A93 Appendix Proposition 2, printed pp. 37–38 / PDF pp. 38–39; A94 equations (5)–(7), printed pp. 666–667 / PDF pp. 9–10. The finite-horizon, uniform-tail, normalized-parameter and zero-boundary details are the explicitly approved reconstruction in architecture §6.1, not a claim that the paper prints this Lean proof. All mandatory predecessor qualifications remain in force. In particular, H06 does not invoke any marginal object, does not identify `rightMarginalValue m 0` with the economic boundary marginal, and makes no stationarity, tightness, drift, invariant-law, or equilibrium claim. REVIEW_READY is not GREEN and does not authorize any later contract.

H07 — Policy order and Lipschitz bounds

Status: GREEN, externally accepted on 2026-09-11; see `reviews/03a_acceptance.md`. **Scope:** core. **Review gate:** M03A.

Target: `Aiyagari1994.policies_order_lipschitz`.

Module: `Aiyagari1994/Household/PolicyOrder.lean`.

Mathematical statement. Both shifted saving A and consumption c are nondecreasing. For $0 \le z_1 \le z_2$,

$$0 \le A(z_2) - A(z_1) \le z_2 - z_1, \qquad 0 \le c(z_2) - c(z_1) \le z_2 - z_1.$$

The public theorem expresses these differences in Real with explicit NNReal coercions. It makes no strict-order or policy-derivative claim.

Actual assumptions and dependencies. BASIC only. H03 supplies concavity of the constructed value, H04 supplies the unique maximizing shifted-asset policy and the consumption budget. The proof uses no H05 lifetime-optimality result. No impatience, smoothness, Inada, income nondegeneracy, mean-one labor, stationary law, asset bound or CoreRegularity enters. The state remains all NNReal and labor retains its general compactly supported probability law.

Reusable concavity inequality. `concave_four_point` proves that, for f concave on $[0, \infty)$,

$$0 \le x < y, \quad x \le u, v \le y, \quad u + v = x + y \quad \Longrightarrow \quad f(x) + f(y) \le f(u) + f(v).$$

For $w = u, v$, write

$$w = \frac{y - w}{y - x} x + \frac{w - x}{y - x} y.$$

Concavity gives the two weighted inequalities. Adding them and using $u + v = x + y$ makes each endpoint coefficient equal one. This helper is proved directly from the definition of concavity, without a comparative-statics assumption.

Savings order. Suppose $A(z_2) < A(z_1)$. Both actions are feasible at both states. The original consumptions are the endpoints $z_1 - A(z_1)$ and $z_2 - A(z_2)$. Swapping savings gives the two inward consumptions, preserving their sum. Apply the four-point inequality to utility. The sum of the two Bellman optimality inequalities has identical continuation terms on either side; together with the concavity inequality it forces equality in the lower-state comparison. The reversed action is therefore also optimal at z_1. H04's unique maximizer makes the two actions equal, contradicting reversal. Strict concavity of utility enters through this already proved uniqueness theorem; the new comparison uses only weak concavity.

Consumption order. This is a separate argument. Suppose $c(z_2) < c(z_1)$. Compare the canonical savings with $b_1 = z_1 - c(z_2)$ and $b_2 = z_2 - c(z_1)$. Both are nonnegative and feasible at their respective states. The two new savings lie between the original savings and preserve their sum. The continuation function is concave by H03, the affine transition and integration. Apply the four-point inequality to continuation, multiply by nonnegative beta, and combine the two optimality inequalities. Current

utilities cancel because the comparison swaps consumption. Equality again gives a second lower-state optimizer, contradicting H04 uniqueness and the assumed consumption reversal.

Lipschitz bounds. The exact budget is $z = A(z) + c(z)$. Nonnegativity of each difference follows from its own monotonicity. Nonnegativity of the other component's difference supplies its upper bound by $z_2 - z_1$. No differentiability or strict monotonicity is inferred.

Source correspondence. Approved locator: A93 Appendix Proposition 2, printed pp. 37–38 / PDF pp. 38–39; A94 equations (5)–(7), printed pp. 666–667 / PDF pp. 9–10. The A93 pages were visually checked again for M03A. The weak-order proof is reconstructed explicitly from architecture section 4; stronger envelope, Euler and strict-order claims remain outside this gate.

Exact elaborated public theorem and axiom output.

```
Aiyagari1994.policies_order_lipschitz (m : Aiyagari1994.HouseholdPrimitives) :
  Monotone (Aiyagari1994.assetPolicy m) ∧
    Monotone (Aiyagari1994.consumptionPolicy m) ∧
      ∀ (z1 z2 : Aiyagari1994.Resources),
        z1 ≤ z2 →
          (0 ≤ ↑(Aiyagari1994.assetPolicy m z2) - ↑(Aiyagari1994.assetPolicy m z1) ∧
              ↑(Aiyagari1994.assetPolicy m z2) - ↑(Aiyagari1994.assetPolicy m z1) ≤ ↑z2 -
               ↪  ↑z1) ∧
            0 ≤ ↑(Aiyagari1994.consumptionPolicy m z2) - ↑(Aiyagari1994.consumptionPolicy m
             ↪  z1) ∧
              ↑(Aiyagari1994.consumptionPolicy m z2) - ↑(Aiyagari1994.consumptionPolicy m z1)
               ↪  ≤ ↑z2 - ↑z1
'Aiyagari1994.policies_order_lipschitz' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

Verification: fresh module build, full build, direct audit and no-sorry/axiom checks pass. Only standard foundational axioms occur. External review accepted H07; see `reviews/03a_acceptance.md`.

H08 — Right marginal value

Status: GREEN, externally accepted on 2026-09-11; see `reviews/03a_acceptance.md`. **Scope:** core. **Review gate:** M03A.

Target: `Aiyagari1994.rightMarginalValue_properties`.

Module: `Aiyagari1994/Household/RightMarginal.lean`.

Mathematical statement. At every positive real state the canonical right marginal exists as a finite Real, is strictly positive, nonincreasing and continuous from the right, and satisfies

$$0 < q(z) \le \frac{V(z) - V(0)}{z} \le \frac{\operatorname{osc}(U)}{(1-\beta)z}.$$

At zero a separate ENNReal marginal preserves the possible infinite limit. Positive-state marginals converge increasingly to this extended value as states decrease to zero.

Actual assumptions and dependencies. BASIC only, through the accepted H03 concavity and strict increase. H02's exact value bounds supply the oscillation estimate. This bound uses existing constructed objects, not a new hypothesis; the manifest dependency tags remain unchanged. The module imports value analysis, not PolicyOrder or lifetime verification. No derivative of utility, value or policy is assumed. No CoreRegularity, impatience, Inada, stationary law or bounded-state assumption enters.

Real-domain representation. `valueExtension m x` is $V(x.\text{toNNReal})$. `NNRealConcave.extension` converts the accepted NNReal convex-combination inequality to real concavity on $[0, \infty)$. The extension is continuous and strictly increasing on that economic domain. Concavity is never asserted for negative arguments.

Secants and canonical construction. Write

$$S(z,y) = \frac{V(y) - V(z)}{y - z}, \qquad y > z \geq 0.$$

The increment form is exactly $S(z, z+h) = [V(z+h) - V(z)]/h$, certified by `rightSecant_increment`. Concavity yields

$$z < y_1 \leq y_2 \quad \Longrightarrow \quad S(z,y_2) \leq S(z,y_1).$$

Apply the pinned Mathlib theorem for a convex function's right secant infimum to $-V$ on $[0,\infty)$ at the interior point $z > 0$. That theorem proves boundedness of the secants using a point to the left and constructs their finite one-sided limit. Define

$$q(z) = -\inf\{S_{-V}(z,y) : y \geq 0,\ y > z\}.$$

Negating the proved limit gives `rightMarginalValue_hasDerivWithinAt` and the explicit secant limit. There is no ordinary `deriv` definition. A separate bridge proves the equivalent, finite supremum representation

$$q(z) = \sup_{y>z} S(z,y).$$

The real function's economic interpretation is restricted to positive states; its value at zero is not the boundary marginal.

Secant inequalities and positivity. Every right secant obeys $S(z,y) \leq q(z)$. For $0 \leq x < z < t$, concavity gives $S(z,t) \leq S(x,z)$. Letting t decrease to z gives $q(z) \leq S(x,z)$. Consequently,

$$q(y) \leq S(z,y) \leq q(z) \qquad (0 < z < y).$$

This proves monotonicity. Strict increase of V makes $S(z, z+1) > 0$; hence $q(z) > 0$ without differentiability assumptions. Choosing $x = 0$ gives the left-secant bound.

Oscillation bound. `utilityOscillation` is precisely `utilitySup - utilityInf`, using H02's bounded utility range. Subtract the lower bound for $V(0)$ from the upper bound for $V(z)$:

$$V(z) - V(0) \leq \frac{U_{\sup} - U_{\inf}}{1 - \beta}.$$

Divide by positive z. This yields the stated estimate on all positive states, with no artificial cutoff.

Right continuity, proved. For fixed $z > 0$, monotonicity bounds $q(x)$ for $x > z$ above by $q(z)$. Let $L = \sup_{x>z} q(x)$. The monotone-limit theorem gives $q(x) \to L$ as $x \downarrow z$ and $L \leq q(z)$. For fixed $y > z$ and nearby $z < x < y$,

$$S(x,y) \leq q(x) \leq L.$$

Continuity of V and the nonzero fixed secant denominator give $S(z,y) \leq L$ on sending $x \downarrow z$. Send $y \downarrow z$ using the already proved secant limit to get $q(z) \leq L$. Thus $L = q(z)$. The public result is `ContinuousWithinAt (rightMarginalValue m) (Ici z) z` for every $z > 0$, not merely semicontinuity; it implies convergence along every positive sequence decreasing to z.

Extended boundary marginal. Define

$$q_0 = \sup_{y>0} \mathrm{ofReal}(S(0,y)) \quad \text{in ENNReal.}$$

Every secant here is positive, so the nonnegative conversion does not discard an economic sign. The antitone secants converge to this complete-order supremum as $y \downarrow 0$. The inequality $q(z) \leq S(0,z)$

bounds every converted positive marginal by q_0. Conversely, fix $y > 0$. For $0 < x < y$, $S(x, y) \le q(x)$; continuity at zero allows $x \downarrow 0$, and then taking the supremum over y proves

$$q_0 = \sup_{z>0} \mathrm{ofReal}(q(z)).$$

Since positive marginals are antitone in state, the complete-order monotone-limit theorem now proves their convergence to q_0 as $z \downarrow 0$, even when the limit is infinity. The theorem `zeroRightMarginal_eq_top_iff` characterizes infinity by arbitrarily large converted positive marginals. No theorem assumes or concludes universal finiteness at zero. This gate does not construct a separate economic example with infinite boundary marginal and does not impose an Inada condition.

Source correspondence. This is new concave-analysis/value-marginal infrastructure under architecture section 4.1, inspired by the CW00 context in the manifest, not a literal numbered Aiyagari theorem or an imported economic result. The actual imported analysis is from the pinned Mathlib source: convex negative-value right secant limits, concave secant order and monotone limits. A93's broader differentiability/envelope statements are not inferred here.

Exact elaborated public theorem and axiom output.

```
Aiyagari1994.rightMarginalValue_properties (m : Aiyagari1994.HouseholdPrimitives) :
  (∀ (z : ℝ),
      0 < z →
        Filter.Tendsto (slope (Aiyagari1994.valueExtension m) z) (nhdsWithin z (Set.Ioi z))
            (nhds (Aiyagari1994.rightMarginalValue m z)) ∧
          0 < Aiyagari1994.rightMarginalValue m z ∧
            ContinuousWithinAt (Aiyagari1994.rightMarginalValue m) (Set.Ici z) z ∧
              Aiyagari1994.rightMarginalValue m z ≤
                  (Aiyagari1994.valueExtension m z - Aiyagari1994.valueExtension m 0) / z ∧
                (Aiyagari1994.valueExtension m z - Aiyagari1994.valueExtension m 0) / z ≤
                  Aiyagari1994.utilityOscillation m / ((1 - m.beta) * z)) ∧
    AntitoneOn (Aiyagari1994.rightMarginalValue m) (Set.Ioi 0) ∧
      Filter.Tendsto (fun (y : ℝ) => ENNReal.ofReal (slope (Aiyagari1994.valueExtension m) 0
      ↪  y))
          (nhdsWithin 0 (Set.Ioi 0)) (nhds (Aiyagari1994.zeroRightMarginal m)) ∧
        Filter.Tendsto (fun (z : ℝ) => ENNReal.ofReal (Aiyagari1994.rightMarginalValue m z))
        ↪  (nhdsWithin 0 (Set.Ioi 0))
          (nhds (Aiyagari1994.zeroRightMarginal m))
'Aiyagari1994.rightMarginalValue_properties' depends on axioms: [propext, Classical.choice,
↪  Quot.sound]
```

Exact canonical definition types.

```
Aiyagari1994.rightMarginalValue (m : Aiyagari1994.HouseholdPrimitives) (z : ℝ) : ℝ
'Aiyagari1994.rightMarginalValue' depends on axioms: [propext, Classical.choice, Quot.sound]
Aiyagari1994.zeroRightMarginal (m : Aiyagari1994.HouseholdPrimitives) : ENNReal
'Aiyagari1994.zeroRightMarginal' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Verification and boundary of the predecessor gate. Both M03A targets and all supporting exports pass fresh builds, the full build, direct `Audit.lean`, no-sorry assertions, the prohibited-pattern scan and transitive axiom checks. Only `propext`, `Classical.choice` and `Quot.sound` occur. The exact signature report contains all secant, right-continuity and boundary-limit interfaces plus the printed definitions. H07 and H08 are GREEN by external review; H01–H05 remain GREEN. Historically, at M03A acceptance, H09 and later targets were UNFORMALIZED. Current statuses are generated in the economic-status overview and individual contract headings. H09 passed independent Astra review; see `reviews/m03b1_acceptance.md` and its structured counterpart. The H10 heading records its current independent-review status. The accepted boundary qualification remains mandatory:

never use `rightMarginalValue m 0` as the economic zero-state marginal; use the separate extended `zeroRightMarginal`.

H09 — Right Marginal Value superharmonic

Status: GREEN. Independent Astra acceptance: `reviews/m03b1_acceptance.md`.

Scope: core. **Milestone:** 03. **Review gate:** M03B1.

Target declaration: `Aiyagari1994.rightMarginalValue_superharmonic`.
Module: `Aiyagari1994/Household/MarginalInequality.lean`.

Mathematical contract. For every z with finite right marginal, q(z)>=beta*R*Integral q(R*A(z)+e) dnu. Establish the extended-integral inequality first and deduce conditional integrability; it is valid without beta*R<1 and without consumption positivity.

Assumption profiles: BASIC. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H04, H08. **Source keys:** CW00.

Source locator: New concave-analysis/value-marginal infrastructure for the Aiyagari claims; inspired by CW00 Sections 2-4. Not a literal numbered Aiyagari theorem.

Exact state-space interface. The new ENNReal-valued marginal is defined on all `Resources`. At `z=0` it is exactly `zeroRightMarginal m`; at positive `z` it converts the accepted finite real marginal with `ENNReal.ofReal`. Thus the economic zero marginal can be infinite, and the real marginal function is never evaluated economically at zero.

The primary extended theorem proves the unconditional `lintegral` inequality. The contracted theorem `rightMarginalValue_superharmonic` assumes only that the left marginal is finite and exposes almost-everywhere continuation finiteness, Bochner integrability of `toReal`, and the real integral inequality.

Actual assumptions and dependencies. BASIC only. The proof uses H04's canonical Bellman optimizer and H08's concave right-secant limits, including its separate extended boundary limit. Positivity of beta and gross return comes from `HouseholdPrimitives`. There is no `beta*R<1`, consumption-positivity, smoothness, Inada, nondegeneracy, stationarity, invariant law, or bounded-asset premise. The state is all `NNReal`, and the labor law is the original general compactly supported probability law.

Readable proof. Fix a state `z`, its canonical shifted saving `A(z)`, and `h>0`. At state `z+h`, choose shifted saving `A(z)+h`. It is feasible because `A(z)<=z`, and it preserves current consumption exactly. Bellman optimality therefore gives

$$V(z+h) - V(z) \geq \beta \int [V(RA(z) + e(\ell) + Rh) - V(RA(z) + e(\ell))], d\nu(\ell).$$

The finite-h integrands are continuous and integrable. Divide by `h`; rewriting each continuation quotient as a secant introduces the exact positive factor `R`:

$$S(z, z+h) \geq \beta R \int S(x(\ell), x(\ell) + Rh), d\nu(\ell), \qquad x(\ell) = RA(z) + e(\ell).$$

Every secant is nonnegative by strict increase of value, so the proved finite-h real integral identity can be converted to a `lintegral` without assigning economic meaning to an unproved marginal integral.

Take `h_n=1/(n+1)`. Concavity makes the nonnegative continuation secants increase pointwise as `n` increases. At a positive continuation state H08 identifies their limit with `ofReal(q)`; at a zero continuation state it identifies the limit with the separate `zeroRightMarginal`. Mathlib's monotone convergence theorem for `lintegral` therefore gives

$$\text{ofReal}(\beta R) \int^{-} \bar{q}(RA(z) + e(\ell)), d\nu(\ell) \leq \bar{q}(z),$$

where `bar q` denotes `extendedRightMarginalValue`. This theorem holds even if either side is infinite.

If `bar q(z)<infinity`, positivity of beta and `R` makes the extended continuation integral finite. Measurability plus this finite `lintegral` proves `bar q(x(ell))<infinity` almost everywhere and integrability of its `toReal`. Only then does `integral_toReal` identify the real integral with the extended integral's finite real value and yield

$$\beta R \int \bar{q}(RA(z) + e(\ell)).\text{toReal}\, d\nu(\ell) \leq \bar{q}(z).\text{toReal}.$$

At positive `z`, H08 makes the right side exactly the finite real `q(z)`. At zero, the premise and conclusion continue to refer to `zeroRightMarginal`.

Source correspondence. CW00 motivates the value-marginal supermartingale inequality in its overview, printed p. 367 / PDF p. 3, and states the corresponding conditional right-value- derivative inequality in Lemma 1(a), printed p. 372 / PDF p. 8. The paper calls the proof standard and omits it; the Lean proof is the new secant/monotone-convergence construction required by architecture section 4.1, not a formalization of a literal numbered Aiyagari theorem.

Exact elaborated contract signature. The full kernel print, including the extended prerequisite theorem, is also in the M03B1 signature report and probe.

```
Aiyagari1994.rightMarginalValue_superharmonic
  (m : Aiyagari1994.HouseholdPrimitives)
  (z : Aiyagari1994.Resources)
  (hz : Aiyagari1994.extendedRightMarginalValue m z < ⊤) :
  Filter.Eventually
    (fun l : m.income.Labor =>
      Aiyagari1994.extendedRightMarginalValue m
        (m.prices.nextResources (Aiyagari1994.assetPolicy m z) l) < ⊤)
    (Measure.ae (m.income.law : Measure m.income.Labor)) ∧
  MeasureTheory.Integrable
    (fun l : m.income.Labor =>
      (Aiyagari1994.extendedRightMarginalValue m
        (m.prices.nextResources (Aiyagari1994.assetPolicy m z) l)).toReal)
    (m.income.law : Measure m.income.Labor) ∧
  m.beta * m.prices.grossReturn * ∫ l : m.income.Labor,
      (Aiyagari1994.extendedRightMarginalValue m
        (m.prices.nextResources (Aiyagari1994.assetPolicy m z) l)).toReal
      ∂(m.income.law : Measure m.income.Labor) ≤
    (Aiyagari1994.extendedRightMarginalValue m z).toReal
'Aiyagari1994.rightMarginalValue_superharmonic' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Audit result. The contracted declaration and every new public helper are checked by `Audit.lean` with `#check`, `assert_no_sorry`, and `#print axioms`. The transitive axiom output contains only `propext`, `Classical.choice`, and `Quot.sound`. Kernel checking is complete. H09 passed independent Astra review and is GREEN; see `reviews/m03b1_acceptance.md` and `reviews/m03b1_acceptance.json`. All qualifications in that acceptance, including the finite-left boundary qualification, remain in force.

H10 — Consumption positive subcritical

Status: GREEN. Independent Astra acceptance: `reviews/m03b2_acceptance.md`.

Scope: core. **Milestone:** 03. **Review gate:** M03B2.

- **Target declaration:** `Aiyagari1994.consumption_positive_subcritical`.
- **Module:** `Aiyagari1994/Household/ConsumptionPositive.lean`.

Mathematical contract. If beta*R<1, then c(z)>0 at every z>0, allowing either a finite or infinite right marginal of U at zero.

Assumption profiles: BASIC, SMOOTH, IMPATIENT. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H04, H08. **Source keys:** A93.

Source locator: A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39; A94 equations (5)-(7), printed pp. 666-667 / PDF pp. 9-10. Lifetime and parameter details are reconstructed explicitly.

Exact state-space and boundary interface. The theorem quantifies over every positive `z : Resources` on the continuous `NNReal` state space. The new object `utilityZeroRightMarginal m : ENNReal` is the supremum of nonnegative utility secants from zero and may equal infinity. This is distinct from the accepted value boundary object `zeroRightMarginal m : ENNReal`. The proof never evaluates `rightMarginalValue m 0` and does not identify the right marginal of value with the right marginal of utility at a corner.

Actual and transitive economic assumptions.

The public signature takes `m : HouseholdPrimitives`, `UtilitySmooth m.utility`, and the strict impatience premise `m.beta * m.prices.grossReturn < 1`. Thus the exposed profiles are exactly BASIC, SMOOTH, and IMPATIENT. The mathematical proof uses BASIC's bounded continuous strictly increasing strictly concave utility, positive beta and return, H04's constructed canonical optimizer, H08's finite positive-state right value marginal, and IMPATIENT. Smoothness is retained in the contracted interface for the source theorem family, although the secant proof establishes this positivity conclusion without differentiating utility. No curvature, nondegeneracy, income density or atom, positive minimum effective income, stationarity, asset bound, or integrability of a marginal expectation is assumed. H09 is accepted but is not needed by this proof; the manifest dependencies remain exactly H04 and H08.

Readable proof. Define the zero utility marginal by

$$L_0 = \sup_{h>0} \mathrm{ofReal}\, \frac{U(h) - U(0)}{h} \in [0, \infty].$$

Strict increase makes $L_0 > 0$, while concavity makes the secants antitone in h and gives their extended-real limit L_0 as $h \downarrow 0$.

First suppose $L_0 < \infty$ and write $L = L_0.\mathrm{toReal}$. Concavity bounds every economic-domain utility increment by $L(y - x)$. Starting from the zero function, finite-horizon induction proves that the Bellman operator preserves the same ordered Lipschitz bound whenever $\beta R \le 1$. If the optimizer at the larger state is feasible at the smaller state, keep that action fixed. Otherwise use the smaller state's consume-zero action: the resource increment splits between current consumption and additional saving, whose discounted continuation contribution is bounded by $\beta R L$ and hence by L. Uniform convergence of Bellman iterates passes the bound to the canonical value function and integration passes it to the continuation value.

If $c(z) = 0$, then H04's budget identity gives $A(z) = z$. For $0 < h \le z$, choosing $z - h$ instead yields current consumption h. Optimality and the continuation Lipschitz bound imply

$$\frac{U(h) - U(0)}{h} \le \beta R L.$$

Taking the extended secant limit gives $L \le \beta R L$, contradicting $L > 0$ and $\beta R < 1$.

If $L_0 = \infty$, retain saving $A(z) = z$ at the larger state $z + h$. This consumes the entire increment without changing continuation value, so

$$\frac{U(h) - U(0)}{h} \le \frac{V(z+h) - V(z)}{h}.$$

The left side tends to infinity in `ENNReal`; H08 identifies the right side's limit with the finite positive-state `rightMarginalValue m z`. This is impossible. This branch uses neither a real conversion of the infinite endpoint marginal nor an integral.

Exact elaborated contract signature. Full helper signatures and kernel prints are in `reports/m03b2_signatures.md` and `Probes/M03B2Signatures.lean`.

```
Aiyagari1994.consumption_positive_subcritical
  (m : Aiyagari1994.HouseholdPrimitives)
  (_hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (hbetaR : m.beta * m.prices.grossReturn < 1)
  (z : Aiyagari1994.Resources) :
  0 < z → 0 < Aiyagari1994.consumptionPolicy m z
'Aiyagari1994.consumption_positive_subcritical' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Source correspondence. A93 Appendix Proposition 2(a), printed p. 37 through the first line of printed p. 38 / PDF pp. 38-39, states positive consumption and gives the source's concavity case argument. A94 equations (5)-(7), printed pp. 666-667 / PDF pp. 9-10, give the Bellman problem, shifted saving policy and resource transition. The finite/infinite endpoint split and finite-horizon Lipschitz construction are the explicit reconstruction required by architecture section 4.2; no lifetime or later envelope/Euler conclusion is claimed here.

Audit result. The contracted theorem and every new public helper are checked in `Audit.lean` by `#check`, `assert_no_sorry`, and `#print axioms`. The only transitive Lean axioms are `propext`, `Classical.choice`, and `Quot.sound`, inherited from the canonical Bellman construction. Kernel checking supports REVIEW_READY only; no economic adequacy or GREEN status is self-awarded.

H11 — Value envelope at positive consumption

Status: GREEN. Independent Astra acceptance: `reviews/m03b3_acceptance.md`.

Scope: core. **Milestone:** M03B3.

Target declaration: `Aiyagari1994.value_envelope_at_positive_consumption`.
Module: `Aiyagari1994/Household/Envelope.lean`.

Mathematical contract. For every canonical household model, positive resource state z, and positive canonical consumption $c(z)$, the real extension of V has derivative $U'(c(z))$ at z. Its previously constructed positive-state right marginal therefore equals the same utility derivative. The theorem is local and has no $\beta R < 1$ premise.

Actual and transitive economic assumptions. `HouseholdPrimitives m` supplies BASIC: $0 < \beta < 1$, bounded continuous strictly increasing and strictly concave utility on nonnegative consumption, a general compact positive labor support and probability law, $R > 0$, $w > 0$, and nonnegative effective income.

`UtilitySmooth m.utility` supplies SMOOTH: C^1 regularity and a positive derivative on strictly positive consumption. The local hypotheses are `0 < z` and `0 < consumptionPolicy m z`. No IMPATIENT, CURVATURE, atom, density, nondegeneracy, stationary-law, asset-bound, endpoint-derivative, or consumption-positivity theorem is assumed.

Dependencies: H03, H04, H08. **Source keys:** BS79, A93.

Source locator: A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39; BS79 Lemma 1, printed p. 728 / PDF p. 3.

Exact elaborated signatures. Lean reports:

```
Aiyagari1994.concave_hasDerivAt_of_lowerTouching {S : Set ℝ} {f g : ℝ → ℝ} {x d : ℝ}
  (hf : ConcaveOn ℝ S f) (hx : x ∈ interior S) (hg : HasDerivAt g d x)
  (heq : g x = f x) (htouch : ∀ᶠ (y : ℝ) in nhds x, g y ≤ f y) : HasDerivAt f d x

Aiyagari1994.value_envelope_at_positive_consumption
  (m : Aiyagari1994.HouseholdPrimitives)
  (hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (z : Aiyagari1994.Resources) (hz : 0 < z)
  (hc : 0 < Aiyagari1994.consumptionPolicy m z) :
  HasDerivAt (Aiyagari1994.valueExtension m)
      (deriv m.utility.utility ↑(Aiyagari1994.consumptionPolicy m z)) ↑z ∧
    Aiyagari1994.rightMarginalValue m ↑z =
      deriv m.utility.utility ↑(Aiyagari1994.consumptionPolicy m z)
```

Readable proof. The generic one-dimensional lemma constructs the finite right derivative of a concave real function at an interior point from its right secants. Concavity orders every left secant above that right derivative. Because the differentiable function touches from below and agrees at the point, its right secants lie below the concave function's right secants, while the concave function's left secants lie below the touching function's left secants. Taking limits identifies the right derivative with the touching derivative and squeezes the left secants to the same value. The two one-sided secant limits give an ordinary derivative.

For the economic wrapper, fix the actual shifted optimizer $A(z)$ and define $W(x) = U(x - A(z)) + \beta\,\text{continuation}(A(z))$. Positive consumption gives $A(z) < z$, so on the open neighborhood $x > A(z)$ the action remains feasible and its consumption remains strictly positive. H04 optimality gives $W(x) \le V(x)$ there and equality at z. SMOOTH differentiates W with derivative $U'(c(z))$. H03 concavity and the lower-touching lemma yield the full derivative of the value extension. Uniqueness of the right secant limit identifies H08's `rightMarginalValue` with that derivative.

Boundary and integrability audit. H11 is stated only at positive z and positive $c(z)$. It never evaluates `rightMarginalValue m 0` and never uses or alters the distinct `zeroRightMarginal : ENNReal`. The continuation term is constant as x varies; no differentiation under an integral and no marginal expectation occurs. Its underlying bounded value integral is the already-integrable H01 continuation object, so no new real-integral economic interpretation is made.

Source correspondence. A93 Appendix Proposition 2(c), printed pp. 37-38 / PDF pp. 38-39, states the envelope equality and explicitly attributes it to Benveniste–Scheinkman. BS79 Lemma 1, printed p. 728 / PDF p. 3, gives the differentiable lower-touching criterion for a concave value function. The Lean helper proves the required one-dimensional version directly from left/right secants rather than importing the paper as an axiom.

Audit result. Both new exported declarations are checked in `Audit.lean` with `#check`, `assert_no_sorry`, and `#print axioms`. Their transitive Lean axioms are exactly `propext`, `Classical.choice`, and `Quot.sound`; the economic theorem inherits choice from the canonical optimizer/fixed point. At the M03B3 REVIEW_READY submission boundary, kernel checking alone

did not certify economic adequacy or GREEN status. Subsequent independent Astra review accepted H11; see `reviews/m03b3_acceptance.md`.

H12 — Euler subcritical

Status: GREEN. Independent Astra acceptance: `reviews/m03c_acceptance.md`.

Scope: core. **Milestone:** M03C.

Target declaration: `Aiyagari1994.euler_subcritical`.
Module: `Aiyagari1994/Household/Euler.lean`.

Exact signature (abridged only by notation). For `m : HouseholdPrimitives`, `hsmooth : UtilitySmooth m.utility`, `hbetaR : m.beta * m.prices.grossReturn < 1`, `z : Resources`, and `hz : 0 < z`, `euler_subcritical` proves: (i) the conditional `eulerNextMarginal` is finite almost everywhere; (ii) its real conversion is integrable; (iii) `beta * R * integral eulerNextMarginal.toReal <= deriv U (c z)`; and (iv), if `0 < assetPolicy m z`, ordinary next-period marginal utility is integrable, agrees pointwise with the explicit extended marginal, and the preceding inequality is an equality. The complete elaborated signature is in `reports/m03c_signatures.md`.

`eulerNextMarginal m x : ENNReal` is `zeroRightMarginal m` when `x=0` and `ENNReal.ofReal (deriv U (c x))` when `x>0`. Thus the conditional expectation has an explicit economic boundary object and never uses `rightMarginalValue m 0` or a default real derivative at zero.

Actual and transitive economic assumptions. BASIC is carried by `HouseholdPrimitives`: $0 < \beta < 1$, bounded continuous strictly increasing and strictly concave utility on nonnegative consumption, a compact positive labor support with a probability law, $R > 0$, and nonnegative effective income. H12 adds SMOOTH (`UtilitySmooth`) and IMPATIENT (`beta*R<1`). It also quantifies over a positive state. Interior equality adds only `0 < assetPolicy m z`. There is no curvature, nondegeneracy, density, atom, positive income floor, stationary law, or bounded-asset assumption.

Dependencies: H09, H10, H11. **Source keys:** A93.

Source locator: A93 Appendix Proposition 2, printed pp. 37-38 / PDF pp. 38-39; A94 equations (5)-(7), printed pp. 666-667 / PDF pp. 9-10. Lifetime and parameter details are reconstructed explicitly.

Readable proof. H10 makes current consumption positive, and H11 identifies the finite positive-state value marginal with current marginal utility. H09 first supplies its unconditional `ENNReal` superharmonic inequality; finiteness of the positive current marginal then proves the conditional `lintegral` finite and only afterward yields a real integrable conditional marginal. The pointwise bridge `eulerNextMarginal_eq_extendedRightMarginalValue` rewrites this certified H09 expectation into H12's utility marginal at positive next states while retaining `zeroRightMarginal` at a zero next state.

For positive shifted savings, every next resource is at least $RA(z) > 0$. The assigned positive-asset continuation helper works on the common neighborhood $a > A(z)/2$: concavity bounds every absolute continuation derivative by the constant $Rq(RA(z)/2)$, which is integrable under the probability law. Differentiation under the integral therefore proves both conditional marginal-utility integrability and the derivative $R\mathbb{E}U'(c(z'))$. Since H10 and H11 apply to every positive next state, no endpoint derivative enters this branch. The canonical optimizer is locally interior because both savings and consumption are positive. Fermat's theorem applied to the actual Bellman objective gives the Euler equality.

Boundary and integrability audit. `zeroRightMarginal : ENNReal` remains the sole value marginal at zero resources and may be infinite. The public theorem exposes its almost-everywhere finiteness under the particular conditional law before calling `.toReal`; the associated real integrability certificate

is returned before the real integral inequality. In the equality branch, positive savings rules out zero next resources pointwise and ordinary marginal utility receives a separate `Integrable` certificate.

Source correspondence. A93 Appendix Proposition 2(c), printed pp. 37-38 / PDF pp. 38-39, states the envelope/Euler inequality and equality for positive next assets. A94 equations (5)-(7), printed pp. 666-667 / PDF pp. 9-10, record the Bellman objective, shifted asset policy, and next resource transition used here. Both exact page ranges were hash-verified, rendered, and visually inspected. The conditional integrability and explicit zero-state reconstruction are formal qualifications supplied by this project rather than assumptions imported from the papers.

Audit result. All six M03C exports pass `assert_no_sorry`. Their printed transitive axioms are exactly `propext`, `Classical.choice`, and `Quot.sound`. At the M03C REVIEW_READY submission boundary, kernel checking alone did not certify economic adequacy or GREEN status. Subsequent independent Astra review accepted H12; see `reviews/m03c_acceptance.md`.

H13 — Binding interval exists

Status: GREEN. Independent Astra acceptance: `reviews/m03d_acceptance.md`.

Scope: core. **Milestone:** 03.

Target declaration: `Aiyagari1994.binding_interval_exists`.
Module: `Aiyagari1994/Household/BorrowingThreshold.lean`.

Mathematical contract. Under beta*R<1 and either e_min>0 or finite U_right_prime(0), there exists zHat>e_min with A(z)=0 for every z in [e_min,zHat].

Assumption profiles: BASIC, SMOOTH, IMPATIENT, THRESHOLD. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H08, H10, H12. **Source keys:** A93.

Source locator: A93 Appendix Proposition 3 and the following note, printed p. 38 / PDF p. 39; A94 threshold discussion, printed p. 667 / PDF p. 10.

Exact signature. The formal endpoint is `minimumEffectiveIncome m : Resources`, the effective income at the lower labor endpoint. The contract theorem is

```
binding_interval_exists
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hbetaR : m.beta * m.prices.grossReturn < 1)
  (hthreshold : Or (0 < minimumEffectiveIncome m)
    (utilityZeroRightMarginal m < OrderTop.top)) :
  Exists fun zHat : Resources =>
    And (minimumEffectiveIncome m < zHat)
      (forall z : Resources, minimumEffectiveIncome m <= z -> z <= zHat ->
        assetPolicy m z = 0)
```

Actual and transitive economic assumptions. `HouseholdPrimitives` supplies BASIC: bounded, continuous, strictly increasing and strictly concave utility; positive beta below one; a general probability law on fixed compact positive labor support; positive wage and gross return; and nonnegative affine effective income. `UtilitySmooth` supplies SMOOTH. The strict inequality `beta*R<1` is IMPATIENT. THRESHOLD is the displayed disjunction: either the derived minimum effective income is positive or the separate utility endpoint marginal `utilityZeroRightMarginal : ENNReal` is finite. There is no curvature, density, atom, nondegeneracy, stationarity, compact asset bound, consumption-positivity premise, or assumed policy threshold. H10 derives consumption positivity. H08 and H12 supply the marginal facts.

Readable proof. Let `e_min` be minimum effective income and let `Q` be its extended value marginal, using `zeroRightMarginal` when `e_min = 0` and the finite positive-state marginal otherwise. If `e_min > 0`, H08 makes `Q` finite. If `e_min = 0` and the utility endpoint marginal is finite, H10's finite-horizon Lipschitz construction proves the global value increment bound. Applying that bound to every secant from zero gives `zeroRightMarginal <= utilityZeroRightMarginal < infinity`. Thus in either branch `0 < Q < infinity`.

Suppose `A(z) > 0`. H10 makes current and next consumption positive. H12 first proves the next marginal-utility integrand is integrable and then gives the interior Euler equality. Every next resource satisfies `R*A(z) + e(l) >= e_min`. H08's antitonicity therefore bounds each next value marginal by `Q`. H11 identifies these positive-state value marginals with marginal utility, so integration and the Euler equality yield `q(z) <= beta*R*Q`. No real integral is used until H12's integrability certificate is available.

Since `0 < beta*R < 1`, `beta*R*Q < Q`. When `e_min > 0`, H08's genuine right continuity gives a right neighborhood of `e_min` on which `q(z) > beta*R*Q`. When `e_min = 0`, H08's convergence of positive-state marginals to the separate finite `zeroRightMarginal : ENNReal` gives the same strict inequality on a punctured right neighborhood. Hence an interior choice is impossible there. At `e_min = 0`, `assetPolicy_zero` handles the endpoint. At `e_min > 0`, the interior upper bound applied at `e_min` itself contradicts `Q <= beta*R*Q`, so the endpoint also binds. Choosing a smaller positive neighborhood endpoint produces the required `zHat` and establishes `A(z)=0` throughout the closed interval.

Boundary and integrability audit. The proof never evaluates `rightMarginalValue m 0`. `zeroRightMarginal : ENNReal` is the sole value marginal at zero and may be infinite before the THRESHOLD branch proves it finite. `utilityZeroRightMarginal` is a distinct utility endpoint object. The only new economic real integral is inherited through H12 together with its explicit integrability proof.

Source correspondence. A93 Appendix Proposition 3, printed p. 38 / PDF p. 39, states the qualified threshold and argues from the interior marginal equality, monotonicity, and `beta*R<1`. A94 printed p. 667 / PDF p. 10 records the shifted policy, transition, and threshold discussion. Both approved pages were hash-verified, rendered, and visually inspected. The formal proof reconstructs the endpoint finiteness and neighborhood argument; it does not certify the unqualified Inada note following Proposition 3.

Audit result. Both M03D exports pass `assert_no_sorry`. Their printed transitive axioms are exactly `propext`, `Classical.choice`, and `Quot.sound`. At the M03D REVIEW_READY submission boundary, kernel checking alone did not certify economic adequacy or GREEN status. Subsequent independent Astra review accepted H13; see `reviews/m03d_acceptance.md`.

H14 — Zero income atom implies nonbinding

Status: GREEN. Independent Astra acceptance: `reviews/m03e_acceptance.md`. **Scope:** diagnostic. **Milestone:** 03.

Target declaration: `Aiyagari1994.zero_income_atom_implies_nonbinding`.
Module: `Aiyagari1994/Household/BorrowingThreshold.lean`.

Mathematical contract. If e_min=0, U has Inada marginal at zero, and Pr(e=0)>0, then A(z)>0 for every z>0. No impatience assumption is needed for this sufficient condition.

Actual assumptions. BASIC is carried by `HouseholdPrimitives`. The public signature separately assumes `UtilitySmooth`, `minimumEffectiveIncome m = 0`, `utilityZeroRightMarginal m = ⊤`, and positive probability of the measurable zero-effective-income event. There is no impatience, curvature, density, nondegeneracy, consumption-positivity, stationarity, or asset-bound premise. Smoothness and the explicit minimum-income equality are retained as contract premises; the proof itself needs only BASIC, utility Inada, and the atom.

Dependencies: H03, H04, H08. **Source keys:** A93.

Source locator: A93 Appendix Proposition 3 and the following note, printed p. 38 / PDF p. 39; A94 threshold discussion, printed p. 667 / PDF p. 10.

Exact elaborated signatures and transitive axioms:

```
Aiyagari1994.zero_income_atom_implies_nonbinding
  (m : Aiyagari1994.HouseholdPrimitives)
  (_hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (_hmin : Aiyagari1994.minimumEffectiveIncome m = 0)
  (hinada : Aiyagari1994.utilityZeroRightMarginal m = ⊤)
  (hatom : 0 < (m.income.law : Measure m.income.Labor)
    {l | m.prices.effectiveIncome l = 0})
  (z : Aiyagari1994.Resources) :
  0 < z → 0 < Aiyagari1994.assetPolicy m z
'Aiyagari1994.zero_income_atom_implies_nonbinding' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Readable proof. First, for every $h > 0$, choosing zero shifted saving at resource state h and comparing with the forced zero action at state zero gives

$$U(h) - U(0) \le V(h) - V(0).$$

Passing to the already-constructed ENNReal secant limits shows `utilityZeroRightMarginal m` $\le$ `zeroRightMarginal m`; Inada therefore implies `zeroRightMarginal m` $= \top$ inside the H14 proof.

Fix $z > 0$ and suppose, for contradiction, that $A(z) = 0$. Let $p > 0$ be the real probability of the event $E = \{e = 0\}$. Concavity and strict increase of utility give a finite positive bound $C = \mathrm{slope}(U; z/2, z)$ on the current-utility cost per unit of any deviation $0 < a < z/2$. Since the zero-state value secants converge to infinity, choose $x > 0$ so small that $a = x/R < z/2$ and

$$\mathrm{slope}(V; 0, x) > \frac{C}{\beta p R} + 1.$$

The bounded value function makes both continuation-value integrands integrable. Their difference is nonnegative for every labor realization by monotonicity of V, and on E it equals $V(x) - V(0)$. Integrating the indicator lower bound yields

$$p[V(x) - V(0)] \le G(a) - G(0).$$

Optimality of the assumed zero action and the utility secant bound then imply

$$\beta p R \, \mathrm{slope}(V; 0, x) \le C,$$

contradicting the choice of x. Hence $A(z) > 0$ for every positive state.

Boundary and integrability audit. The economic zero-state object is exclusively `zeroRightMarginal : ENNReal`; the proof never evaluates `rightMarginalValue m 0`. It takes no real integral of a marginal. The only real integrals are finite continuation-value differences; their integrability is proved before `integral_mono` is used. The atom event is measurable because effective income is continuous, and its ENNReal probability is converted to a positive real only after its finiteness under the probability law is proved.

Source correspondence. A93 Appendix Proposition 3 and its following note, printed p. 38 / PDF p. 39, state the qualified binding result and then the unqualified Inada/zero-minimum-income nonbinding

note. A94 printed p. 667 / PDF p. 10 records the shifted policy, transition, and threshold discussion. Both approved pages were hash-verified, rendered, and visually inspected. H14 proves the architecture's corrected atom-sufficient statement; it does not validate the unqualified note without an atom or certify D01.

Audit result. The M03E export passes `assert_no_sorry`; its printed transitive axioms are exactly `propext`, `Classical.choice`, and `Quot.sound`. At the M03E REVIEW_READY submission boundary, kernel checking alone did not certify economic adequacy or GREEN status. Subsequent independent Astra review accepted H14; see `reviews/m03e_acceptance.md`.

D01 — Inada without atom binding example

Status: GREEN. Independent Astra acceptance: `reviews/m03f_acceptance.md`.

Scope: diagnostic. **Milestone:** 03F.

Target declaration: `Aiyagari1994.inada_without_atom_binding_example`.
Module: `Aiyagari1994/Diagnostics/InadaCounterexample.lean`.

Mathematical contract. In the exact continuous-state uniform-income model in EXACT_DIAGNOSTIC, all core primitive conditions hold and A(z)=0 whenever 0<z<=1/100, despite Inada and e_min=0.

Actual and transitive assumptions. The theorem has no arguments: it constructs the EXACT_DIAGNOSTIC witness. Its definitions fix $\beta = 1/2$, $R = w = 3/2$, $r = 1/2$, $\phi = 2$, labor as the continuous pushforward of uniform $e \in [0,1]$ under $\ell = 2/3 + (2/3)e$, and $U(c) = \sqrt{c}/(1+\sqrt{c})$. The returned `CoreRegularity` proof supplies BASIC, SMOOTH, CURVATURE, NONDEGENERATE, and labor mean one. No consumption-positivity, impatience premise, atom, stationarity, compact asset bound, or numerical-grid premise occurs.

Dependencies: H02, H03, H04. **Source keys:** A93.

Source locator: A93 Appendix Proposition 3 and the following note, printed p. 38 / PDF p. 39; A94 threshold discussion, printed p. 667 / PDF p. 10.

Exact elaborated signature and transitive axioms:

```
Aiyagari1994.inada_without_atom_binding_example :
  CoreRegularity exactDiagnosticModel ∧
  exactDiagnosticOriginalPrices.netRate = 1 / 2 ∧
  exactDiagnosticOriginalPrices.wage = 3 / 2 ∧
  exactDiagnosticOriginalPrices.debtLimit = 2 ∧
  exactDiagnosticModel.beta = 1 / 2 ∧
  exactDiagnosticModel.prices.grossReturn = 3 / 2 ∧
  (∀ l, exactDiagnosticModel.prices.effectiveIncome l = 3 / 2 * (l : ℝ) - 1) ∧
  Measure.map (fun l => exactDiagnosticModel.prices.effectiveIncome l)
    exactDiagnosticIncome.law = volume.restrict (Set.Icc 0 1) ∧
  exactDiagnosticModel.prices.effectiveIncome exactDiagnosticLowerLabor = 0 ∧
  (Measure.map (fun l => exactDiagnosticModel.prices.effectiveIncome l)
    exactDiagnosticIncome.law) {0} = 0 ∧
  exactDiagnosticUtility.utility 0 = 0 ∧
  (∀ c : ℝ, 0 < c → deriv exactDiagnosticUtility.utility c =
    exactDiagnosticMarginal c) ∧
  Tendsto exactDiagnosticMarginal (nhdsWithin 0 (Set.Ioi 0)) atTop ∧
  (∀ z, 0 ≤ valueFunction exactDiagnosticModel z ∧
    valueFunction exactDiagnosticModel z ≤ 2) ∧
  (∀ z, 0 < z → z ≤ (1 / 100 : Resources) →
    assetPolicy exactDiagnosticModel z = 0)
```

```
'Aiyagari1994.inada_without_atom_binding_example' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Readable proof. Direct differentiation on $c > 0$ gives

$$U'(c) = \frac{1}{2\sqrt{c}(1+\sqrt{c})^2}, \qquad -\frac{cU''(c)}{U'(c)} = \frac{1}{2} + \frac{\sqrt{c}}{1+\sqrt{c}} < \frac{3}{2}.$$

The displayed derivative tends to infinity from the right. The affine labor construction sends uniform $e \in [0,1]$ exactly to labor uniform on $[2/3, 4/3]$, has mean one, and sends effective income back to e. Hence effective income is uniform on $[0,1]$, its minimum is zero, and zero has probability zero.

The accepted Bellman bounds, together with $0 \le U < 1$ and $\beta = 1/2$, give $0 \le V \le 2$. Continuity makes every continuation integrand integrable before its real integral is used. After the proved pushforward and interval-integral changes of variables,

$$G(a) = \int_0^1 V(3a/2 + e)\, de.$$

A generic moving-endpoint fundamental-theorem lemma proves the exact identity

$$G'_+(0) = \frac{3}{2}[V(1) - V(0)].$$

The sliding-interval identity and $0 \le V \le 2$ yield $G(a) - G(0) \le 3a$. Thus the discounted continuation gain is at most $3a/2$. For $0 < z \le 1/100$, the exact formula gives $U'(z) > 3/2$. Concavity of U therefore gives $U(z-a) + (3/2)a \le U(z)$ for every $0 \le a \le z$. Combining the inequalities shows that the Bellman objective at any feasible a is no larger than at zero. The accepted uniqueness of the canonical maximizer then proves $A(z) = 0$ throughout the stated interval.

Boundary and scope audit. The proof never uses `rightMarginalValue`, at zero or elsewhere; in particular it does not replace the economic `zeroRightMarginal : ENNReal`. No expected marginal is represented by a real integral. Assets and income are continuous, and the result is an exact analytic witness rather than a computation or finite-grid certificate.

Source correspondence and adequacy note. A93 Appendix Proposition 3 and its following note, printed p. 38 / PDF p. 39, and A94's threshold discussion, printed p. 667 / PDF p. 10, were hash-verified, rendered, and visually inspected. D01 is a proposed correction to the unqualified note after Proposition 3, not to Proposition 3 itself. At the M03F REVIEW_READY submission boundary, kernel checking alone did not certify the counterexample or its source-correction interpretation. Subsequent independent Astra review accepted D01, including the compatibility of the continuous atom-free witness with A93's maintained assumptions and the qualified correction to the unqualified note after Proposition 3; see `reviews/m03f_acceptance.md`.

D02 — Marginal Utility ratio bound

Status: GREEN. Independent Astra acceptance: `reviews/m04b_acceptance.md`. **Scope:** core. **Milestone:** 04.

Target declaration: `Aiyagari1994.marginalUtility_ratio_bound`.
Module: `Aiyagari1994/Analysis/Curvature.lean`.

Kernel-checked statement. For every `m : HouseholdPrimitives` with `hs : UtilitySmooth m.utility` and `hc : UtilityCurvature m.utility`, there are a positive integer `n` and a positive real

threshold `C` such that, whenever `C ≤ c1 ≤ c2`,

$$\frac{U'(c_1)}{U'(c_2)} \le \left(\frac{c_2}{c_1}\right)^n.$$

The exact public signature is:

```
Aiyagari1994.marginalUtility_ratio_bound (m : HouseholdPrimitives)
  (hs : UtilitySmooth m.utility) (hc : UtilityCurvature m.utility) :
  ∃ n : ℕ, 0 < n ∧ ∃ C > (0 : ℝ), ∀ {c1 c2 : ℝ},
    C ≤ c1 → c1 ≤ c2 →
      deriv m.utility.utility c1 / deriv m.utility.utility c2 ≤ (c2 / c1) ^ n
```

Actual and transitive economic assumptions. `HouseholdPrimitives m` supplies BASIC, while `UtilitySmooth m.utility` supplies a strictly positive first derivative on positive consumption and `UtilityCurvature m.utility` supplies C^2 regularity on positive consumption together with an eventual finite upper bound on $-cU''(c)/U'(c)$. The proof uses no impatience, income nondegeneracy, atom, density, invariant-law, bounded-state, policy, envelope, Euler, or endpoint differentiability assumption. BASIC's utility and income fields enter the contracted wrapper but are not needed by the reusable analytic helper.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** SE77, A93.

Source locator: A93 Appendix Proposition 4, printed pp. 38-39 / PDF pp. 39-40; SE77 Theorems 3.8-3.9, printed pp. 161-162 / PDF pp. 11-12.

Readable proof. Take the curvature witnesses $C > 0$ and $M \in \mathbb{R}$, and use the Archimedean property to choose a positive integer $n > \max\{M, 0\}$. For $c > C$, positivity of $U'(c)$ lets the RRA inequality be rearranged to

$$0 \le nU'(c) + cU''(c).$$

Consequently

$$\frac{d}{dc}\big(c^n U'(c)\big) = c^{n-1}\big(nU'(c) + cU''(c)\big) \ge 0.$$

Continuity at the threshold and differentiability on its interior make $c^n U'(c)$ nondecreasing on $[C, \infty)$. Thus $c_1^n U'(c_1) \le c_2^n U'(c_2)$ for $C \le c_1 \le c_2$. Since $c_1, c_2, U'(c_1), U'(c_2)$ are positive, division gives the stated ratio bound.

The reusable helper `power_mul_deriv_monotoneOn` states the intermediate monotonicity result. It is confined to the authorized helper file `Analysis/M04B/PowerMarginal.lean`.

Boundary, source, and adequacy note. This is a positive-consumption analytic statement. It does not evaluate `rightMarginalValue m 0`, replace `zeroRightMarginal`, form a real expected marginal integral, or derive D03's drift and invariant interval. The proof implements the authorized Architecture §6.2 reconstruction corresponding to A93 Appendix Proposition 4, printed pp. 38–39 / PDF pp. 39–40, and SE77 Theorems 3.8–3.9, printed pp. 161–162 / PDF pp. 11–12. The source pages were not re-inspected for this gate; source correspondence relies on the supplied capsule and accepted repository design. Kernel checks establish formal validity, not independent mathematical or source adequacy.

Audit. `#check`, `assert_no_sorry`, and `#print axioms` cover both new public declarations. Their transitive axiom output is exactly `propext`, `Classical.choice`, and `Quot.sound`.

D03 — Uniform upper drift

Status: GREEN. Independent Astra acceptance: `reviews/m04c_acceptance.md`. **Scope:** core. **Milestone:** 04.

Target declaration: `Aiyagari1994.uniform_upper_drift`.
Module: `Aiyagari1994/Household/UpperDrift.lean`.

Kernel-checked statement. Fix the utility, discount factor, compact iid labor law, and a specified set `Q` of admissible normalized price triples. Assume explicit common bounds `0<Rmin≤R≤Rmax`, `beta*R≤gammaStar<1`, `e_max≤EStar`, and `e_max-e_min≤DeltaStar`, with nonnegative `EStar` and `DeltaStar`. Then `uniform_upper_drift` constructs one real `B>0`, independent of `q∈Q`, such that `e_max(q)≤B`, `R(q)*A_q(z)+e_max(q)≤z` whenever `B≤z`, and every transition from `z∈[e_min(q),B]` remains in `[e_min(q),B]`.

Actual assumptions. `HouseholdPrimitives` supplies BASIC. `UtilitySmooth` supplies positive marginal utility on positive consumption, and `UtilityCurvature` supplies the eventual RRA bound used through D02. LOCAL_IMPATIENT is represented exactly by the displayed primitive uniform bounds on the specified set `Q`; no continuity or compactness of `Q`, continuously selected pointwise cap, income density, atom, nondegeneracy, invariant law, or stationary integrability premise is added.

Dependencies: H07, H08, H10, H12, D02. **Source keys:** SE77, A93.

Source locator: A93 Appendix Proposition 4, printed pp. 38-39 / PDF pp. 39-40; SE77 Theorems 3.8-3.9, printed pp. 161-162 / PDF pp. 11-12.

Readable proof. D02 gives a positive integer `n` and tail threshold `C0`. Since `gammaStar<1`, the limit of `gammaStar*(1+DeltaStar/C)^n` as `C` tends to infinity is `gammaStar`, so choose `C≥C0` with this expression below one. H08's secant bound and H10/H11's positive-state envelope identity imply that consumption exceeds `C` above one common resource level `L`. Choose `K` with `Rmin*K>L` and then `B>Rmax*K+EStar`.

For `z≥B`, if `A_q(z)≤K`, the maximal next resource is at most `B≤z`. Otherwise every next resource exceeds `L`. H07 bounds the difference between extreme next consumptions by the income span. D02 therefore bounds every next marginal utility by `(1+DeltaStar/C)^n` times the marginal at the maximal next state. H12's interior Euler equality and `beta*R≤gammaStar` make the current marginal strictly smaller than that maximal-next marginal. H08's positive-state antitonicity then forces the maximal next state below `z`. Finally H07's monotonicity of `A_q`, together with endpoint income bounds, proves forward invariance below `B`.

Boundary and scope audit. All uses of `rightMarginalValue` occur at proved positive resource states. The proof never evaluates `rightMarginalValue m 0`; `zeroRightMarginal : ENNReal` remains the economic boundary object and may be infinite. H07 is used only in its accepted weak order/Lipschitz form. Weak downward drift is not strengthened to strict drift everywhere and no finite-time entry, stationary law, tightness, convergence, asset supply, or equilibrium result is claimed.

Audit. All nine new public declarations in the authorized M04C helper and target modules are covered by `#check`, `assert_no_sorry`, and `#print axioms` in `Audit.lean` and `Probes/M04CSignatures.lean`. Their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

S01 — Household Kernel feller monotone

Status: GREEN. Independent Astra acceptance: `reviews/m05a_acceptance.md`. **Scope:** core. **Milestone:** 05.

Target declaration: `Aiyagari1994.householdKernel_feller_monotone`.
Module: `Aiyagari1994/Stationary/Kernel.lean`.

Mathematical contract. The pushforward of nu under l↦R*A(z)+e(l) is a probability Markov kernel, is Feller, and preserves stochastic order. Establish the kernel test-function integral formula.

Assumption profiles: BASIC. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H04, H07. **Source keys:** A93, SLP89.

Source locator: A93 Appendix Proposition 5 and proof, printed pp. 39-40 / PDF pp. 40-41; SLP89 Section 12.4. The explicit primitive-to-crossing construction is reconstructed here.

Checked statement. For every household primitive object `m`, `householdKernel m` is a probability Markov kernel. For every bounded continuous real test `f`, the map `z ↦ ∫ y, f y ∂ householdKernel m z` is continuous. If such an `f` is nondecreasing, this map is nondecreasing. More generally, for every measurable real test `f`,

```
∫ y, f y ∂ householdKernel m z
  = ∫ l, f (m.prices.nextResources (assetPolicy m z) l) ∂ m.income.law.
```

The last identity therefore covers the contracted bounded-measurable test class and is slightly more general at the API level.

Readable proof. The authorized helper defines the jointly continuous map `householdTransition m (z,l) = R*A(z)+e(l)`. Continuity follows from H04's continuity of `A` and the affine transition. The kernel takes the product of the identity Dirac kernel with the constant labor-law kernel and maps that product through `householdTransition`; Mathlib's product and map instances make it Markov. The product-kernel and measure-map formulas reduce integration against the kernel to integration over the labor law. Dominated continuity, with the uniform bound supplied by the bounded continuous test, gives the Feller conclusion. For `z1 <= z2`, H07 gives `A(z1) <= A(z2)`; positivity of `R` makes the transition pointwise ordered under the same labor draw, and integral monotonicity gives stochastic monotonicity.

Boundary and scope audit. The construction uses the general compactly supported probability law in `IncomeData`, not the finite witness. It uses H07 only at weak order strength. It introduces no differentiability, strict order, crossing, invariant-law, convergence, moment, asset-supply, or equilibrium claim. It does not use any marginal-value object, so the required distinction between `zeroRightMarginal` and `rightMarginalValue m 0` is untouched.

Audit. The six new public declarations are:

- `householdTransition` and `householdTransition_continuous`;
- `householdKernel` and `householdKernel_isMarkov`;
- `householdKernel_integral`; and
- `householdKernel_feller_monotone`.

Each has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and the M05A signature probe. Their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

S02 — Lower transition iterates tendsto

Status: GREEN. Independent Astra acceptance: `reviews/m05b_acceptance.md`. **Scope:** core. **Milestone:** 05.

Target declaration: `Aiyagari1994.lower_transition_iterates_tendsto`.
Module: `Aiyagari1994/Stationary/LowerTransition.lean`.

Mathematical contract. Under beta*R < 1, h_min(e_min)=e_min, h_min(z) < z for z > e_min, and h_min iterated from any finite upper bound decreases to e_min.

Actual economic assumptions. `m : HouseholdPrimitives` supplies BASIC: bounded strictly increasing and strictly concave utility on nonnegative consumption, continuous resources, a general compact iid labor law, positive wage and gross return, and nonnegative effective income. The explicit premises are `hsmooth : UtilitySmooth m.utility` and `hbetaR : m.beta * m.prices.grossReturn < 1`. No density, endpoint atom, finite labor support, nondegeneracy, invariant interval, or upper asset bound is assumed.

Dependencies: H04, H08, H12. **Source keys:** A93.

Source locator: A93 Appendix Proposition 5 and proof, printed pp. 39-40 / PDF pp. 40-41; SLP89 Section 12.4. The explicit primitive-to-crossing construction is reconstructed here.

Exact elaborated signature.

```
Aiyagari1994.lower_transition_iterates_tendsto
    (m : Aiyagari1994.HouseholdPrimitives)
    (hsmooth : Aiyagari1994.UtilitySmooth m.utility)
    (hbetaR : m.beta * m.prices.grossReturn < 1) :
  Aiyagari1994.lowerTransition m (Aiyagari1994.lowerEffectiveIncome m) =
      Aiyagari1994.lowerEffectiveIncome m ∧
  (∀ z, Aiyagari1994.lowerEffectiveIncome m < z →
      Aiyagari1994.lowerTransition m z < z) ∧
  ∀ B, Aiyagari1994.lowerEffectiveIncome m ≤ B →
    Antitone (fun n ↦ (Aiyagari1994.lowerTransition m)^[n] B) ∧
    Filter.Tendsto (fun n ↦ (Aiyagari1994.lowerTransition m)^[n] B)
      Filter.atTop (nhds (Aiyagari1994.lowerEffectiveIncome m))
```

Readable proof. Define `lowerEffectiveIncome` by evaluating affine effective income at the lower endpoint of labor support, and define `lowerTransition z = R*A(z)+e_min`. It is continuous by H04 policy continuity and the affine transition formula, and it always lies weakly above `e_min`.

Suppose at a positive state with positive shifted saving that `z ≤ lowerTransition z`. Every labor realization then gives next resources at least `lowerTransition z`, hence at least `z`. H12 gives the interior Euler equality. H10 supplies positive consumption and H11 identifies its marginal utility with H08's positive-state right value marginal. H08 antitonicity therefore bounds every next marginal by the current strictly positive marginal. Probability normalization bounds the integral by that same marginal. Multiplication by `beta*R < 1` makes the Euler equality strictly smaller than its right side, a contradiction. If saving is zero, the lower transition is exactly `e_min`. Thus it is strictly below every

`z>e_min`. At the endpoint, zero saving gives the fixed-point identity directly; positive saving is ruled out by the same Euler contradiction (with `A(0)=0` handling a zero endpoint).

For any `B≥e_min`, every iterate remains above `e_min`, while endpoint fixation and strict drift make the sequence antitone. Its real coercion is bounded below and hence converges. Continuity of the lower transition makes the limit a fixed point. Strict drift excludes every fixed point above `e_min`, so the limit equals `e_min`.

Boundary and integrability audit. Euler reasoning is used only at positive current resources and only after positive saving is proved. H12 supplies integrability of the ordinary next marginal under positive saving. The proof never evaluates `rightMarginalValue m 0`; the zero endpoint is handled by `assetPolicy_zero`, preserving `zeroRightMarginal` as the distinct extended boundary object. The convergence is deterministic convergence of the lower-shock transition iterates, not kernel convergence, invariant-law existence, mixing, or moment convergence.

Audit. The six new public declarations are:

- `lowerEffectiveIncome`;
- `lowerTransition`;
- `lowerTransition_continuous`;
- `lowerEffectiveIncome_le_lowerTransition`;
- `M05B_lower_transition_iterates_tendsto`; and
- `lower_transition_iterates_tendsto`.

Each has `#check`, `assert_no_sorry`, and `#print axioms` coverage in the global audit and the M05B signature probe. Their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. The implementation is kernel checked and submitted as REVIEW_READY only. No adequacy certification or GREEN status is self-awarded.

S03 — Economic crossing condition

Status: GREEN. Independent Astra acceptance: `reviews/m05c_acceptance.md`. **Scope:** core. **Milestone:** 05.

Target declaration: `Aiyagari1994.economic_crossing_condition`.
Module: `Aiyagari1994/Stationary/Crossing.lean`.

Mathematical contract. On a compact invariant interval, construct $d \in (e_{\min}, e_{\max})$, $N \geq 1$, and $\varepsilon > 0$ with $P^N(e_{\min}, [d, B]) \geq \varepsilon$ and $P^N(B, [e_{\min}, d]) \geq \varepsilon$ from endpoint-neighborhood probabilities.

Actual economic assumptions. `m : HouseholdPrimitives` supplies BASIC, including the general compact labor law, positive wage and return, and nonnegative effective income. `hsmooth : UtilitySmooth m.utility` and `hbetaR : m.beta * m.prices.grossReturn < 1` are exactly the SMOOTH and IMPATIENT premises inherited through S02. `hnd : IncomeNondegenerate m.income` gives distinct essential endpoints and positive mass in every relative endpoint neighborhood; it requires neither atoms nor a density. IID is implemented by powers of the S01 policy-induced Markov kernel. The local compact-interval premises are `upperEffectiveIncome m ≤ B` and forward invariance of `[lowerEffectiveIncome m,B]` for every labor realization; these are exactly the price-specific consequences supplied by D03's common-bound conclusion. No invariant probability law, mixing theorem, finite labor support, density, endpoint atom, or upper fixed-point uniqueness is assumed.

Dependencies: D03, S01, S02. **Source keys:** A93, SLP89.

Source locator: A93 Appendix Proposition 5 and proof, printed pp. 39-40 / PDF pp. 40-41; SLP89 Section 12.4. The explicit primitive-to-crossing construction is reconstructed here.

Exact elaborated signature.

```
Aiyagari1994.economic_crossing_condition
    (m : Aiyagari1994.HouseholdPrimitives)
    (hsmooth : Aiyagari1994.UtilitySmooth m.utility)
    (hnd : Aiyagari1994.IncomeNondegenerate m.income)
    (hbetaR : m.beta * m.prices.grossReturn < 1)
    (B : Aiyagari1994.Resources)
    (hUpperB : Aiyagari1994.upperEffectiveIncome m ≤ B)
    (hInvariant : ∀ z, Aiyagari1994.lowerEffectiveIncome m ≤ z → z ≤ B →
      ∀ l,
        Aiyagari1994.lowerEffectiveIncome m ≤
            m.prices.nextResources (Aiyagari1994.assetPolicy m z) l ∧
          m.prices.nextResources (Aiyagari1994.assetPolicy m z) l ≤ B) :
  ∃ d N eps, Aiyagari1994.lowerEffectiveIncome m < d ∧
    d < Aiyagari1994.upperEffectiveIncome m ∧ 1 ≤ N ∧ 0 < eps ∧
    eps ≤ (Aiyagari1994.householdKernel m ^ N)
      (Aiyagari1994.lowerEffectiveIncome m) (Set.Icc d B) ∧
    eps ≤ (Aiyagari1994.householdKernel m ^ N) B
      (Set.Icc (Aiyagari1994.lowerEffectiveIncome m) d)
```

Readable proof. Let $e_{\min}$ and $e_{\max}$ be effective income at the two labor-support endpoints and choose their midpoint d. Wage positivity and endpoint distinctness give $e_{\min} < d < e_{\max}$, while the invariant-interval premises give $e_{\min} \le B$. S02 implies that the iterated minimum-shock transition from B eventually lies below d; choose such an $N \ge 1$. Continuity of the finite constant-shock path provides a labor endpoint neighborhood whose constant-shock path also lies below d. Monotonicity of the canonical asset policy in resources and monotonicity of affine income in labor imply that any sequence of N shocks in that neighborhood ends in $[e_{\min}, d]$. NONDEGENERATE gives this neighborhood mass $p_- > 0$. A kernel-power induction, using the exact S01 pushforward kernel, proves

$$P^N(B, [e_{\min}, d]) \ge p_-^N > 0.$$

For the other direction, every transition starting in the invariant interval remains there. A final shock above the labor midpoint gives effective income above d, regardless of prior wealth, and the invariant upper bound keeps the next state below B. Its mass is $p_+ > 0$ by upper-endpoint nondegeneracy. Since $N \ge 1$, the first $N-1$ transitions stay in the interval with probability one and the last-step kernel bound yields

$$P^N(e_{\min}, [d, B]) \ge p_+.$$

Taking $\varepsilon = \min\{p_-^N, p_+\}$ supplies both inequalities at the same horizon.

Boundary, probability, and scope audit. The proof uses no marginal-value object and in particular never evaluates `rightMarginalValue m 0`; all zero-boundary reasoning remains inside accepted S02. Kernel powers, rather than an assumed infinite product, encode the finite iid shock sequence. Every measure in the conclusion is a probability-kernel value, and no real-valued expectation or new integrability claim occurs. The result is only endpoint crossing on the supplied invariant interval. It proves no invariant-law existence or uniqueness, weak or total-variation convergence, moment convergence, stationary integrability, asset supply, or equilibrium.

Audit. The three new public declarations are:

```
upperEffectiveIncome
M05C_economic_crossing_condition
economic_crossing_condition
```

Each has all three required audit commands in the global audit and the dedicated M05C signature probe: `#check`, `assert_no_sorry`, and `#print axioms`. Their transitive axiom output contains only `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. The implementation is kernel checked and submitted as REVIEW_READY only. No adequacy certification, GREEN status, or later-stage conclusion is self-awarded.

S04 — Compact monotone feller stability

Status: GREEN. Independent Astra acceptance: `reviews/m05d_acceptance.md`. **Scope:** core. **Milestone:** 05.

Target declaration: `Aiyagari1994.compact_monotone_feller_stability`.
Module: `Aiyagari1994/Analysis/MonotoneFeller.lean`.

Mathematical contract. A monotone Feller kernel on a nonempty compact real interval satisfying the common-horizon endpoint-crossing condition has exactly one invariant probability law; every initial probability law converges weakly to it. Prove endpoint invariant existence and oscillation contraction.

Exact signature (readable form). For real endpoints `a ≤ b`, a Markov kernel `k` on `Icc a b`, a Feller proof, and preservation of nondecreasing bounded-continuous tests, fix `d ∈ Icc a b`, `N ≥ 1`, and `0 < eps ≤ 1`. Assume the common-horizon crossing inequalities

```
eps * f(d) + (1-eps) * f(a) ≤ T^[N] f(a)
T^[N] f(b) ≤ eps * f(d) + (1-eps) * f(b)
```

for every nondecreasing bounded-continuous `f`, where `T = testStep k hFeller`. Then there is exactly one probability law `pi` fixed by `lawStep k`; for every initial probability law `mu`, `(lawStep k)^[n] mu` tends weakly to `pi`. In addition,

```
(T^[N])^[m] f(b) - (T^[N])^[m] f(a)
  ≤ (1-eps)^m * (f(b)-f(a)).
```

Assumption profiles: No economic profile. The actual premises are the Markov, Feller, stochastic-monotonicity, compact-interval, positive common-horizon, and crossing hypotheses displayed in the exact signature. The theorem does not assume an invariant law or convergence.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** SLP89.

Source locator: SLP89 Assumption 12.1, Lemma 12.11 and Theorem 12.12, printed pp. 381-383 / PDF pp. 391-393.

Readable proof. `lawStep` pushes probability laws through the kernel, while `testStep` is the Feller Markov operator on bounded-continuous tests. Compactness of probability laws gives cluster points of the lower- and upper-endpoint orbits. For every continuous increasing test, endpoint integrals are respectively monotone increasing and decreasing and bounded by endpoint values. A convergent subsequence therefore identifies the limit of the full scalar sequence. Polynomial approximation proves that continuous increasing tests determine laws and that convergence on this class is weak convergence. Continuity of `lawStep` then makes both endpoint limits invariant.

Applying crossing to each block iterate gives the displayed geometric oscillation contraction. Hence the lower and upper invariant limits agree on continuous increasing tests and therefore as laws. Every initial law is sandwiched between the endpoint laws on those tests, giving weak convergence to their common limit. An invariant initial law has a constant orbit, proving uniqueness. This is weak convergence only; no total-variation or moment convergence is asserted.

Public declarations.

- `RealInterval`; `CI`; `lawStep`; `testStep`.
- `continuous_increasing_tests_determine`.
- `tendsto_probabilityMeasure_of_increasing_tests`.
- `M05D_compact_monotone_feller_stability`.
- `compact_monotone_feller_stability`.

Audit. Every public declaration has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M05DSignatures.lean`. Each transitive axiom print contains only `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. Generic mathematical theorem. The crossing premise is the test-function form of the common-horizon endpoint crossing bound; S05 must discharge it from S03 and stochastic monotonicity. The implementation is submitted as REVIEW_READY only. No GREEN status or later stationary/equilibrium conclusion is self-awarded.

S05 — Stationary Law exists unique global

Status: GREEN. Independent Astra acceptance: `reviews/m05e_acceptance.md`. **Scope:** core. **Milestone:** 05.

Target declaration: `Aiyagari1994.stationaryLaw_exists_unique_global`.
Module: `Aiyagari1994/Stationary/GlobalStability.lean`.

Mathematical contract. For the core strictly impatient household, construct one invariant probability law supported on [e_min,B]. It is the only invariant probability law on all NNReal, and every initial probability law converges weakly to it.

Exact signature (readable form). Given `m : HouseholdPrimitives`, `UtilitySmooth m.utility`, `UtilityCurvature m.utility`, `IncomeNondegenerate m.income`, and `m.beta * m.prices.grossReturn < 1`, there exist `B : Resources` and `pi : ProbabilityMeasure Resources` such that `upperEffectiveIncome m <= B`, `pi (Icc (lowerEffectiveIncome m) B) = 1`, `householdLawStep m pi = pi`, every full-space invariant probability law equals `pi`, and for every full-space probability law `mu`, `(householdLawStep m)^[n] mu` tends to `pi` in the weak topology.

Assumption profiles: BASIC is carried by `HouseholdPrimitives`; SMOOTH is `UtilitySmooth`; CURVATURE is `UtilityCurvature`; NONDEGENERATE is `IncomeNondegenerate`; IID is built into `householdKernel` from the fixed labor probability law; IMPATIENT is the displayed strict inequality `beta*R<1`. No positive minimum income, atom, density, finite-state, invariant-law, or convergence premise is assumed.

Dependencies: D03, S01, S03, S04. **Source keys:** A93, SLP89.

Source locator: A93 Appendix Proposition 5 and proof, printed pp. 39-40 / PDF pp. 40-41; SLP89 Section 12.4. The explicit primitive-to-crossing construction is reconstructed here.

Readable proof. D03 is specialized to the household's singleton price set to construct `B`, maximal-shock weak drift above `B`, and forward invariance of `[e_min,B]`. The restricted economic kernel is built on the real compact interval by projecting resources; on an invariant interval it is proved equal to the exact measurable-embedding restriction. S01 supplies Feller continuity and stochastic monotonicity. S03 supplies finite ENNReal endpoint-event probabilities. Their finiteness is proved from the Markov bound before conversion with `ENNReal.toReal`; elementary integral bounds then discharge S04's real increasing-test crossing interface. S04 yields a compact invariant law and weak convergence from every compact-interval law.

The compact law is pushed forward to `NNReal`. For any initial state above `B`, the interval is enlarged to `[e_min,max B z]`; D03's maximal-shock drift proves its invariance. The base compact invariant law lifted into that interval is fixed, so compact convergence identifies the enlarged-interval limit with the base law. A state below `e_min` enters `[e_min,B]` after one step, proved directly from policy monotonicity and income bounds; no finite-time entry from states above `B` is claimed. Pointwise weak convergence is integrated against an arbitrary initial probability law using bounded-continuous tests and dominated convergence. A full-space invariant law has a constant orbit, so global convergence makes it equal to the constructed law.

Public support declarations: `householdLawStep`; the `M05E` restricted-kernel, embedding, invariant-interval, compact-stability, pointwise-stability, and dominated-convergence helpers under `Aiyagari1994/Analysis/M05E/`; and `stationaryLaw_exists_unique_global`.

Audit. Every new public declaration has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M05ESignatures.lean`.

Adequacy note. The conclusion is weak convergence only. It does not assert total-variation convergence, moment convergence, finite-time absorption above `B`, stationary marginal integrability, asset-supply continuity, or equilibrium. The implementation is submitted as REVIEW_READY only; no GREEN status or later contract is self-awarded.

S06 — Stationary Law weakly continuous

Status: GREEN. Independent Astra acceptance: `reviews/m06a_acceptance.md`. **Scope:** core. **Milestone:** 06.

Target declaration: `Aiyagari1994.stationaryLaw_weakly_continuous`.
Module: `Aiyagari1994/Stationary/ParameterContinuity.lean`.

Mathematical contract. With fixed U, nu and beta, the canonical invariant law is weakly continuous at every strictly impatient normalized price vector. Prove that each subsequential limit is invariant on a common compact state interval.

Exact signature (readable form). Given `m : HouseholdPrimitives`, `UtilitySmooth m.utility`, `UtilityCurvature m.utility`, and `IncomeNondegenerate m.income`, the map from the subtype of admissible normalized prices satisfying `m.beta * q.grossReturn < 1` to `ProbabilityMeasure Resources` is continuous in the weak topology. The selected law is the S05-unique invariant law for `m.withPrices q`.

Assumption profiles: BASIC is carried by `HouseholdPrimitives`; SMOOTH, CURVATURE, and NONDEGENERATE are explicit premises. IID is built into `householdKernel` from the fixed labor probability law. IMPATIENT is the defining predicate of `M06A.ImpatientPrices m`. Utility, beta, and the income law are definitionally fixed by `withPrices`.

Dependencies: H06, D03, S05. **Source keys:** A93, SLP89.

Source locator: A93 Appendix Proposition 5; SLP89 Theorem 12.13, printed pp. 384-385 / PDF pp. 394-395.

Readable proof. S05 supplies a canonical invariant law and full-space uniqueness at every strictly impatient price. For any convergent price sequence, strict impatience at the limit gives a neighborhood with a uniform positive return floor, finite return ceiling, impatience margin, upper-income bound, and income-span bound. D03 applied to that neighborhood produces one finite `B`; the interval uses the fixed lower endpoint zero, avoiding identification of price-dependent economic subtype spaces. Compact economic-kernel existence plus S05 uniqueness proves that every tail stationary law gives mass one to `[0,B]`.

For every bounded continuous test `f`, H06 and the affine transition imply joint continuity of `(q,z)` $\mapsto$ $\int$ `f(z') P_q(z,dz')`. Restricting `z` to `[0,B]` turns this into uniform-norm convergence of the one-step test operators. Prokhorov compactness gives a weakly convergent subsequence of any subsequence of stationary laws. The invariance identity passes to its limit: weak convergence handles the fixed limiting test operator, while uniform convergence on `[0,B]` handles the varying operator. The limit is invariant for the limiting full-space kernel and therefore equals the S05-unique invariant law. The unique-cluster-point criterion yields convergence of the whole sequence, and sequential continuity gives continuity on the strictly impatient price subtype.

Public support declarations: `M06A.ImpatientPrices`, the canonical law and invariance/uniqueness selectors, parameterized test-operator continuity and integral identities, an explicit subsequential-limit invariance lemma, common-support lemmas, the sequential convergence theorem, and the S06 target declaration.

Audit. Every new public declaration has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M06ASignatures.lean`. The target prints only `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. The conclusion is weak continuity only. The common bound is local around a strictly impatient limit and is not asserted to remain valid as `beta*R` approaches one. No moment convergence, stationary marginal integrability, asset-supply continuity, total-variation continuity, finite-time entry, density, or continuum law of large numbers is claimed. The implementation is submitted as REVIEW_READY only; no GREEN status or later contract is self-awarded.

A01 — Resource asset labor law bridge

Status: GREEN. Independent Astra acceptance: `reviews/m06b_acceptance.md`. **Scope:** core. **Milestone:** 06.

Target declaration: `Aiyagari1994.resource_asset_labor_law_bridge`.
Module: `Aiyagari1994/Aggregate/CrossSection.lean`.

Kernel-checked statement. For a household `m`, debt shift `phi`, resource probability law `pi`, and real net-asset law `rho`, assume exactly `rho = M06B.netAssetLaw m phi pi`, where the right side is `(A-phi)#pi`. Then the resource image of the independent product `rho × nu`, formed by `M06B.assetLaborLaw`, equals `pi` if and only if `householdLawStep m pi = pi`. The resource map first restores shifted assets by `Real.toNNReal (a+phi)` and then applies the normalized transition. On the induced net-asset law this restoration is exactly `A`.

Actual assumptions: BASIC is contained in `HouseholdPrimitives`. IID is represented exactly by the product with the fixed labor probability law `m.income.law`. The bridge itself is valid for every probability law and every real shift and therefore needs no impatience premise; IMPATIENT enters only when S05 is used to supply the canonical invariant `pi`. No smoothness, curvature, nondegeneracy, moment, density, finite-support, or mean-one assumption enters A01.

Exact public target signature:

```
theorem resource_asset_labor_law_bridge (m : HouseholdPrimitives) (phi : ℝ)
    (pi : ProbabilityMeasure Resources) (rho : ProbabilityMeasure ℝ)
    (hrho : rho = M06B.netAssetLaw m phi pi) :
    (M06B.assetLaborLaw m rho).map
        (M06B.resourceFromAssetLabor_continuous m phi).measurable.aemeasurable = pi ↔
      householdLawStep m pi = pi
```

The dedicated signature probe prints:

```
'Aiyagari1994.resource_asset_labor_law_bridge' depends on axioms:
[propext, Classical.choice, Quot.sound]
```

Dependencies: P01, S05. **Source keys:** A93, A94.

Source locator: A94 equation (8) and stationary aggregation discussion, printed pp. 667-670 / PDF pp. 10-13; A93 Proposition 5 for invariant-law dependence.

Readable proof. First prove the nonstationary identity `M06B.resourceImage_eq_lawStep`: map `pi` by `z ↦ A(z)-phi`, take its product with `nu`, and map `(a,l)` by the normalized resource transition after restoring `a+phi`. Functoriality of measure pushforwards reduces the composite map pointwise to `(z,l) ↦ nextResources (A(z)) l`. The product measure is the composition-product of `pi` with the

constant labor kernel, so mapping this transition is definitionally the household kernel composed with `pi`, namely `householdLawStep m pi`. Substituting the assumed identity for `rho` and rewriting by this exact law identity proves both directions.

P01 supplies the original-to-normalized budget interpretation of the shift; A01 does not strengthen P01 to No-Ponzi. S05 supplies an invariant law under its accepted additional premises, but A01 does not reprove or strengthen invariant existence, uniqueness, convergence, or moment claims.

Public support declarations: `M06B.netAsset`, `M06B.netAsset_continuous`, `M06B.netAssetLaw`, `M06B.assetLaborLaw`, `M06B.resourceFromAssetLabor`, `M06B.resourceFromAssetLabor_continuous`, and `M06B.resourceImage_eq_lawStep`.

Audit. Every new public declaration has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M06BSignatures.lean`.

Adequacy note. This is a probability-law identity, not a continuum law of large numbers. No real expectation is formed, so no integrability claim is implicit. No boundary marginal, weak-limit argument, stationary moment, asset-supply statement, or later aggregate identity enters. The implementation is submitted as REVIEW_READY only; no GREEN status or later contract is self-awarded.

A02 — Stationary budget identity

Status: GREEN. Independent Astra acceptance: `reviews/m06c_acceptance.md`.

Scope: core. **Milestone:** 06. **Gate:** M06C.

Target declaration: `Aiyagari1994.stationary_budget_identity`.
Module: `Aiyagari1994/Aggregate/AssetSupply.lean`.

Kernel-checked statement. Fix household primitives `m`, utility smoothness and curvature, income endpoint nondegeneracy, and strict impatience `m.beta * m.prices.grossReturn < 1`. Let `p : OriginalPrices m.income` and assume the explicit normalization bridge `m.prices = p.normalized`. For the S05-selected canonical stationary law `pi = M06C.stationaryLaw m hsmooth hcurvature hnd hbetaR`, the following functions are integrable:

```
fun z : Resources => (z : ℝ)
fun z : Resources => (assetPolicy m z : ℝ)
fun z : Resources => (assetPolicy m z : ℝ) - p.debtLimit
fun z : Resources => (consumptionPolicy m z : ℝ)
m.prices.effectiveIncome
```

The theorem then proves, in order,

$$\int z\,d\pi = R\int A(z)\,d\pi + \int e(\ell)\,d\nu, \qquad S = \int A(z)\,d\pi - \phi,$$

and

$$\int c(z)\,d\pi = rS + w\int \ell\,d\nu,$$

Here the notation is:

- `R = m.prices.grossReturn` and `r = p.netRate`;
- `w = p.wage` and `phi = p.debtLimit`;
- `S = stationaryAssetSupply m p.debtLimit pi`.

Actual assumptions and scope. BASIC is carried by `HouseholdPrimitives` and `OriginalPrices`. The canonical-law construction uses SMOOTH, CURVATURE, NONDEGENERATE, and the strict IMPATIENT inequality exactly as S05 requires. IID is the product of the resource law and `m.income.law`; no continuum law of large numbers is asserted. The original-coordinate formulas are conditional on

`m.prices = p.normalized`, preserving P01's normalization qualification. No mean-one labor premise is used, so the labor mean remains explicit. Resources and assets remain continuous; the labor law remains an arbitrary compactly supported probability law.

Dependencies: S05, A01. **Source keys:** A93, A94.

Source locator: A94 equation (8) and stationary aggregation discussion, printed pp. 667-670 / PDF pp. 10-13; A93 Proposition 5 for invariant-law dependence.

Readable proof. `M06C.stationaryLaw` and `M06C.stationaryBound` select the two S05 witnesses only after S05 proves existence. `M06C.stationaryLaw_properties` recovers compact support and invariance. Restricting the stationary probability measure to that compact interval proves the resource, policy, and consumption integrability claims; compact labor support proves effective income integrability. No moment is passed through weak convergence.

For the resource identity, `M06C.resourceImage_eq_lawStep` expresses one transition as the image of the product of current resources and an independent current labor draw. Invariance identifies that image with `pi`. The product integral is integrable from the shifted-asset and effective-income first moments, so Fubini and probability normalization give `E_pi z = R * E_pi A + E_nu e`. This is proved before either later accounting equality.

`stationaryAssetSupply` is the real integral of `A-phi`; integral linearity gives `S=E_pi A-phi`. The pointwise policy budget gives `E_pi c=E_pi z-E_pi A`. Finally the explicit normalization bridge gives `R=1+r` and `e(l)=w*l-r*phi`; substitution and ring algebra yield `E_pi c=r*S+w*E_nu l`.

The additional theorem `M06C.stationary_budget_of_invariant` proves the same three identities for any supplied invariant law with explicit resource, shifted-asset, and consumption integrability premises. This is the legitimate interface for a later equilibrium record: construction derives the moments here, while an arbitrary equilibrium may carry them as required fields.

Public support declarations:

- `stationaryAssetSupply`;
- `M06C.stationaryBound` and `M06C.stationaryLaw`;
- `M06C.stationaryLaw_properties`;
- `M06C.resource_integrable_of_compact_support`;
- `M06C.asset_integrable_of_compact_support`;
- `M06C.consumption_integrable_of_compact_support`;
- `M06C.effectiveIncome_integrable`;
- `M06C.resourceImage_eq_lawStep`;
- `M06C.stationary_resource_identity_of_invariant`;
- `M06C.stationaryAssetSupply_eq_mean_shifted_sub`;
- `M06C.mean_effectiveIncome_of_original`;
- `M06C.stationary_budget_of_invariant`.

Audit. Every new public declaration has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M06CSignatures.lean`.

Adequacy note. S05 supplies compact support but no moment theorem; A02 derives its moments directly from that support. A01's independent asset/labor interpretation remains compatible with the direct resource/labor product used for the expectation calculation. There is no stationary marginal-utility claim, boundary-marginal substitution, moment convergence, parameter continuity, mean-one specialization, equilibrium assertion, or numerical model. The implementation is submitted as REVIEW_READY only; no GREEN status or later contract is self-awarded.

A03 — Stationary Asset Supply continuous

Status: GREEN. Independent Astra acceptance: `reviews/m06dr_acceptance.md`.

Scope: core. **Milestone:** 06. **Gate:** M06DR.

Target declaration: `Aiyagari1994.stationaryAssetSupply_continuous`.
Module: `Aiyagari1994/Aggregate/ParameterContinuity.lean`.

Mathematical contract. With utility, beta, and the compact iid labor law fixed, let `q` range over strictly impatient admissible normalized prices and let `phi` vary independently over `Real`. The map `(q,phi) |-> stationaryAssetSupply (m.withPrices q) phi pi_q`, where `pi_q` is the canonical S05 invariant resource law, is jointly continuous on `M06A.ImpatientPrices m × Real`. Consequently every continuous graph `q |-> (q, phi q)` gives the previously exposed price-indexed continuity statement.

Assumption profiles: BASIC, SMOOTH, CURVATURE, NONDEGENERATE, IID, IMPATIENT. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H06, D03, S06, A02. **Source keys:** A93, C90.

Source locator: A94 equation (8) and stationary aggregation discussion, printed pp. 667-670 / PDF pp. 10-13; A93 Proposition 5 for invariant-law dependence.

Exact signatures (readable form). The unchanged contract anchor takes:

- `m : HouseholdPrimitives`;
- `hsmooth : UtilitySmooth m.utility`;
- `hcurvature : UtilityCurvature m.utility`;
- `hnd : IncomeNondegenerate m.income`;
- `phi : M06A.ImpatientPrices m → ℝ`; and
- `hphi : Continuous phi`.

The theorem `stationaryAssetSupply_continuous` proves:

```
Continuous
  (M06D.stationaryAssetSupplyAtPrice
    m hsmooth hcurvature hnd phi).
```

The new export `stationaryAssetSupply_joint_continuous` takes `m`, `hsmooth`, `hcurvature`, and `hnd` and proves:

```
Continuous (fun qphi : M06A.ImpatientPrices m × Real =>
  stationaryAssetSupply (m.withPrices qphi.1.1) qphi.2
    (M06A.stationaryLawAtPrice m hsmooth hcurvature hnd qphi.1)).
```

Actual and transitive economic assumptions. BASIC is carried by `m`; SMOOTH, CURVATURE, NONDEGENERATE, IID, and strict IMPATIENT are exactly those needed to construct and continuously vary the canonical stationary law. Normalized effective income contains only `(R-1)*phi`, so at zero net interest it cannot identify `phi`; the joint interface therefore retains the debt shift as an independent coordinate. For economic original-coordinate use, that coordinate must be a compatible nonnegative debt limit. No density, positive minimum income, finite labor support, moment-convergence premise, or uniform bound at the impatience boundary is assumed.

Readable proof. For a convergent price sequence, D03 supplies a common compact support `[0,B]` for the tail stationary laws, and a second constant-sequence application supplies support for the limiting law on the same enlarged interval. H06 makes the shifted policy continuous jointly in price and resources, hence uniformly convergent on `[0,B]`. The proof clips the limiting policy at `B` to obtain a globally bounded continuous test. Weak continuity from S06 then handles the fixed clipped test, while the uniform norm controls the varying-policy integral. Feasibility `A_q(z) ≤ z` makes clipping exact on every supported law. Thus `Integral A_q d pi_q` is continuous. Compact support establishes

integrability, and A02's accepted decomposition gives `S(q,phi)=Integral A_q d pi_q-phi`. The accepted `stationaryMeanShiftedAssets_continuous` theorem composed with the first projection, minus the continuous second projection, proves genuine joint continuity. The earlier graph theorem and all accepted analytic proof bodies remain byte-for-byte unchanged.

State-space, boundary, and integrability audit. Every stationary law remains a probability measure on the fixed full resource space; no varying subtype is silently identified. Weak convergence is applied only to the bounded continuous clipped policy, never directly to the unbounded policy. The common bound is local to a strictly impatient limit and is not asserted near `beta*R=1`. No marginal-value object or zero-state derivative is used, so all inherited `zeroRightMarginal` qualifications remain untouched.

Source correspondence. The proof follows architecture section 8's common-compact split and the accepted A93 Proposition 5 / SLP Theorem 12.13 stationary-law dependence supplied by S06. A94 equation (8) and its stationary aggregation discussion identify net asset supply as shifted saving less the effective debt limit. This gate used the authorized capsule extracts and inherited accepted source qualifications; it does not claim a fresh source-PDF inspection.

Audit result. The unchanged contract anchor and the new joint export appear in the exact `Probes/M06DRSignatures.lean` probe. The new public declaration is covered by `#check`, `assert_no_sorry`, and `#print axioms` there and in `Audit.lean`. The implementation introduces no `sorry`, `admit`, project axiom, unsafe bypass, or numerical model. A03 is submitted as REVIEW_READY only; no GREEN status or later contract is self-awarded.

Adequacy note. Kernel checking establishes the formal statement above, not independent economic adequacy. Review is requested for A03 only.

A04 — Certainty stationary assets at limit

Status: GREEN. Independent Astra acceptance: `reviews/m09c1_acceptance.md`. **Scope:** core. **Milestone:** 09 / gate M09C1.

Target declaration: `Aiyagari1994.certainty_stationary_assets_at_limit`.
Module: `Aiyagari1994/Equilibrium/CertaintyBenchmark.lean`.

Elaborated interface (notation normalized for readability). Given a BASIC household `base`, `hsmooth : UtilitySmooth base.utility`, certainty original prices `p` of type

```
OriginalPrices (M09C1.certaintyIncome (M09C1.meanLabor base)),
```

and `hbetaR : base.beta * p.normalized.grossReturn < 1`, let `m` be the constructed certainty household, `ebar := lowerEffectiveIncome m`, and `delta := M09C1.pointMass ebar`. Then `certainty_stationary_assets_at_limit` proves:

```
householdKernel m compMeasure (delta : Measure Resources) = delta
and for every probability law mu,
  certaintyLawStep m mu = householdKernel m compMeasure mu
and for every probability law mu,
  Tendsto (fun n => (certaintyLawStep m)^[n] mu) atTop (nhds delta)
and for every probability law mu,
  householdKernel m compMeasure mu = mu -> mu = delta
and assetPolicy m ebar = 0
and Integrable (fun z => (assetPolicy m z : Real) - p.debtLimit) delta
and integral (fun z => (assetPolicy m z : Real) - p.debtLimit) delta
      = -p.debtLimit.
```

The exact kernel-rendered signature is recorded by `Probes/M09C1Signatures.lean` and the global audit.

Assumption profiles: BASIC, SMOOTH, IMPATIENT. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H04, H10, H12, S02. **Source keys:** A93, A94.

Source locator: A94 mean-income certainty and precautionary-assets discussion, printed pp. 669-670 / PDF pp. 12-13, especially notes 22-23; deterministic benchmark on p. 671 / PDF p. 14.

Readable proof. `M09C1.meanLabor` integrates labor under the complete supplied risky law. Its strict positivity follows by integrating the pointwise lower-support bound and using the positive labor floor. `M09C1.certaintyIncome` has identical lower and upper endpoints and is a Dirac law at this actual mean. `certaintyHousehold` inherits beta and utility from `base` and uses `p.normalized`, so its affine intercept is exactly `-p.netRate * p.debtLimit`; the debt limit is therefore the certainty limit supplied by the branch constructing `p`, not the risky minimum- income limit.

Every labor realization in the degenerate subtype equals mean labor. Hence the canonical household kernel at state `z` is the Dirac measure at the deterministic map `h(z)=lowerTransition m z = R*A(z)+ebar`, and its law update is pushforward by `h`. S02, whose strict descent proof uses H04, H10 and H12, gives `h(ebar)=ebar`, `h(z)<z` above `ebar`, and convergence of iterates begun at every `z>=ebar`. If an initial finite state lies below `ebar`, one application of `h` lies weakly above `ebar`; the same result applied to that next state and a one-period shift proves convergence from every finite resource state.

For an arbitrary initial probability law and every bounded continuous test `f`, continuity gives `f(h^[n](z)) -> f(ebar)` pointwise. The uniform bound by `norm f` is integrable under the initial probability law. Dominated convergence and the pushforward integral identity therefore give weak convergence of the iterated canonical laws to `delta_ebar`. The point mass is invariant because `h(ebar)=ebar`. If any law is invariant, all of its iterates are constant at that law; uniqueness of topological limits and global convergence identify it with `delta_ebar`.

Finally the fixed-point identity is `R*A(ebar)+ebar=ebar`. Positive gross return implies `A(ebar)=0`. Integrating the net-asset map `A-phi` under `delta_ebar` is explicitly proved integrable and yields `-p.debtLimit`. This is shifted saving zero and net assets minus the correct certainty debt shift.

Assumptions and exclusions. The signature uses BASIC through `base`, SMOOTH through `hsmooth`, and strict impatience through `hbetaR`. It assumes no `IncomeNondegenerate`, curvature, mixing, crossing, moment, or compact-support premise on the initial resource law. The conclusion is weak convergence tested by bounded continuous functions; it does not assert moment convergence for arbitrary initial laws. No marginal object at zero is used, so all inherited `zeroRightMarginal` qualifications remain intact. The finite-history lifetime interpretation is unchanged and no No-Ponzi claim is introduced.

Source correspondence and audit. A94 printed pp. 669-671 / PDF pp. 12-14 motivates the mean-income certainty comparison and deterministic benchmark. The complete statewise and arbitrary-law convergence argument is the project reconstruction authorized by architecture §11.4, not a claim that the paper prints this Lean proof. A93 evidence is inherited through the approved resolver and predecessor interfaces. The public wrapper and its gate-owned public construction objects are covered by `#check`, `assert_no_sorry`, and `#print axioms` in both the gate signature probe and `Audit.lean`; only `propext`, `Classical.choice`, and `Quot.sound` occur. At the M09C1 submission this implementation was REVIEW_READY and made no A05, G04/G05, later comparison, or stage-advancement claim. A04 is now GREEN under its recorded independent acceptance; that later status does not enlarge the theorem. A05 is addressed separately below.

A05 — Risky assets above certainty near impatience

Status: GREEN. Independent Astra acceptance: `reviews/m09c2_acceptance.md`. **Scope:** core. **Milestone:** 09 / gate M09C2.

Target declaration: `Aiyagari1994.risky_assets_above_certainty_near_impatience`.
Module: `Aiyagari1994/Aggregate/CertaintyComparison.lean`.

Mathematical contract. At fixed positive wages and the same institutional cap, or in the two respective natural-limit families, risky stationary mean assets are weakly above the mean-income certainty level at every strictly impatient admissible rate. They are strictly above it for all rates sufficiently close to lambda from below. The weak bound follows from nonnegative shifted assets and the ordering of effective debt limits; the strict near-boundary result follows from B02. Do not assume convex marginal utility or assert a general ordering across two risky distributions.

Assumption profiles: BASIC, SMOOTH, CURVATURE, NONDEGENERATE, IID, IMPATIENT, FINITE_CAP, NATURAL_CAP. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: P02, A02, A04, B02. **Source keys:** A94.

Source locator: A94 mean-income certainty and precautionary-assets discussion, printed pp. 669-670 / PDF pp. 12-13, especially notes 22-23; deterministic benchmark on p. 671 / PDF p. 14.

Elaborated signature (notation normalized for the ledger). Given a risky household `m`, `UtilitySmooth`, `UtilityCurvature`, `IncomeNondegenerate`, an institutional cap `b >= 0`, and a fixed wage `w > 0`, the theorem returns four clauses. For every `r : M09C2.SubcriticalRate m` it proves `finiteRiskSupply r >= finiteCertaintySupply r`; for every `r : M09C2.PositiveSubcriticalRate m` it proves `naturalRiskSupply r >= naturalCertaintySupply r`. It also proves both strict inequalities eventually in the respective pulled-back neighborhood filters `comap coe (nhds (M09C2.criticalRate m))`. The exact parser output is recorded in `Probes/M09C2Signatures.lean`.

The finite-cap domain contains every rate with `-1 < r` and `beta*(1+r) < 1`, including the nonpositive-rate branch. Its risky and certainty shifts are respectively `effectiveLimit b lower w r` and `effectiveLimit b meanLabor w r`; hence the formula is `b` on the nonpositive branch and `min(b,w*labor/r)` on the positive branch. The natural domain separately requires `0 < r` and uses `w*lower/r` and `w*meanLabor/r`. Thus the theorem compares the two respective debt-rule families and never substitutes risky minimum labor for certainty mean labor.

Readable proof. Compact-support integrability gives `lower <= integral labor = meanLabor`. Monotonicity of multiplication and division at positive rates gives `phiRisk <= phiCertainty`; for the finite-cap rule, unfolding the definition treats the nonpositive branch separately and applies monotonicity of `min` only when `r>0`.

At every admissible strictly impatient rate, A02 supplies integrability and the identity `S_risky = integral A - phiRisk`. Since shifted saving `A` is NNReal-valued, its integrable real mean is nonnegative, so `S_risky >= -phiRisk >= -phiCertainty`. A04 is applied to the separately constructed deterministic household at actual risky mean labor and identifies its actual stationary integral with `-phiCertainty`. This proves both weak clauses without Jensen, convex marginal utility, or any risky-distribution order.

For strictness, fix the wage and let rates approach `lambda=1/beta-1` within each strictly impatient rate subtype. For every convergent sequence in the pulled-back boundary filter, the corresponding admissible normalized prices converge to the critical fixed-wage price. The finite-cap shifts converge by P02 continuity on `w>0, r>-1`; the natural shifts converge by the quotient formula and `lambda>0`. B02 therefore sends risky stationary net assets to `+infinity`. Consequently they are positive throughout

one neighborhood of the boundary. A04 gives certainty assets `-phiCertainty <= 0`, yielding strict comparison for all rates sufficiently close to lambda from below in each family.

Assumptions and exclusions. BASIC is inherited from `m`; SMOOTH, CURVATURE and NONDEGENERATE are explicit. IID is inherited through the canonical stationary construction, and strict impatience is carried by each rate subtype. FINITE_CAP and NATURAL_CAP are separate constructed branches. The theorem is partial equilibrium at fixed positive wage. It asserts no global strict inequality, convex-marginal-utility/Jensen premise, risky-versus-risky stochastic order, equilibrium existence, capital comparison, or saving comparison.

Source correspondence and audit. A94 printed pp. 669-671 / PDF pp. 12-14 motivates the mean-income certainty comparison, borrowing-limit distinction, and qualified near-boundary conclusion. The integrability, debt-shift ordering, filter specialization, and complete Lean proof are project reconstruction. All thirteen new public declarations have `#check`, `assert_no_sorry`, and `#print axioms` coverage in both the gate probe and `Audit.lean`; only `propext`, `Classical.choice`, and `Quot.sound` occur. This submission stops at REVIEW_READY and does not self-award GREEN or advance to G04/G05.

Adequacy note. This contracts the qualified precautionary-assets claim; it is not the deferred Sibley/Miller aggregate risk-order question.

N01 — Stationary zero state marginal resolved

Status: GREEN. Independent Astra acceptance: `reviews/m07a1_acceptance.md`. **Scope:** core. **Milestone:** 07a / gate M07A1.

Target declaration: `Aiyagari1994.stationary_zero_state_marginal_resolved`.
Module: `Aiyagari1994/Stationary/ZeroState.lean`.

Elaborated signature (notation normalized for PDF).

```
Aiyagari1994.stationary_zero_state_marginal_resolved
  (m : Aiyagari1994.HouseholdPrimitives)
  (_hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (hnd : Aiyagari1994.IncomeNondegenerate m.income)
  (pi : MeasureTheory.ProbabilityMeasure Aiyagari1994.Resources)
  (hinv : householdKernel m compMeasure (pi : Measure Resources) = pi) :
  (zeroRightMarginal m = top -> pi {0} = 0) and
  for pi-almost every z:
    extendedRightMarginalValue m z < top,
    0 < extendedRightMarginalValue m z,
    the next marginal is finite for income-law-almost every l,
    its conditional real-valued function is Integrable, and
    beta * grossReturn * conditionalIntegral <=
      (extendedRightMarginalValue m z).toReal
```

The notation-normalized display above is line-wrapped for the ledger PDF; the exact parser output is recorded by `Probes/M07A1Signatures.lean`.

Mathematical statement and assumptions. BASIC is contained in `HouseholdPrimitives`; SMOOTH is the explicit `UtilitySmooth` premise; NONDEGENERATE is the explicit endpoint-neighborhood premise. The probability law `pi` is arbitrary subject to the displayed invariance equation, so the statement holds at every admissible `grossReturn > 0` and does not use S05 or strict impatience. The boundary object is `zeroRightMarginal : ENNReal`, never `rightMarginalValue m 0`. The conclusion first resolves a possibly infinite boundary marginal, then proves finite positive marginals almost everywhere and H09's conditional finiteness, integrability, and real inequality. It does not assert `Integrable q pi` or finite stationary `E_pi[q]`.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H04, H08, H09, S01. **Source keys:** CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Readable proof plan: Architecture §9.1.

Readable proof. Let `p0` be the labor probability of zero effective income. Endpoint nondegeneracy gives positive probability to an upper-endpoint neighborhood. Because the wage is strictly positive and the lower-endpoint effective income is nonnegative, that neighborhood has strictly positive effective income; hence `p0 < 1`.

Suppose the boundary marginal is infinite. If `p0 > 0` and a positive resource state chose zero shifted saving, every zero-income realization would lead to zero next resources and hence an infinite next marginal. H09 says that from a positive current state the next marginal is finite almost surely, a contradiction. Thus positive states save strictly positively and cannot reach zero. At the current zero state, feasibility gives `A(0)=0`, so the kernel's zero-state mass is exactly `p0`. Invariance on the singleton therefore yields `pi{0}=p0*pi{0}`. Since `p0<1`, this forces `pi{0}=0`. If `p0=0`, every state's zero-transition mass is bounded by `p0`, so invariance immediately gives `pi{0}=0`. This proves both atom and no-atom branches.

Away from zero the extended marginal is the finite `ENNReal.ofReal` of the positive-state value marginal. If the boundary marginal is finite, finiteness therefore holds everywhere; if it is infinite, the preceding singleton result makes its sole possible infinite state null. H08 gives strict positivity at positive states and at the economic boundary. Filtering H09 over this almost-everywhere finite set gives conditional next-state finiteness, integrability of the conditional real marginal, and the displayed real inequality. No integration of `q` against `pi` occurs.

Public helper declarations. `M07A1.invariant_zero_measure_of_boundary_infinite`, `M07A1.extendedRightMarginalValue_pos`, and `M07A1.invariant_extendedMarginal_finite_ae` live in the assigned capsule helper directory `Aiyagari1994/Analysis/M07A1/`.

Axiom and audit output. The target and all three public helpers have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07A1Signatures.lean`. Each transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, project axiom, `native_decide`, unsafe bypass, or numerical model is used.

Adequacy note. This is the architecture-approved new bounded-stationarity reconstruction motivated by the approved sources, not a literal theorem attributed to CW00 or A94. It proves N01 only. N02–N07, Stage 07b, and Stage 08 remain unformalized. Independent adequacy review is requested; no GREEN status is self-awarded.

N02 — Stationary bounded jensen equality

Status: GREEN. Independent Astra acceptance: `reviews/m07a2_acceptance.md`. **Scope:** core. **Milestone:** 07a / gate M07A2.

Target declaration: `Aiyagari1994.stationary_bounded_jensen_equality`.
Module: `Aiyagari1994/Analysis/BoundedJensen.lean`.

Elaborated signature (notation normalized for PDF).

```
Aiyagari1994.stationary_bounded_jensen_equality
  {Omega : Type} [MeasurableSpace Omega]
```

```
(P : Kernel Omega Omega) [IsMarkovKernel P]
(pi : Measure Omega) [IsProbabilityMeasure pi]
(q : Omega -> Real) (gamma : Real)
(q_measurable : Measurable q)
(q_positive : forall z, 0 < q z)
(q_conditionally_integrable : forall z, Integrable q (P z))
(one_le_gamma : 1 <= gamma)
(superharmonic : forall z, gamma * Integral q (P z) <= q z)
(invariant : P compMeasure pi = pi) :
gamma = 1 and
  for (pi compProd P)-almost every (z,z'), q z' = q z
```

The notation-normalized display is line-wrapped for the ledger PDF; the exact parser output is recorded by `Probes/M07A2Signatures.lean`.

Mathematical statement and assumptions. `q` is globally positive and real-valued, hence finite, and measurable. Its integral is required separately under every conditional probability law `P z`. The real scalar `gamma` is at least one, the displayed superharmonic inequality holds pointwise, and `pi` is a supplied invariant probability measure. The conclusion rules out `gamma > 1` by proving `gamma = 1`, then proves equality of `q` at the current and next states under the stationary one-step joint law `pi compProd P`. No integral of `q` against `pi` occurs in the signature or proof.

Assumption profiles: Generic mathematical hypotheses stated in the contract. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Readable proof plan: Architecture §9.2.

Readable proof. Set `psi(x)=x/(1+x)`, `m(z)=Integral q (P z)`, and `e(z)=Integral (psi o q) (P z)`. Positivity and conditional integrability make every `m(z)` strictly positive. The transform is measurable, nonnegative and bounded above by one on the positive half-line, so `psi o q`, `psi o m`, `psi o (gamma*m)`, and `e` are integrable under the stationary law even though `q` need not be.

For each state, the proof verifies the exact tangent identity

```
psi(m) + (x-m)/(1+m)^2 - psi(x)
  = (x-m)^2 / ((1+m)^2 * (1+x)).
```

Its nonnegative right side gives `e(z) <= psi(m(z))`. Monotonicity of `psi`, `gamma>=1`, and the superharmonic inequality give the remaining chain `psi(m(z)) <= psi(gamma*m(z)) <= psi(q(z))`. Stationarity and boundedness identify the integrals of the two end terms: Fubini for `pi compProd P` gives `Integral e pi = Integral (psi o q) pi`. Equality throughout the integral chain therefore holds almost everywhere. Since `m(z)>0`, equality of `psi(m(z))` and `psi(gamma*m(z))` forces `gamma=1`.

At the critical value, equality in the first Jensen step makes the conditional integral of the displayed tangent gap zero. The gap is nonnegative and integrable, so it vanishes conditionally almost surely. Its strictly positive denominator forces `q(z')=m(z)` for `P z`-almost every next state. Equality in the last monotonicity step similarly gives `q(z)=m(z)` for `pi`-almost every current state. Combining these statements yields `q(z')=q(z)` under `pi compProd P`.

Axiom and audit output. The target has `#check`, `assert_no_sorry`, and `#print axioms` coverage in both `Audit.lean` and `Probes/M07A2Signatures.lean`. Its transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, project axiom, `native_decide`, unsafe bypass, or numerical model is used.

Adequacy note. This is the architecture-approved bounded strict-Jensen reconstruction motivated by the approved source context, not a literal theorem attributed to CW00 or A94. It proves N02 only. N03–N07, Stage 07b, and Stage 08 remain unformalized. Independent adequacy review is requested; no GREEN status is self-awarded.

N03 — No invariant supercritical

Status: GREEN. Independent Astra acceptance: `reviews/m07a3_acceptance.md`. **Scope:** core. **Milestone:** 07a.

Target declaration: `Aiyagari1994.no_invariant_supercritical`.
Module: `Aiyagari1994/Stationary/Supercritical.lean`.

Exact elaborated signature.

```
Aiyagari1994.no_invariant_supercritical
  (m : Aiyagari1994.HouseholdPrimitives)
  (hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (hnd : Aiyagari1994.IncomeNondegenerate m.income)
  (hsupercritical : 1 < m.beta * m.prices.grossReturn) :
  not (exists pi : MeasureTheory.ProbabilityMeasure Aiyagari1994.Resources,
    (pi : MeasureTheory.Measure Aiyagari1994.Resources).bind
      (Aiyagari1994.householdKernel m) = pi)
```

Thus the candidate invariant law is quantified only inside a contradiction. No moment, bounded-support, consumption-positivity, curvature, density, atom, or finite-state hypothesis is present.

Actual economic assumptions. `HouseholdPrimitives` supplies BASIC, including bounded strictly increasing and strictly concave utility, a compact iid labor law, positive normalized prices and the canonical policy and kernel. `UtilitySmooth` is explicit, as required by the contract and N01, although N01's final wrapper does not use its fields. `IncomeNondegenerate` gives essential lower and upper endpoints. The branch premise is the strict supercritical inequality `1 < m.beta * m.prices.grossReturn`.

Dependencies: H09, S01, N01, N02. **Source keys:** A93, A94, CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Readable proof. Suppose an invariant probability law `pi` exists. Define the real marginal placeholder

```
q(z) = if extendedRightMarginalValue m z = top then 1
       else (extendedRightMarginalValue m z).toReal.
```

This definition is measurable and strictly positive at every state. It uses `extendedRightMarginalValue`, whose zero branch is `zeroRightMarginal : ENNReal`; it never uses `rightMarginalValue m 0`. N01 proves that the extended marginal is finite `pi`-almost everywhere and, at those current states, proves almost-everywhere finiteness of the next marginal, integrability of its real value under the labor law, and H09's real superharmonic inequality. Consequently the placeholder agrees with the real extended marginal at the current state and for almost every next labor draw.

The exact household-kernel pushforward formula transports that conditional labor integrability to `Integrable q (householdKernel m z)`. The S01 integral formula transports H09's inequality to

```
m.beta * m.prices.grossReturn * Integral q (householdKernel m z) <= q z
```

for `pi`-almost every state. This is conditional integrability only: the proof never forms `Integral q pi`.

The gate-local helper `Aiyagari1994.M07A3.stationary_bounded_jensen_ae_gamma_eq_one` is the almost-everywhere interface to N02's approved bounded-Jensen argument. It permits conditional

integrability and the superharmonic inequality to hold almost everywhere, while retaining pointwise positivity and measurability of the finite placeholder. Its exact signature quantifies over an arbitrary measurable state space, Markov kernel `P`, probability measure `pi`, measurable pointwise-positive real `q`, and real `gamma`; it assumes

```
(forall-ae z under pi, Integrable q (P z))
1 <= gamma
(forall-ae z under pi, gamma * Integral q (P z) <= q z)
P compMeasure pi = pi
```

and concludes `gamma = 1`. With `psi(x)=x/(1+x)`, conditional mean `avg=Pq`, and `boundedAvg=P(psi o q)`, its proof establishes almost everywhere

```
boundedAvg <= psi(avg) <= psi(gamma * avg) <= psi(q).
```

The first inequality uses the same exact tangent-gap identity as N02. All four displayed quantities are bounded and integrable under `pi`, even though `q` need not be. Stationarity equates the endpoint integrals. Equality of the middle terms, together with the strictly positive conditional mean on the almost-everywhere integrability set, forces `gamma=1`. Applied with `gamma=m.beta*m.prices.grossReturn`, this contradicts the strict supercritical premise.

Axiom and audit output. Both the helper and N03 have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07A3Signatures.lean`. Each has transitive axiom set exactly `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, `admit`, project axiom, `native_decide`, unsafe bypass, numerical model, candidate-law existence theorem, or stationary marginal-moment assumption is used.

Adequacy note. This is the architecture-approved bounded-transform reconstruction motivated by A94 printed p. 669 / PDF p. 12 and CW00 background, not a literal source theorem. It proves N03 only. At the N03 submission boundary, N04, N05–N07, Stage 07b, and Stage 08 remained unformalized; the separate N04 entry below records its current REVIEW_READY submission.

N04 — Critical stationary consumption constant

Status: GREEN. Independent Astra acceptance: `reviews/m07a4_acceptance.md`. **Scope:** core. **Milestone:** 07a.

Target declaration: `Aiyagari1994.critical_stationary_consumption_constant`.
Module: `Aiyagari1994/Stationary/CriticalConsumption.lean`.

Kernel-checked statement. Given BASIC household primitives, SMOOTH utility, NONDEGENERATE compact iid income, a supplied invariant probability law `pi`, and `m.beta * m.prices.grossReturn = 1`, consumption agrees almost everywhere under the stationary one-step joint law. For every natural `n`, it also agrees between the two coordinates under `pi.compProd (householdKernel m ^ n)`. Thus the endpoints of every finite stationary history have the same consumption, including zero-consumption corners.

The exact public signature is:

```
Aiyagari1994.critical_stationary_consumption_constant
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hnd : IncomeNondegenerate m.income) (pi : ProbabilityMeasure Resources)
  (hinv : householdKernel m compMeasure (pi : Measure Resources) = pi)
  (hcritical : m.beta * m.prices.grossReturn = 1) :
  (forall-ae (z,z') under pi.compProd (householdKernel m), c(z') = c(z)) and
  forall n, forall-ae (z,z') under pi.compProd (householdKernel m ^ n), c(z') = c(z)
```

Actual assumptions. `HouseholdPrimitives` supplies BASIC and the compact iid labor law; `UtilitySmooth` supplies positive-state differentiability and positive marginal utility;

`IncomeNondegenerate` is used through N01; invariance and criticality are explicit hypotheses. The theorem does not construct an invariant law at the critical return and assumes no bounded support, stationary resource moment, or stationary marginal moment.

Dependencies: H11, H09, N01, N02. **Source keys:** CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Readable proof. N01 makes the extended economic value marginal finite and positive almost everywhere under the candidate law and supplies conditional next-marginal integrability plus H09's real inequality. The proof uses the same finite placeholder as N03 only on N01's null infinite branch. A gate-local almost-everywhere version of N02's exact bounded tangent-gap argument, specialized to the critical coefficient, proves equality of the placeholder across the stationary one-step joint law without forming `Integral q pi`.

On the full-measure set of finite marginals, placeholder equality is equality of `extendedRightMarginalValue`. At positive consumption H11 identifies this with `U'(c)`, whose strict antitonicity follows from strict concavity. At zero consumption, the Bellman comparison that consumes an extra increment while retaining saving proves `utilityZeroRightMarginal <= extendedRightMarginalValue`; every positive-consumption marginal is strictly below `utilityZeroRightMarginal`. Hence equal finite economic marginals force equal consumption even at corners. A generic stationary-kernel induction then propagates one-step equality to `householdKernel m ^ n` for every finite `n`.

Boundary and integrability audit. At zero the proof uses `extendedRightMarginalValue m 0 = zeroRightMarginal m : ENNReal`; it never evaluates `rightMarginalValue m 0`. H11 is invoked only after positive consumption proves positive resources. Conditional integrability is not promoted to stationary `E[q]`. H10 and H12 are not imported or used, so no strict-impatience argument is applied at a critical corner.

Axiom and export audit. The target and three public gate-local helpers have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07A4Signatures.lean`. Their transitive axiom sets are exactly `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. This is the approved §9.3 reconstruction motivated by A94 printed p. 669 / PDF p. 12 and CW00 background, not a literal theorem copied from those sources. It proves N04 only. N05–N07, Stage 07b, Stage 08, and all later contracts remain unformalized. Independent adequacy review is requested; no GREEN status is self-awarded.

N05 — Two independent histories contradiction

Status: GREEN. Independent Astra acceptance: `reviews/m07b1_acceptance.md`. **Scope:** core. **Milestone:** 07b.

Target declaration: `Aiyagari1994.two_independent_histories_contradiction`.
Module: `Aiyagari1994/Analysis/TwoShockStrings.lean`.

Mathematical contract. For R>1, a state-dependent consumption rule c(z), and resource recursion Z_next=R*(Z-c(Z))+e_next with nonnegative resources and bounded nondegenerate iid income, a stationary probability law under which consumption remains constant along finite histories is impossible. Couple the same initial state with two independent future shock strings and contradict a strictly positive discounted-shock variance. No state moments are assumed.

Assumption profiles: NONDEGENERATE, IID. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Exact elaborated target signature.

```
Aiyagari1994.two_independent_histories_contradiction
  {E : Type} [MeasurableSpace E]
  (nu : Measure E) [IsProbabilityMeasure nu]
  (shock : E -> Real) (hshock : Measurable shock)
  (B : Real) (hB : 0 <= B) (hbounded : forall e, |shock e| <= B)
  (hnonconstant : not (exists a,
    Filter.EventuallyEq (Measure.ae nu) shock (fun _ => a)))
  (R : Real) (hR : 1 < R)
  (c : NNReal -> NNReal) (hc : Measurable c)
  (step : NNReal -> E -> NNReal)
  (hstep : Measurable (fun p : NNReal x E => step p.1 p.2))
  (hrecursion : forall z e,
    (step z e : Real) = R * ((z : Real) - (c z : Real)) + shock e)
  (pi : Measure NNReal) [IsProbabilityMeasure pi]
  (hinv : Measure.bind pi (M07B1.resourceKernel nu step hstep) = pi)
  (hconstant : Filter.Eventually
    (fun zz' => c zz'.2 = c zz'.1)
    (Measure.ae (pi.compProd (M07B1.resourceKernel nu step hstep)))) : False
```

Actual mathematical assumptions. `nu` and `pi` are probability measures. The measurable shock is uniformly bounded and not almost everywhere constant. The gross return satisfies `R>1`. The primitive measurable inputs are a state-dependent consumption rule `c`, a one-step resource map `step`, and the exact recursion `step(z,e)=R*(z-c(z))+shock(e)`. The induced resource kernel has invariant probability law `pi`, and `c` is unchanged almost everywhere under its stationary one-step joint law. There is no state moment, bounded state support, stationary resource moment, stationary marginal-utility moment, strict impatience, positive-consumption premise, or invariant-law existence theorem.

Readable proof. `M07B1.kernel_ae_to_innovation_ae` first transfers the N04-shaped kernel almost-everywhere equality to the product of the stationary current state and one fresh IID innovation. `M07B1.coupledHistoryLaw` then constructs, separately for every finite horizon, a common initial state and two independent innovation strings by recursive finite products. `coupled_endpoint_laws` inducts on stationarity to prove that both actual recursively generated terminal resources have law `pi`. `coupled_consumption_constant` repeatedly pulls back the one-step full-measure event and intersects the finitely many events; it never asserts equality on null histories. `coupled_telescope` combines those a.e. equalities with `hrecursion` to derive the exact discounted identity `D_n=R^(-n)*(Z_n-Z'_n)` on the coupled finite product.

The signed discounted shock difference is uniformly bounded by `2*B/(1-R^(-1))`. Tightness of the arbitrary probability law `pi` gives a finite upper-tail cutoff. If `R^(-n) |Z_n-Z'_n|` exceeds a fixed threshold, then one terminal state exceeds that cutoff; the union bound therefore makes this probability small for a large finite `n`. Splitting the bounded square on that event makes its second moment strictly smaller than a fixed positive number, without using a state moment and without putting different horizons on one probability space.

On the same finite product, independence and identical distribution give the exact identity `E[D_n^2] = 2*Var(shock)*sum_(j=1)^n R^(-2j)`. Boundedness supplies `L2`; non-a.e.-constancy and the variance-zero characterization prove `Var(shock)>0`. For `n >= 1`, the exact moment is at least `2*Var(shock)*R^(-2)>0`, contradicting the tightness bound.

N04 compatibility. The public premise `hconstant` has exactly N04's stationary one-step almost-everywhere form for the induced resource kernel. The finite-product bridge and discounted telescope are proved inside N05, rather than deferred to N06. This does not require equality on every shock history. N05 remains generic: it does not import N04, S05, household primitives, or any invariant-law existence theorem.

Exports and axioms. The target and thirty-eight public finite-product, bridge, recursion, moment, and tightness declarations in `Aiyagari1994/Analysis/M07B1/` have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07B1Signatures.lean`. Every printed transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. This is the approved new §9.4 finite-history reconstruction, not a theorem copied from A94 or CW00. It proves N05 only. N06, N07, Stage 08, and later gates remain unformalized. Independent adequacy review is requested; no GREEN status is self-awarded.

N06 — No invariant critical

Status: GREEN. Independent Astra acceptance: `reviews/m07b2_acceptance.md`. **Scope:** core. **Milestone:** 07b.

Target declaration: `Aiyagari1994.no_invariant_critical`.
Module: `Aiyagari1994/Stationary/Critical.lean`.

Mathematical contract. When beta*R=1 and effective income is nondegenerate, the canonical household kernel has no invariant probability law on NNReal, including laws of infinite first moment.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE, IID. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: S01, N04, N05. **Source keys:** A93, A94, CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Exact elaborated target signature.

```
Aiyagari1994.no_invariant_critical (m : Aiyagari1994.HouseholdPrimitives)
  (hsmooth : Aiyagari1994.UtilitySmooth m.utility)
  (hnd : Aiyagari1994.IncomeNondegenerate m.income)
  (hcritical : m.beta * m.prices.grossReturn = 1) :
  ¬ ∃ pi : MeasureTheory.ProbabilityMeasure Aiyagari1994.Resources,
    MeasureTheory.Measure.bind
      (pi : MeasureTheory.Measure Aiyagari1994.Resources)
      (Aiyagari1994.householdKernel m) =
      (pi : MeasureTheory.Measure Aiyagari1994.Resources)
```

Actual mathematical assumptions. The household primitives retain `0<beta<1`, bounded strictly increasing and strictly concave utility, compact positive labor support, positive wage and gross return, and nonnegative effective income. The explicit additional premises are positive-domain utility smoothness, essential-endpoint income nondegeneracy, and `beta*grossReturn=1`. IID innovations are represented by the primitive labor probability law and the finite products constructed inside N05. No invariant law, state moment, stationary support bound, positive-consumption statement, strict-impatience inequality, or S05 existence result is assumed.

Readable proof. Assume only for contradiction that a probability law `pi` is invariant for the canonical household kernel. Criticality together with `0<beta<1` gives `grossReturn>1`. The compact labor interval bounds the affine effective-income shock pointwise. Essential positive mass in separated endpoint

neighborhoods, combined with the strictly positive wage, proves that effective income is not almost everywhere constant; variance positivity is then supplied by N05, not assumed here.

Instantiate N05 with `c=consumptionPolicy` and `step(z,l)=nextResources(assetPolicy(z),l)`. The accepted budget identity `c(z)+A(z)=z` (in its real coercion form) and the definition of `nextResources` derive the exact recursion `step(z,l)=R*(z-c(z))+effectiveIncome(l)`. The induced generic resource kernel is definitionally the canonical `householdKernel`. N04, applied to the contradictory candidate `pi`, supplies the stationary one-step almost-everywhere equality of consumption. N05 legitimately transfers that a.e. statement to finite innovation products, derives terminal stationary laws and the exact discounted telescope, and obtains the tightness/variance contradiction without a state moment.

Exports and axioms. `no_invariant_critical` and the four public household bridge declarations in `Aiyagari1994/Analysis/M07B2/HouseholdBridge.lean` have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07B2Signatures.lean`. Every printed transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. This is the approved new §9.4 household instantiation of the finite-history reconstruction, not a theorem copied from A94 or CW00. It proves N06 only. N07, Stage 08, and later gates remain unformalized. Independent adequacy review is requested; no GREEN status is self-awarded.

N07 — No invariant at or above impatience

Status: GREEN. Independent Astra acceptance: `reviews/m07b3_acceptance.md`. **Scope:** core. **Milestone:** 07b.

Target declaration: `Aiyagari1994.no_invariant_at_or_above_impatience`.
Module: `Aiyagari1994/Stationary/NoInvariant.lean`.

Mathematical contract. For beta*R>=1 and nondegenerate iid effective income, no invariant probability law exists for the canonical household kernel.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE, IID. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: N03, N06. **Source keys:** A93, A94, CW00.

Source locator: A94 printed p. 669 / PDF p. 12, notes 20-21, distinguishes pathwise claims from the stationary implication. CW00 supplies background; the bounded-Jensen/two-string stationary proof here is a new reconstruction, not a cited source theorem.

Exact Lean signature.

```
theorem no_invariant_at_or_above_impatience (m : HouseholdPrimitives)
    (hsmooth : UtilitySmooth m.utility) (hnd : IncomeNondegenerate m.income)
    (hatOrAbove : 1 ≤ m.beta * m.prices.grossReturn) :
    ¬ ∃ pi : ProbabilityMeasure Resources,
      (pi : Measure Resources).bind (householdKernel m) = (pi : Measure Resources)
```

Readable proof. Split `1 ≤ beta*R` into equality or strict inequality. In the equality branch, reverse the equality and apply the accepted N06 declaration `no_invariant_critical`. In the strict branch, apply the accepted N03 declaration `no_invariant_supercritical`. Thus N07 is only an assembly theorem: it introduces no invariant law, moment premise, boundary conversion, or additional economic assumption, and preserves the exact canonical kernel conclusion of both dependencies.

Exports and axioms. `no_invariant_at_or_above_impatience` has `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Audit.lean` and `Probes/M07B3Signatures.lean`. Its printed transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. The critical two-string argument remains the accepted new reconstruction in N05/N06, not a theorem copied from A94 or CW00; N07 merely combines it with accepted N03. This entry claims nonexistence of invariant probability laws only, not pathwise divergence, stationary moment divergence, or equilibrium. Independent adequacy review is requested; no GREEN status is self-awarded, and Stage 08 remains unformalized.

B01 — Tight kernel invariant limit

Status: GREEN. Independent Astra acceptance: `reviews/m08a_acceptance.md`. **Scope:** core. **Milestone:** 08.

Target declaration: `Aiyagari1994.tight_kernel_invariant_limit`.
Module: `Aiyagari1994/Analysis/TightKernelLimit.lean`.

Mathematical contract. For Feller probability Markov kernels P_n and P on NNReal, suppose invariant probability laws mu_n form a tight family and converge weakly along a subsequence, and P_n f converges locally uniformly to P f for each bounded continuous f. Then the limit law is invariant for P. Prove the compact/tail split and use continuity of P f when passing the fixed test through weak convergence.

Actual mathematical hypotheses. `Pseq n` and `P` are Markov probability kernels on `NNReal`. Every kernel is Feller in bounded-continuous test form. For every bounded continuous real test, the corresponding `Pseq n` expectations tend locally uniformly to the `P` expectation. The probability laws `muSeq n` are uniformly tight, converge weakly to `mu`, and are invariant for their respective `Pseq n`. There is no economic assumption profile and no household object.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** SLP89.

Source locator: New generic tight-kernel limit lemma; related to the compact parameter argument in SLP89 Theorem 12.13, but the noncompact tail passage is proved here.

Exact elaborated signature.

```
Aiyagari1994.tight_kernel_invariant_limit
  (Pseq : ℕ → ProbabilityTheory.Kernel NNReal NNReal)
  (P : ProbabilityTheory.Kernel NNReal NNReal)
  [hPseq : ∀ (n : ℕ), ProbabilityTheory.IsMarkovKernel (Pseq n)]
  [hP : ProbabilityTheory.IsMarkovKernel P]
  (muSeq : ℕ → MeasureTheory.ProbabilityMeasure NNReal)
  (mu : MeasureTheory.ProbabilityMeasure NNReal)
  (hFellerSeq : ∀ (n : ℕ) (f : BoundedContinuousFunction NNReal ℝ),
    Continuous fun z => ∫ y, f y ∂(Pseq n) z)
  (hFeller : ∀ (f : BoundedContinuousFunction NNReal ℝ),
    Continuous fun z => ∫ y, f y ∂P z)
  (hLocal : ∀ (f : BoundedContinuousFunction NNReal ℝ),
    TendstoLocallyUniformly
      (fun n z => ∫ y, f y ∂(Pseq n) z)
      (fun z => ∫ y, f y ∂P z) Filter.atTop)
  (hTight : MeasureTheory.IsTightMeasureSet
    {nu | ∃ n, nu = (muSeq n : Measure NNReal)})
  (hWeak : Filter.Tendsto muSeq Filter.atTop (nhds mu))
  (hInvariant : ∀ n, (muSeq n : Measure NNReal).bind (Pseq n) = muSeq n) :
  (mu : Measure NNReal).bind P = mu
```

The directly necessary helper `M08A.kernelTest` packages a Feller expectation as a bounded continuous function; `M08A.kernelTest_norm_le` proves its pointwise norm is at most the test's sup norm. Both helpers live in the capsule-authorized helper directory.

Readable proof. Fix a bounded continuous test `f`, write `g_n=P_n f` and `g=P f`, and use tightness with `eta = eps / (4 * (norm f + 1))` to select a compact `K` whose complement has `mu_n`-mass at most `eta` for every `n`. Local uniform convergence makes `norm (g_n-g)<eps/2` on `K` eventually. Hence the norm of the integral over `K` is at most `eps/2`, since `mu_n(K)≤1`. The Markov property gives `norm(g_n z)≤norm f` and `norm(g z)≤norm f`; therefore the complement integral is at most `2*norm f*eta<eps/2`. Splitting the integral over `K` and its complement proves `integral (g_n-g) d mu_n → 0`.

The limiting Feller hypothesis makes `g` a fixed bounded continuous test, so weak convergence gives `integral g d mu_n → integral g d mu`. Weak convergence also gives the corresponding limit for `f`. Kernel integration and each invariance identity identify `integral g_n d mu_n` with `integral f d mu_n`. Uniqueness of real limits therefore yields `integral g d mu = integral f d mu` for every bounded continuous `f`. Bounded-continuous separation of finite Borel measures proves `P compMeasure mu = mu`.

Audit. `M08A.kernelTest`, `M08A.kernelTest_norm_le`, and `tight_kernel_invariant_limit` each have `#check`, `assert_no_sorry`, and `#print axioms` in the gate signature probe and global audit. Their transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`.

Adequacy note. This is a generic noncompact-state lemma, not a primitive closure assumption. It assumes neither globally uniform kernel convergence, compact state space, bounded support nor moments. B02, now separately submitted as REVIEW_READY, derives every B01 hypothesis from its declared dependencies. The B01 implementation remains GREEN under its independent acceptance; no Stage 08 advancement or later contract is self-awarded here.

B02 — Asset Supply tends To infinity at impatience

Status: GREEN. Independent Astra acceptance: `reviews/m08b_acceptance.md`. **Scope:** core. **Milestone:** 08.

Target declaration: `Aiyagari1994.assetSupply_tendsTo_infinity_at_impatience`.
Module: `Aiyagari1994/Aggregate/UpperBoundary.lean`.

Mathematical contract. If $r_n < \lambda$, $r_n \to \lambda$, $w_n \to w_* > 0$ and $\phi_n \to \phi_* < \infty$ through admissible normalized/original-coordinate prices, then stationary net asset supply tends to $+\infty$. In normalized coordinates this is stated as $\beta R_n < 1$, $(q_n, \phi_n) \to (q_*, \phi_*)$, $\beta R_* = 1$, and $k_n = -(R_n - 1)\phi_n$. Convergence of the full admissible price vector includes the wage convergence, and admissibility of q_* supplies $w_* > 0$.

Assumption profiles: BASIC, SMOOTH, CURVATURE, NONDEGENERATE, IID. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: H06, A02, B01, N07. **Source keys:** A94, C90.

Source locator: C90 Proposition 2.4, printed p. 548 / PDF p. 7 (stated without proof); A94 note 19 and upper-boundary discussion. The proof route in architecture Section 10.1 is a reconstruction.

Checked sequential declaration. `Aiyagari1994.assetSupply_tendsTo_infinity_at_impatience` takes a fixed primitive household `m`, `UtilitySmooth`, `UtilityCurvature`, and `IncomeNondegenerate`; admissible normalized price sequences `qseq` and critical `qstar`; real shifts `phiSeq` and `phiStar`; nonnegative shifts; the normalization identity; strict subcriticality; exact criticality; and convergence of prices and shifts. Its conclusion is

```
Tendsto (fun n =>
  stationaryAssetSupply (m.withPrices (qseq n)) (phiSeq n)
    (M06C.stationaryLaw (m.withPrices (qseq n)) hsmooth hcurvature hnd
      (hsubcritical n))) atTop atTop.
```

The contracted one-sided theorem `Aiyagari1994.assetSupply_tendsto_at_impatience` packages the nonnegative-shift, normalization, and strict-subcritical premises in `M08B.UpperBoundaryPrices`. It

proves the corresponding `Tendsto` result from the comap of the neighborhood filter at `(qstar,phiStar)` to `atTop`.

Readable proof. Negating divergence produces infinitely many indices with asset supply below a fixed real bound. Convergence of the finite shift, gross return, and maximum effective income is combined with those indices before extracting a strictly increasing subsequence. A02 supplies, for every selected stationary law, integrability of resources and shifted saving, the identity $E[A] = S + \phi$, and the stationary resource identity $E[z] = RE[A] + E[e]$. Nonnegativity of shifted saving and effective income, together with the selected upper bounds, gives a single finite upper bound for all selected resource first moments.

The helper `M08B.tight_of_uniform_first_moment` proves Markov tightness directly on noncompact `NNReal`: for each positive tail tolerance it chooses a finite interval and bounds its complement by the uniform first moment. Prokhorov then supplies a weakly convergent further subsequence. The helper `M08B.householdKernel_tendstoLocallyUniformly` derives, rather than assumes, B01's compact-local bounded-test convergence: H06's joint policy continuity and the canonical kernel integral formula give uniform convergence on every compact resource interval. S01 supplies the Feller/Markov interfaces. B01 therefore makes the weak limit invariant under the exactly critical canonical kernel. N07 excludes that invariant law because $\beta R_* = 1$, a contradiction.

No unbounded moment is passed through weak convergence. Tightness, resource means, local kernel convergence, Feller continuity, and invariance are all derived before B01 is invoked. The proof uses neither S06 at the boundary, a common compact support, global kernel convergence, exploding supports, nor pathwise divergence.

Audit. The eight new public declarations in `Aiyagari1994/Analysis/M08B/UpperBoundary.lean` and `Aiyagari1994/Aggregate/UpperBoundary.lean` have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M08BSignatures.lean` and `Audit.lean`. Their transitive axiom set is exactly `propext`, `Classical.choice`, and `Quot.sound`.

Source and review boundary. C90 Proposition 2.4, printed p. 548 / PDF p. 7, states the related boundary conclusion without proof. A94 supplies the upper-boundary discussion. The compact/tail and critical-invariant contradiction are the authorized project reconstruction, not a literal source proof. All predecessor qualifications remain operative. This entry is submitted as REVIEW_READY only; no GREEN status, B03, Stage 09, or later result is self-awarded.

B03 — Natural Asset Supply tends To neg infinity

Status: GREEN. Independent Astra acceptance: `reviews/m08c_acceptance.md`. **Scope:** core. **Milestone:** 08.

Target declaration: `Aiyagari1994.naturalAssetSupply_tendsTo_neg_infinity`.
Module: `Aiyagari1994/Aggregate/LowerBoundary.lean`.

Mathematical contract. Under the natural limit, r_n>0 tends to zero and w_n→w0>0 imply S_n→-infinity. Uniformly bound shifted saving in normalized effective-income coordinates before subtracting phi_n→infinity.

Assumption profiles: BASIC, SMOOTH, CURVATURE, NONDEGENERATE, IID, NATURAL_CAP. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: D03, A02. **Source keys:** A93, A94, C90.

Source locator: A94 natural-limit boundary discussion, printed p. 673 / PDF p. 16; C90 Proposition 2.4, printed p. 548 / PDF p. 7.

Checked sequential declaration. Given fixed household primitives `m`, smoothness, curvature, and income nondegeneracy, let `rseq n > 0` and `wseq n > 0`. If `rseq` tends to zero and `wseq` tends to a positive `w0`, then the target, with no all-index strict-impatience premise, proves

```
Tendsto (fun n =>
  M08C.naturalAssetSupplyExtension m hsmooth hcurvature hnd
    (rseq n) (wseq n) (hrpos n) (hwpos n))
  atTop atBot.
```

`M08C.naturalAssetSupplyExtension` is the actual canonical stationary asset supply whenever `m.beta * (1+r) < 1` and is defined as zero otherwise. The fallback branch is only a total finite-prefix extension: it asserts no invariant-law existence at critical or supercritical prefix prices. The proof derives that the extension is eventually equal to actual stationary asset supply.

The normalized price is exactly `(R,w,k) = (1+r, w, -w*m.income.lower)`, so effective income is `w*(1-m.income.lower)`. The raw natural shift `M08C.naturalDebtShift m r w = w*m.income.lower/r` is a separate argument to asset-supply accounting and never enters household continuity or drift parameters. The contracted lower-boundary theorem packages positivity and strict impatience in `M08C.LowerBoundaryPrices` and proves the corresponding comap-neighborhood limit at `(0,w0)`.

Readable proof. From `rseq -> 0`, the products `m.beta*(1+rseq n)` tend to `m.beta`, which is strictly below one by the primitive discount-factor assumption. Thus the proof chooses `N` such that every index `n >= N` is strictly impatient. It applies the constrained all-index helper to the reindexed sequences `rseq (n+N)` and `wseq (n+N)`. `tendsto_add_atTop_iff_nat` then transfers the tail limit back to the original full-index extension, whose values at `n+N` simplify by the proved tail inequality. This is the explicit eventual-equality/reindexing bridge; it makes no stationary claim about the finite prefix.

On that tail, the bounded normalized family converges to `(1,w0,-w0*m.income.lower)`. The gate-local common-support lemma chooses uniform return, impatience, income-level, and income-span bounds in a neighborhood and invokes D03 to obtain one finite upper endpoint `B`. D03 supplies only weak drift plus forward invariance. Respecting that qualification, the proof constructs an invariant law on the economic compact interval and uses the accepted full-space uniqueness theorem to identify it with the canonical stationary law; it does not claim finite-time entry. Thus every tail stationary law is supported on `[0,B]`.

For each tail index, A02 is instantiated with the explicitly reconstructed natural original prices. It proves shifted-saving integrability and the exact identity `S_n = integral A_n d pi_n - phi_n`. On the common support, feasibility gives `A_n(z) <= z <= B`, and integrability justifies integral monotonicity, hence `integral A_n d pi_n <= B`. Positivity of `w0`, the primitive positive labor minimum, wage convergence, positive `r_n`, and `r_n -> 0` prove `phi_n = w_n*m.income.lower/r_n -> +infinity`. Therefore, for every real lower target `b`, eventually `S_n <= B-phi_n <= b`, which is convergence to `-infinity`.

Scope and dependency audit. The proof uses D03 and A02 (including their accepted transitive stationary-law construction) and no B02, B01, N07, monotonicity in prices, equilibrium object, finite institutional cap, support-expansion surrogate, or raw-parameter continuity theorem. IID remains the fixed labor probability law used by the canonical kernel. No zero-state marginal object occurs, so all inherited distinctions between `rightMarginalValue m 0` and `zeroRightMarginal` remain untouched. The contracted filter theorem explicitly invokes the full sequential theorem and uses its subtype's strict-impatience field only to identify the extension with `M08C.lowerBoundaryAssetSupply`. All predecessor qualifications remain operative.

Audit. The eleven new public declarations in the M08C helper and aggregate modules have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M08CSignatures.lean` and `Audit.lean`. Their transitive axiom set is limited to `propext`, `Classical.choice`, and `Quot.sound`.

Source and review boundary. A94 printed p. 673 / PDF p. 16 supplies the natural-limit discussion, and C90 Proposition 2.4, printed p. 548 / PDF p. 7, states a related result without proof. This complete normalized-family and weak-drift argument is the authorized project reconstruction, not a literal source proof. The implementation is submitted as REVIEW_READY only; no GREEN status, Stage 09 work, or later result is self-awarded.

F01 — Capital Demand wage constructed

Status: GREEN. Independent Astra acceptance: `reviews/m09a1_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.capitalDemand_wage_constructed`.
Module: `Aiyagari1994/Firms/Neoclassical.lean`.

Mathematical contract. For r>-delta, construct the unique K(r)>0 satisfying f_prime(K)=r+delta and w(r)=f(K)-K*f_prime(K)>0. Prove firm optimization, continuity of both functions and strict decrease of K.

Assumption profiles: PRODUCTION. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: Primitive mathematics / installed Mathlib. **Source keys:** A94.

Source locator: A94 firm-side general-equilibrium discussion, printed pp. 670-671 / PDF pp. 13-14; explicit primitive assumptions and consistency witness are supplied here.

Actual assumptions and exact interface. `ProductionData` contains only the output function and depreciation; it contains no demand or wage field. `ProductionRegularity p` states exactly `f(0)=0`, continuity on `Ici 0`, `ContDiffOn Real 2 f (Ioi 0)`, positive first derivative, negative second derivative, the zero Inada limit, vanishing marginal product at infinity, and `0<delta<1`. Rates have type `FirmRate p = Ioi (-p.depreciation)`. The audit anchor is:

```
capitalDemand_wage_constructed (p : ProductionData) (hp : ProductionRegularity p) :
  (forall r : FirmRate p,
    deriv p.output (capitalDemand p hp r) = (r : Real) + p.depreciation and
    0 < firmWage p hp r and
    forall K in Ici (0 : Real), K != capitalDemand p hp r ->
      firmProfit p r K < firmProfit p r (capitalDemand p hp r)) and
  Continuous (capitalDemand p hp) and StrictAnti (capitalDemand p hp) and
  Continuous (firmWage p hp)
```

Readable proof. Negative second derivative and C^2 regularity make the marginal product continuous and strictly decreasing on positive capital. For every positive target `r+delta`, the zero Inada limit supplies a small capital with marginal product weakly above the target, while the limit at infinity supplies a positive large capital below it. The intermediate value theorem gives a solution and strict decrease gives uniqueness. `capitalDemandPositive` selects only this proved witness. The inverse is strictly decreasing and surjective onto positive capital; viewed as a monotone map into the order dual, its inverse-range theorem proves continuity.

Strict concavity follows from negative second derivative. Its strict secant inequalities prove that the constructed ratio beats every other nonnegative ratio in unit-labor profit, including the zero boundary; thus the result is global optimization, not a first-order-condition claim. Applying the strict right-endpoint secant inequality between zero and demanded capital, together with `f(0)=0`, gives `f(K)-K f'(K)>0`. Wage continuity follows from demand continuity, positive- capital continuity of output, and the marginal equation.

Nonvacuity witness. `sqrtProduction` is exactly `f(K)=sqrt(K)`, `delta=1/2`. `sqrtProduction_regular` proves both derivatives, both limits, smoothness and all signs. `fullEquilibriumPrimitives_nonempty` combines this production witness with P03's constructed household witness and retains its exact beta, utility and income identities. P03 is not a dependency of the general firm theorem and imposes no finite-support restriction on it.

Source layout, scope and audit. The contracted wrapper remains in `Aiyagari1994/Firms/Neoclassical.lean`; all supporting proof declarations and witnesses are in the capsule helper directory `Aiyagari1994/Analysis/M09A1/`. No household stationarity, equilibrium, lower bracket, asset-supply result, G01–G08, F02 or Stage 10 declaration is used or proved. The source locator is A94 printed pp. 670–671 / PDF pp. 13–14; the exact primitive package and square-root consistency witness are project constructions. Every new public declaration is covered by `#check`, `assert_no_sorry`, and `#print axioms` in `Probes/M09A1Signatures.lean` and `Audit.lean`. The transitive axiom output is limited to the repository-approved foundational axioms. This entry requests review only and does not self-award GREEN.

F02 — Finite Cap lower bracket

Status: GREEN. Independent Astra acceptance: `reviews/m09a3_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.finiteCap_lower_bracket`.
Module: `Aiyagari1994/Equilibrium/LowerBracket.lean`.

Mathematical contract. For every finite b>=0, derive r_L in (-delta,0) with stationary asset supply S(r_L,w(r_L))<K(r_L), using f(K)/K→0 and E c=w+rS>=0. Do not assume an excess-supply sign.

Actual assumptions and exact interface. The theorem takes `p : ProductionData` and exactly `hp : ProductionRegularity p`; a household `m : HouseholdPrimitives`; `UtilitySmooth m.utility`, `UtilityCurvature m.utility`, `IncomeNondegenerate m.income`, and `LaborMeanOne m.income`; and a finite real institutional cap `b` with `0 <= b`. IID is the constructed finite-history product law in BASIC. No average-product limit, lower excess-supply sign, invariant law, or asset-supply bound is assumed. The conclusion supplies positive `K_L` and `r_L : FirmRate p`, proves `capitalDemand p hp r_L = K_L`, `r_L<0`, `f'(K_L)<delta`, `f(K_L)<delta*K_L`, the exact finite-cap identity `phi=b`, `1+r_L>0`, `beta*(1+r_L)<1`, and the strict canonical stationary-supply inequality.

The contracted wrapper elaborates as follows (proof-only arguments inside `let` terms are shown in the source and signature probe):

```
Aiyagari1994.finiteCap_lower_bracket
  (p : ProductionData) (hp : ProductionRegularity p)
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hcurvature : UtilityCurvature m.utility)
  (hnd : IncomeNondegenerate m.income) (hmean : LaborMeanOne m.income)
  (b : Real) (hb : 0 <= b) :
  exists K_L > 0, exists r_L : FirmRate p,
    capitalDemand p hp r_L = K_L and exists hrneg : (r_L : Real) < 0,
    deriv p.output K_L < p.depreciation and
    p.output K_L < p.depreciation * K_L and
    let prices := M09A3.lowerOriginalPrices m p hp b hb r_L
    let household := M09A3.lowerHousehold m p hp b hb r_L
    let hbetaR : household.beta * household.prices.grossReturn < 1 := ...
    prices.debtLimit = b and 0 < 1 + (r_L : Real) and
    household.beta * household.prices.grossReturn < 1 and
    stationaryAssetSupply household prices.debtLimit
      (M06C.stationaryLaw household hsmooth hcurvature hnd hbetaR) < K_L
```

Dependencies: P02, A02, F01. **Source keys:** A94.

Source locator: New derived lower-bracket lemma supporting A94 equilibrium existence; not a theorem explicitly proved by the source.

Readable proof. `production_average_tendsto_zero` fixes a positive reference capital whose marginal product is below half an arbitrary epsilon. Strict concavity bounds later output by the tangent at that reference point. Dividing the fixed tangent intercept by capital and taking capital large gives `f(K)/K<epsilon`; strict concavity from `f(0)=0` and positive marginal product gives the matching lower bound. Thus average-product decay is derived solely from PRODUCTION.

Apply both the derived average-product limit and the assumed marginal-product limit at infinity to choose positive `K_L` with `f(K_L)<delta*K_L` and `f'(K_L)<delta`. Set `r_L=f'(K_L)-delta`. Positivity of marginal product gives `r_L>-delta`; the second selected inequality gives `r_L<0`. F01's unique marginal-product solution identifies `capitalDemand(r_L)=K_L`, and F01 gives the positive wage. Since `delta<1`, `r_L>-delta` implies `1+r_L>0`; combining `r_L<0` with `0<beta<1` yields `beta*(1+r_L)<1` before stationarity is used.

P02's finite-cap constructor gives admissible original prices and, because the rate is nonpositive, the exact debt limit `phi=b`; the gate-local household changes prices only. A02 then supplies the canonical stationary law, all required integrability, and `E[c]=r_L*S+w_L*E[l]`. LABOR_MEAN_ONE reduces this to `E[c]=w_L+r_L*S`. Pointwise nonnegative consumption makes its integrable expectation nonnegative, so division by `-r_L>0` gives `S<=w_L/(-r_L)`. Substituting `w_L=f(K_L)-K_L*f'(K_L)` and `r_L=f'(K_L)-delta`, the selected strict output inequality gives `w_L/(-r_L)<K_L`. Hence `S<K_L=K(r_L)` with the sign derived rather than assumed.

Scope, dependencies, and audit. The contracted wrapper is in `Aiyagari1994/Equilibrium/LowerBracket.lean`; all proof helpers are in the assigned `Aiyagari1994/Analysis/M09A3/` directory. The proof uses only P02, A02, and F01. It does not use B02, B03, G02, G03, equilibrium existence, asset-supply continuity/divergence, or a numerical model. The result is a project construction supporting the A94 lower bracket, not a theorem literally stated in A94. All six new public declarations have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M09A3Signatures.lean` and `Audit.lean`; only the approved foundational axioms occur. This entry requests review only and does not self-award GREEN.

G01 — Equilibrium resource asset iff

Status: GREEN. Independent Astra acceptance: `reviews/m09a2_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.equilibrium_resource_asset_iff`.
Module: `Aiyagari1994/Equilibrium/Definition.lean`.

Mathematical contract. Define stationary equilibrium with firm optimization, canonical lifetime-optimal policy, invariant cross-sectional law, integrability and market clearing. Prove resource-law and net-asset/current-labor formulations equivalent. The definition allows all r>-delta and does not require r<lambda.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE, IID, LABOR_MEAN_ONE, PRODUCTION. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: P01, H05, S01, A01, F01. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Actual assumptions and equilibrium object. `EquilibriumCore p hp` carries the full PRODUCTION package, an untruncated `rate : FirmRate p = Ioi (-delta)`, BASIC household primitives, SMOOTH, NONDEGENERATE, the constructed finite-history IID property, LABOR_MEAN_ONE, and compatible `OriginalPrices` whose normalization is exactly the household price object. Since `delta<1`, the firm rate condition already implies the ordinary original-price requirement `r>-1`; no additional rate truncation is introduced. In particular, the object has no `r<lambda`, `beta*(1+r)<1`, positive-rate, asset-supply monotonicity, uniqueness, existence-constructor, or debt-rule field.

The witness supplies `resourceLaw : ProbabilityMeasure Resources`, integrability of resources and of induced net assets, and capital clearing

```
integral (netAsset household debtLimit) resourceLaw = capitalDemand p hp rate.
```

It also records the accepted F01 global firm certificate, H05 canonical lifetime optimality against every measurable full-history feasible plan, S01's actual policy-induced Markov kernel, and P01's exact original/shifted budget-and-feasibility equivalence. Thus the definition is not circular through the strictly impatient stationary-law constructor. `StationaryEquilibrium` extends this core only with resource-law invariance.

Exact contracted interface. The audit anchor is:

```
Aiyagari1994.equilibrium_resource_asset_iff
  {p : Aiyagari1994.ProductionData} {hp : Aiyagari1994.ProductionRegularity p}
  (e : Aiyagari1994.EquilibriumCore p hp) :
  Aiyagari1994.resourceLawForm e <-> Aiyagari1994.assetLaborLawForm e
```

Readable proof. The asset/labor formulation existentially supplies a net-asset law `rho`, requires it to equal the pushforward of the resource law under the canonical net-asset policy, and pairs that predetermined marginal with the fresh labor law via the product probability measure. Its image under the normalized resource transition is required to equal the supplied resource law. A01 says exactly that this image equality is equivalent to one household-law step fixing the resource law. The forward implication chooses the induced `rho` and applies A01; the reverse implication uses the formulation's equality identifying its `rho` with the induced law and applies A01 in the other direction.

This is an exact cross-sectional law identity, not a continuum law of large numbers. Resource law is not asset law. Only predetermined assets are independent of the fresh draw; no independence of the contemporaneous saving choice and contemporaneous income is assumed. Economic integrals used for clearing have explicit integrability fields.

Source, scope and audit. A94 printed pp. 670–671 / PDF pp. 13–14 supplies the general-equilibrium and certainty-comparison context, especially notes 24–27. The complete primitive package and exact resource/asset-law equivalence are authorized project constructions, not literal source claims. No G02/G03, A04/A05, G04–G08, N07/B02, equilibrium existence, uniqueness, or Stage 10 result is implemented. Every new public declaration is covered by `#check`, `assert_no_sorry`, and `#print axioms` in `Probes/M09A2Signatures.lean` and `Audit.lean`. The transitive axiom output is limited to `propext`, `Classical.choice`, and `Quot.sound`. This entry requests review only and does not self-award GREEN.

G02 — Finite Cap equilibrium exists

Status: GREEN. Independent Astra acceptance: `reviews/m09b1_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.finiteCap_equilibrium_exists`.
Module: `Aiyagari1994/Equilibrium/Existence.lean`.

Mathematical contract. For every finite b>=0 and the core economic primitives, there exists a stationary equilibrium with r in (-delta,lambda). Derive both signs from F02 and B02 and apply continuity/IVT. Do not assert uniqueness or positive interest.

Actual assumptions and exact interface. The theorem takes `p : ProductionData` with `hp : ProductionRegularity p`; `m : HouseholdPrimitives`; `UtilitySmooth m.utility`, `UtilityCurvature m.utility`, `IncomeNondegenerate m.income`, and `LaborMeanOne m.income`; and a finite real institutional cap `b` with `0 <= b`. BASIC includes the compact positive labor support, bounded strictly increasing and strictly concave utility, `0<beta<1`, and admissible price data. IID is the constructed finite-history product law. The conclusion is an actual unchanged G01 object, not a scalar root. Its existential predicate retains the supplied beta, utility, entire `IncomeData` (and hence the entire income law), and the finite cap used at the endogenous wage and rate. It deliberately does not equate the equilibrium household's endogenous normalized prices with the input `m.prices`. Both exported declarations have the following exact signature:

```
Aiyagari1994.M09B1.finiteCap_equilibrium_exists_core
  (p : ProductionData) (hp : ProductionRegularity p)
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hcurvature : UtilityCurvature m.utility)
  (hnd : IncomeNondegenerate m.income)
  (hmean : LaborMeanOne m.income) (b : Real) (hb : 0 <= b) :
  exists e : StationaryEquilibrium p hp,
    e.household.beta = m.beta and
    e.household.utility = m.utility and
    e.household.income = m.income and
    e.originalPrices.debtLimit =
      effectiveLimit b m.income.lower (firmWage p hp e.rate) (e.rate : Real) and
    -p.depreciation < (e.rate : Real) and
    (e.rate : Real) < 1 / m.beta - 1

Aiyagari1994.finiteCap_equilibrium_exists
  (p : ProductionData) (hp : ProductionRegularity p)
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hcurvature : UtilityCurvature m.utility)
  (hnd : IncomeNondegenerate m.income)
  (hmean : LaborMeanOne m.income) (b : Real) (hb : 0 <= b) :
  exists e : StationaryEquilibrium p hp,
    e.household.beta = m.beta and
    e.household.utility = m.utility and
    e.household.income = m.income and
    e.originalPrices.debtLimit =
      effectiveLimit b m.income.lower (firmWage p hp e.rate) (e.rate : Real) and
    -p.depreciation < (e.rate : Real) and
    (e.rate : Real) < 1 / m.beta - 1
```

Dependencies: P02, A03, B02, F02, G01. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Readable proof. On the full firm domain, define the actual finite-cap original prices using P02 at `K(r)` and `w(r)` from F01. The corresponding normalized gross return is `1+r`, the wage is positive, and P02 proves nonnegative effective income and continuity of the debt shift through zero. Restricting to `r<lambda=1/beta-1`, positivity of beta proves `beta*(1+r)<1` before selecting the canonical stationary law. A03's repaired joint theorem, composed with the continuous normalized-price and independent debt-shift paths, proves continuity of canonical stationary net asset supply. Subtracting continuous F01 capital demand gives the actual excess path `X(r)=S(r)-K(r)`; no continuity or monotonicity field is assumed.

F02 supplies `r_L` in `(-delta,0)` on this same path with `S(r_L)<K(r_L)`, hence `X(r_L)<0`. For the upper sign, take an explicit sequence `r_n` increasing to `lambda` from below while remaining in the firm domain. F01 continuity proves `w(r_n)->w(lambda)>0` and `K(r_n)->K(lambda)`, hence capital demand stays finite. P02 continuity proves that the finite-cap debt shift converges to its finite value at `(w(lambda),lambda)`; the normalized gross return, wage, and intercept therefore converge to critical admissible prices. B02 then makes canonical stationary net asset supply tend to positive infinity. Selecting a term above the finite limiting capital demand gives an actual `r_U<lambda`, with `r_L<r_U` and `X(r_U)>0`.

Parameterize the compact bracket by the unit interval. Continuity and the intermediate value theorem give a strict interior root because both endpoint signs are strict. At that root construct the P02 original prices, the corresponding household, and the canonical stationary resource law. Because that household is definitionally `m.withPrices` at the endogenous normalized prices, its beta, utility, and complete income data are exactly those of `m`; only its prices are replaced. Because the original-price witness is definitionally `finiteCapPrices`, its debt limit is exactly `effectiveLimit b m.income.lower (firmWage p hp e.rate) e.rate`. These identities are returned in the existential predicate rather than left implicit in the construction. The proof supplies its actual-kernel invariance, compact-support resource integrability, accepted stationary asset integrability and hence net-asset integrability, mean-one labor and finite-history IID, exact P01 budget normalization, H05 lifetime optimality, S01 kernel certificate, F01 firm optimization, and capital clearing from `X=0`. Thus every field of G01's unchanged `StationaryEquilibrium` is filled. The result does not claim a positive rate, uniqueness, monotonicity, comparison results, or a statement about every equilibrium.

Source, scope and audit. A94 printed pp. 670–671 / PDF pp. 13–14 motivates the firm capital curve, household asset-supply curve, their steady-state intersection, and notes 24–25's explicit absence of monotonicity and uniqueness guarantees. The complete finite-cap continuity, endpoint limits, IVT bridge, and G01 record construction are project proofs, not literal source proofs. The wrapper is in `Aiyagari1994/Equilibrium/Existence.lean`; the proof engine is confined to `Aiyagari1994/Analysis/M09B1/FiniteCapExistence.lean`. Both public declarations have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M09B1Signatures.lean` and `Audit.lean`; only `propext`, `Classical.choice`, and `Quot.sound` occur. G03 and all later contracts remain unimplemented. This entry requests review only and does not self-award GREEN.

G03 — Natural Cap equilibrium exists

Status: GREEN. Independent Astra acceptance: `reviews/m09b2_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.naturalCap_equilibrium_exists`.
Module: `Aiyagari1994/Equilibrium/Existence.lean`.

Mathematical contract. For the natural-limit model and the core economic primitives, a stationary equilibrium exists with 0<r<lambda. Derive endpoint signs from B03 and B02 along the firm wage schedule.

Actual assumptions and exact interface. The theorem takes `p : ProductionData` with `hp : ProductionRegularity p`; `m : HouseholdPrimitives`; `UtilitySmooth m.utility`, `UtilityCurvature m.utility`, `IncomeNondegenerate m.income`, and `LaborMeanOne m.income`. BASIC supplies bounded strictly increasing and strictly concave utility, `0<beta<1`, and a compact positive labor support; IID is the existing finite-history product-law construction. Both exports have the exact signature:

```
Aiyagari1994.M09B2.naturalCap_equilibrium_exists_core
  (p : ProductionData) (hp : ProductionRegularity p)
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hcurvature : UtilityCurvature m.utility)
  (hnd : IncomeNondegenerate m.income)
  (hmean : LaborMeanOne m.income) :
  exists e : StationaryEquilibrium p hp,
    e.household.beta = m.beta and
    e.household.utility = m.utility and
    e.household.income = m.income and
    e.originalPrices.debtLimit =
      naturalLimit m.income.lower (firmWage p hp e.rate) (e.rate : Real) and
    0 < (e.rate : Real) and (e.rate : Real) < 1 / m.beta - 1

Aiyagari1994.naturalCap_equilibrium_exists
  (p : ProductionData) (hp : ProductionRegularity p)
  (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
  (hcurvature : UtilityCurvature m.utility)
  (hnd : IncomeNondegenerate m.income)
  (hmean : LaborMeanOne m.income) :
  exists e : StationaryEquilibrium p hp,
    e.household.beta = m.beta and
    e.household.utility = m.utility and
    e.household.income = m.income and
    e.originalPrices.debtLimit =
      naturalLimit m.income.lower (firmWage p hp e.rate) (e.rate : Real) and
    0 < (e.rate : Real) and (e.rate : Real) < 1 / m.beta - 1
```

Dependencies: P02, A03, B02, B03, F01, G01. **Source keys:** A93, A94.

Source locator: A94 natural-limit existence discussion and note 30, printed p. 673 / PDF p. 16.

Readable proof. Put `lambda=1/beta-1>0` and restrict the actual firm schedule to `0<r<lambda`. At every such rate, P02's `naturalCapPrices` has debt limit `phi=w(r)l_min/r`, normalized gross return `1+r`, wage `w(r)>0`, and intercept `-w(r)l_min`; its effective income is `w(r)(l-l_min)>=0`. Positivity of beta and `r<lambda` prove `beta*(1+r)<1` before the canonical stationary law is invoked. F01 and P02 make the normalized-price and debt-shift paths continuous on this open positive-rate domain. Composing both coordinates with A03's repaired joint interface makes actual canonical stationary net asset supply continuous; subtracting continuous F01 capital demand gives `X_N(r)=S_N(r)-K(r)`.

For the lower sign, an explicit positive sequence tends to zero. F01 continuity proves `w(r_n)->w(0)>0` and `K(r_n)->K(0)`, a finite real value. B03 then sends actual natural-limit stationary net asset supply to negative infinity. Comparing it with the convergent capital-demand sequence selects a finite `r_L>0` with `X_N(r_L)<0`; no natural debt limit is evaluated at zero.

For the upper sign, a second explicit positive sequence tends to `lambda` from below. F01 gives `w(r_n)->w(lambda)>0` and `K(r_n)->K(lambda)`. The normalized prices converge to the critical admissible triple `(1+lambda,w(lambda),-w(lambda)l_min)`, while P02's raw shift converges to the finite

w(lambda)l_min/lambda. B02 therefore sends stationary net asset supply to positive infinity. Choosing a sufficiently late term also ensures r_L<r_U<lambda and X_N(r_U)>0.

Parameterizing [r_L,r_U] by the unit interval and applying IVT yields a root of the continuous actual excess path. At the root the proof constructs P02 natural original prices, the corresponding household, and the canonical stationary resource law. It supplies actual-kernel invariance, compact-support resource integrability, accepted asset integrability and hence net-asset integrability, mean-one labor, finite-history IID, H05 lifetime optimality, S01's kernel certificate, P01 budget normalization, F01 firm optimization, and exact capital clearing from X_N=0. Thus every field of the unchanged G01 equilibrium is filled while retaining the supplied beta, utility, and complete income data.

Source, scope and audit. A94 printed p. 673 / PDF p. 16, note 30, with adjacent PDF p. 15, motivates natural-limit asset-supply divergence and equilibrium existence. The structured A93 source is resolved through its approved priority context and hash. The complete firm-path limits, continuity composition, endpoint selections, IVT, and G01 record construction are project proofs, not literal source proofs. The gate-local engine is in Aiyagari1994/Analysis/M09B2/NaturalCapExistence.lean; the public wrapper is appended to Aiyagari1994/Equilibrium/Existence.lean without changing G02's declaration or proof. Both new exports have #check, assert_no_sorry, and #print axioms coverage in Probes/M09B2Signatures.lean and Audit.lean; their transitive axioms are only propext, Classical.choice, and Quot.sound. No F02 or G02 lower bracket is used. The result proves one equilibrium only, not uniqueness, monotonicity, comparative statics, every-equilibrium positivity or subcriticality, G04, or stage advancement. This entry requests review and does not self-award GREEN.

G04 — Every equilibrium rate below impatience

Status: GREEN. Independent Astra acceptance: reviews/m09c3_acceptance.md. **Scope:** core. **Milestone:** 09.

Target declaration: Aiyagari1994.every_equilibrium_rate_below_impatience.
Module: Aiyagari1994/Equilibrium/MainTheorem.lean.

Mathematical contract. Every stationary equilibrium satisfying the untruncated definition has r<lambda, because its invariant law contradicts N07 if beta*(1+r)>=1. This is not restricted to the particular equilibria constructed by G02/G03.

Exact elaborated Lean signature.

```
Aiyagari1994.every_equilibrium_rate_below_impatience
  (p : Aiyagari1994.ProductionData)
  (hp : Aiyagari1994.ProductionRegularity p)
  (e : Aiyagari1994.StationaryEquilibrium p hp) :
  (e.rate : ℝ) < 1 / e.household.beta - 1
```

Actual assumptions and unrestricted quantifier. BASIC, SMOOTH, NONDEGENERATE, IID, LABOR_MEAN_ONE, and PRODUCTION are carried by the arbitrary accepted G01 equilibrium witness. The complete EquilibriumCore definition has a FirmRate p, household primitives, smoothness, nondegenerate income, mean-one labor, finite-history IID, exact original/normalized prices, the actual resource probability law and its first moment, integrable net assets, clearing, firm and household certificates, and budget normalization. StationaryEquilibrium extends it only with resource-law stationarity. Neither structure contains impatience or a rate upper bound.

Dependencies: N07, G01. **Source keys:** A93, A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Readable proof. Fix any `e : StationaryEquilibrium p hp`. Coercing its `resource_stationary` equality to measures exhibits `e.resourceLaw` as an invariant probability measure for the actual kernel `householdKernel e.household`. If `1 <= e.household.beta * e.household.prices.grossReturn`, N07 says that no such probability law exists, contradicting this witness. Therefore `beta*R<1`.

The unchanged G01 price fields and definitional normalization give the chain

```
R = e.originalPrices.normalized.grossReturn
  = 1 + e.originalPrices.netRate
  = 1 + e.rate.
```

Because household beta is positive, elementary ordered-field algebra yields `e.rate < 1/e.household.beta - 1`. The proof neither constructs an equilibrium nor invokes G02 or G03, so it applies to every witness of the unrestricted type.

Source, scope, and audit. A94 printed pp. 670–671 / PDF pp. 13–14 motivates the general-equilibrium restriction. The exact invariant-law contradiction against the Lean G01 record is a project reconstruction. The gate-local proof is `Aiyagari1994.M09C3.every_equilibrium_rate_below_impatience_core`; the public theorem is a thin wrapper. Both declarations have `#check`, `assert_no_sorry`, and `#print axioms` coverage in the M09C3 probe and global audit. Their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

This theorem proves no equilibrium existence or positivity, uniqueness, capital result, asset or saving comparison, G05–G08 result, or stage advancement. It is submitted as REVIEW_READY and does not self-award GREEN.

G05 — Certainty benchmark verified

Status: GREEN. Independent Astra acceptance: `reviews/m09c4_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.certainty_benchmark_verified`.
Module: `Aiyagari1994/Equilibrium/CertaintyBenchmark.lean`.

Exact elaborated wrapper signature.

```
Aiyagari1994.certainty_benchmark_verified
  (base : Aiyagari1994.HouseholdPrimitives)
  (hsmooth : Aiyagari1994.UtilitySmooth base.utility)
  (hmean : Aiyagari1994.LaborMeanOne base.income)
  (prod : Aiyagari1994.ProductionData)
  (hprod : Aiyagari1994.ProductionRegularity prod) :
  Aiyagari1994.M09C4.CertaintyBenchmarkStatement base hsmooth hmean prod hprod
```

The statement defines `lambda=1/base.beta-1`, constructs `rFI : FirmRate prod` with real value `lambda` and explicitly records `lambda > -prod.depreciation`. With `K=capitalDemand prod hprod rFI`, `wage=firmWage prod hprod rFI`, and `cFI=prod.output K-prod.depreciation*K`, it proves

```
0<K, 0<wage, deriv prod.output K=lambda+prod.depreciation,
cFI=wage+lambda*K, 0<cFI.
```

It also records that the supplied labor mean is one. For every nonnegative certainty debt shift `phi` satisfying the exact mean-one effective-income condition `lambda*phi<=wage`, it constructs the degenerate labor-one household, original and normalized prices, shifted benchmark saving `astar=K+phi`, consumption `cstar=cFI`, resources `zstar=cstar+astar`, and a constant feasible plan. This quantified debt interface includes the certainty finite-cap value `min(b,wage/lambda)` for every `b>=0` and the certainty natural value `wage/lambda`; it does not use the risky minimum-labor limit.

Actual assumptions: `HouseholdPrimitives`, `UtilitySmooth`, `LaborMeanOne`, and `ProductionRegularity`; no nondegeneracy, curvature, stationary-law, equilibrium-existence, or No-Ponzi premise is added.

Dependencies: P01, H05, F01. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Readable proof. Positivity of beta and `beta<1` give `lambda>0`, hence `lambda>-depreciation`. F01 supplies positive `K`, positive wage and `f'(K)=lambda+depreciation`. Expanding the wage definition yields `f(K)-depreciation*K=wage+lambda*K>0`.

For any admitted certainty debt shift, the original-price intercept is `-lambda*phi`, so labor one has nonnegative effective income. The constant shifted action is `K+phi`; its original asset is exactly `K`, hence it satisfies `-phi<=K`. Its resource level is `cFI+K+phi`. P01's accepted original/shifted bridge is applied at date zero to prove the original budget and borrowing inequality with these initial resources.

For an arbitrary admitted measurable full-history plan, the degenerate labor law makes every finite history equal to the labor-one history. The proof telescopes the normalized feasibility budgets exactly:

```
sum_{t=0}^N beta^t (c_t-cFI) = beta^N (astar-a_N)
                                    <= beta^N astar.
```

This is a derived finite present-value budget. It uses only nonnegative shifted actions, not a primitive transversality condition. Global concavity and differentiability at positive `cFI` give the supporting-line inequality `u(c)-u(cFI)<=u'(cFI)*(c-cFI)`. Summing it and sending `N` to infinity makes the right side vanish because `beta<1`. Thus the constant plan dominates every admitted feasible plan in the accepted finite-history lifetime-utility sense. H05 is then used in the reverse comparison with the canonical plan to identify the constant plan's lifetime utility with the value function; it is not used as an Euler or stationary-law shortcut.

Source, scope, and audit. A94 printed pp. 670–671 / PDF pp. 13–14 motivates the full-information certainty benchmark and competitive factor prices. The finite-budget and supporting-line lifetime verification is an explicit project reconstruction. The gate proves no risky equilibrium existence, critical stationary law, capital comparison, saving comparison, risky uniqueness, G06–G08 result, No-Ponzi theorem, or Stage-10 result.

The public helper statement, degenerate income definition, core theorem, and wrapper each have `#check`, `assert_no_sorry`, and `#print axioms` coverage in the M09C4 probe and global audit. Their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`. This entry is submitted as REVIEW_READY and does not self-award GREEN or advance the stage.

G06 — Equilibrium capital above certainty

Status: GREEN. Independent Astra acceptance: `reviews/m09d1_acceptance.md`.

Scope: core. **Milestone:** 09 / gate M09D1.

Target declaration: `Aiyagari1994.equilibrium_capital_above_certainty`.
Module: `Aiyagari1994/Equilibrium/MainTheorem.lean`.

Mathematical contract. Every risky stationary equilibrium has K>K_FI=K(lambda).

Exact elaborated Lean signature.

```
Aiyagari1994.equilibrium_capital_above_certainty
  (p : Aiyagari1994.ProductionData)
  (hp : Aiyagari1994.ProductionRegularity p)
  (e : Aiyagari1994.StationaryEquilibrium p hp) :
  ∃ rFI : Aiyagari1994.FirmRate p,
```

```
(rFI : ℝ) = 1 / e.household.beta - 1 ∧
Aiyagari1994.capitalDemand p hp rFI <
  ∫ z, Aiyagari1994.M06B.netAsset e.household
    e.originalPrices.debtLimit z ∂e.resourceLaw
```

Actual assumptions and universal scope. BASIC, SMOOTH, NONDEGENERATE, IID, LABOR_MEAN_ONE, and PRODUCTION are carried by an arbitrary `e : StationaryEquilibrium p hp`. The quantifier is not restricted to either G02/G03 existence witness. No positive-rate, uniqueness, asset-supply monotonicity, or A05 premise is added.

Dependencies: F01, G04, G05. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Readable proof. Apply G05's accepted certificate to `e.household`, its smoothness and mean-one fields, and the supplied production economy. Unpacking it yields a firm-domain rate `rFI` with the exact identity `(rFI : ℝ)=lambda=1/e.household.beta-1`; hence `capitalDemand p hp rFI` is the certified certainty capital `K_FI=K(lambda)`. G04 applies to the arbitrary equilibrium and gives `e.rate<lambda`, so `e.rate<rFI` after the benchmark identity. F01's strict antitonicity reverses this inequality:

`capitalDemand p hp rFI < capitalDemand p hp e.rate.`

Finally G01's `capital_clearing` field identifies the right side with the equilibrium's actual integral of net assets. This proves actual risky equilibrium capital strictly exceeds the certainty benchmark, with the required orientation.

Source, dependencies, and audit. Dependencies are exactly F01, G04, and G05. A94 printed pp. 670–671 / original PDF pp. 13–14, especially notes 24–27, motivates the equilibrium and certainty capital comparison; the exact Lean composition is a project reconstruction. The gate-local theorem and thin public wrapper both have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M09D1Signatures.lean` and `Audit.lean`; their transitive axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

This gate proves no rate positivity, equilibrium existence or uniqueness, G07 saving-share result, G08 goods-clearing result, or stage advancement. It requests independent review for G06 only and does not self-award GREEN.

G07 — Equilibrium gross saving share above certainty

Status: GREEN. Independent Astra acceptance: `reviews/m09d2_acceptance.md`. **Scope:** core. **Milestone:** 09.

Target declaration: `Aiyagari1994.equilibrium_gross_saving_share_above_certainty`.
Module: `Aiyagari1994/Equilibrium/Saving.lean`.

Mathematical contract. Every risky stationary equilibrium has delta*K/f(K)>delta*K_FI/f(K_FI). Prove that the gross investment share is strictly increasing using f(K)-K*f_prime(K)>0. Net saving is not the outcome.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE, IID, LABOR_MEAN_ONE, PRODUCTION. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: F01, G06. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Lean statement. For arbitrary `p`, `hp`, and `e : StationaryEquilibrium p hp`, the theorem returns a certainty firm rate `rFI` with `(rFI : ℝ)=1/e.household.beta-1` and proves `delta*K_FI/f(K_FI) < delta*K/f(K)`, where `K_FI=capitalDemand p hp rFI` and `K` is the integral of actual equilibrium net assets under `e.resourceLaw`.

Checked proof. The gate-local definition `Aiyagari1994.M09D2.grossReplacementShare p K = p.depreciation*K/p.output K` records the gross replacement-investment share. For each `K>0`, F01's surjectivity of positive capital demand selects a firm rate whose demanded capital is `K`; F01's positive wage then gives `f(K)-K*f'(K)>0`. Marginal-product positivity implies `f(K)>0`, so the denominator and its square are strictly positive. The quotient rule proves `g'(K)=delta*(f(K)-K*f'(K))/f(K)^2>0`. Mathlib's `strictMonoOn_of_deriv_pos` applies on the convex open interval `Ioi 0`, with continuity proved from the production assumptions and the established nonzero denominator, and yields strict monotonicity rather than merely a pointwise derivative claim.

G06 supplies `K_FI<K` for the arbitrary equilibrium. Both capitals are positive: F01 gives positivity of `capitalDemand` at `rFI`, while G01's accepted capital-clearing field rewrites the actual net-asset integral as `capitalDemand p hp e.rate`. Applying the checked strict monotonicity theorem in this direction proves the displayed share comparison.

Source, dependencies, and audit. Dependencies are exactly F01 and G06. A94 printed pp. 670–671 / original PDF pp. 13–14, especially notes 24–27, motivates the capital and investment-share comparison; the derivative and mean-value proof are project reconstructions. The gate-local declarations and thin public wrapper have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M09D2Signatures.lean` and `Audit.lean`.

The conclusion concerns gross replacement investment divided by output. It asserts no positive stationary net saving, net asset accumulation, capital growth, interest-rate monotonicity, equilibrium uniqueness, G08 goods clearing, or stage advancement. This entry requests independent review for G07 only and does not self-award GREEN.

G08 — Equilibrium goods market clears

Status: GREEN. Independent Astra acceptance: `reviews/m09d3_acceptance.md`. **Scope:** core. **Milestone:** 09 / gate M09D3.

Target declaration: `Aiyagari1994.equilibrium_goods_market_clears`.
Module: `Aiyagari1994/Equilibrium/Saving.lean`.

Mathematical contract. At every stationary equilibrium, E c + delta*K=f(K), deriving goods clearing from household stationarity and competitive factor payments. Establish required moment statements for a general equilibrium law.

Exact elaborated wrapper statement. For every production economy `p`, regularity witness `hp`, and unrestricted `e : StationaryEquilibrium p hp`, `integral c de.resourceLaw + p.depreciation * integral netAsset de.resourceLaw = p.output (integral netAsset de.resourceLaw)`, where consumption is the canonical `consumptionPolicy e.household` and net assets are `M06B.netAsset e.household e.originalPrices.debtLimit`.

Assumption profiles: BASIC, SMOOTH, NONDEGENERATE, IID, LABOR_MEAN_ONE, PRODUCTION. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: G01, F01. **Source keys:** A94.

Source locator: A94 general-equilibrium and certainty comparison, printed pp. 670-671 / PDF pp. 13-14, especially notes 24-27.

Checked proof. The accepted equilibrium witness supplies integrability of resources and net assets. Adding the constant debt-limit shift proves integrability of shifted saving; subtracting that shifted saving from resources and using the canonical identity `c=z-A` proves consumption integrability. Labor is integrable by compact support. The proof pushes the resource coordinate through the actual household transition under the product of the current resource law and a fresh iid labor draw. Resource-law invariance then gives `E z = R E A + E effectiveIncome`.

Exact original-price normalization gives `R=1+r` and effective income `w*l-r*phi`. Integrating the canonical current budget, using mean-one labor, and rewriting `S=E(A-phi)` yields `E c = r*S+w`. G01's net-asset clearing field identifies `S` with the firm's `capitalDemand p hp e.rate`. F01 gives `f'(K)=r+delta`, while the competitive wage is definitionally `f(K)-K*f'(K)`. Substitution and ring normalization prove `E c + delta*K=f(K)`.

Source, dependencies, and audit. Dependencies are exactly G01 and F01. No A02 stationary-law constructor, G02/G03 witness, G04 impatience theorem, or G06/G07 comparison is imported or used. A94 printed pp. 670–671 / original PDF pp. 13–14, especially notes 24–27, motivates factor pricing and aggregate accounting; the exact moment and product-transition proof is a project reconstruction. The gate-local theorem and thin public wrapper have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M09D3Signatures.lean` and `Audit.lean`. At the M09D3 submission this entry requested independent review for G08 only and did not self-award GREEN or advance beyond Stage 09; the later controller acceptance is recorded above.

NP01 — Discounted budget telescope

Status: GREEN. Independent Astra acceptance: `reviews/m10a1_acceptance.md`. **Scope:** core. **Milestone:** 10 / gate M10A1.

Target declaration: `Aiyagari1994.discounted_budget_telescope`.
Module: `Aiyagari1994/Budget/NoPonzi.lean`.

Exact elaborated signature.

```
Aiyagari1994.discounted_budget_telescope
    (R w : Real) (a c labor : Nat -> Real) (hR : 0 < R)
    (hbudget : forall t : Nat, c t + a (t + 1) = R * a t + w * labor t)
    (T : Nat) :
  Finset.sum (Finset.range (T + 1)) (fun t => (Inv.inv R) ^ t * c t) =
    R * a 0 +
      Finset.sum (Finset.range (T + 1)) (fun t => (Inv.inv R) ^ t * (w * labor t)) -
        (Inv.inv R) ^ T * a (T + 1)
```

Actual assumptions and interpretation. The finite identity uses only the BASIC component `0 < R` and the original-coordinate budget-equality component of PATH_FEASIBILITY. It is a deterministic statement on each path. Wage positivity, consumption nonnegativity, measurability, adaptedness, IID income, a natural floor, No-Ponzi, optimality, convergence and integrability are unnecessary for this algebraic result. Here `a` is the original/net asset path, not shifted assets. For `R > 0`, `(Inv.inv R) ^ t` is exactly the displayed discount factor `R^(-t)`.

Dependencies: P01. **Source keys:** A93.

Source locator: A93 Appendix Proposition 1, printed p. 37 / PDF p. 38; A94 p. 666 / PDF p. 9, note 16.

Readable proof. Induct on `T`. At `T=0`, the statement is the date-zero original budget rearranged. At the successor step, split both sums at date `T+1` and substitute the induction hypothesis. Positivity of `R` gives `R != 0`, hence `(R^-1)^(T+1) * R = (R^-1)^T`. Multiplying the date-`T+1` budget by `(R^-1)^(T+1)`

cancels the prior terminal asset and leaves `-(R^-1)^(T+1) * a (T+2)`. Thus at horizon `T` the terminal asset is exactly `-(R^-1)^T * a (T+1)`, not exponent `T+1`.

Source, dependency and audit. Dependency is exactly P01. Its normalization module fixes the meaning of the original budget and distinguishes original assets from shifted assets; NP01 starts from that original budget equality and proves no P01 No-Ponzi conclusion. A93 Appendix Proposition 1, printed p. 37 / original PDF p. 38, displays this finite identity in its proof. A94 printed p. 666 / original PDF p. 9, note 16, states the corresponding present-value and borrowing-limit relationship. The explicit Lean induction is a project reconstruction. The gate-local theorem and thin public wrapper have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M10A1Signatures.lean` and `Audit.lean`. Both axiom inventories are exactly `propext`, `Classical.choice`, and `Quot.sound`. At the M10A1 submission this entry requested independent review of NP01 only and did not self-award GREEN or authorize stage advancement; the later controller acceptance is recorded above. That acceptance did not prove NP02 or NP03.

NP02 — Natural Bound implies no Ponzi

Status: GREEN. Independent Astra acceptance: `reviews/m10a2_acceptance.md`. **Scope:** core. **Milestone:** 10 / gate M10A2.

Target declaration: `Aiyagari1994.naturalBound_implies_noPonzi`.
Module: `Aiyagari1994/Budget/NoPonzi.lean`.

Exact elaborated signature.

```
Aiyagari1994.naturalBound_implies_noPonzi.{u_1}
    {Omega : Type u_1} [MeasurableSpace Omega]
    (mu : MeasureTheory.Measure Omega)
    (r w laborMin laborMax : Real) (a c labor : Nat -> Omega -> Real)
    (hr : 0 < r) (hw : 0 < w) (hlaborMin : 0 < laborMin)
    (hlaborLower : forall t : Nat, almost everywhere omega with respect to mu,
      laborMin <= labor t omega)
    (hlaborUpper : forall t : Nat, almost everywhere omega with respect to mu,
      labor t omega <= laborMax)
    (hc : forall t : Nat, almost everywhere omega with respect to mu,
      0 <= c t omega)
    (hbudget : forall t : Nat, almost everywhere omega with respect to mu,
      c t omega + a (t + 1) omega = (1 + r) * a t omega + w * labor t omega)
    (hfloor : forall t : Nat, almost everywhere omega with respect to mu,
      -Aiyagari1994.naturalLimit laborMin w r <= a t omega) :
  almost everywhere omega with respect to mu, exists L : Real, 0 <= L and
    Filter.Tendsto (fun T : Nat => (Inv.inv (1 + r)) ^ T * a (T + 1) omega)
      Filter.atTop (nhds L)
```

Actual assumptions and interpretation. The signature uses `r > 0`, `w > 0`, a positive lower labor endpoint, datewise almost-sure lower and upper labor bounds, nonnegative consumption, the original one-period budget, and the exact natural original-asset floor `a_t >= -naturalLimit laborMin w r = -w*laborMin/r`. It requires no IID law, stationarity, optimality, Euler equation, equilibrium premise, or zero-limit/transversality condition. Measurability and nonanticipation from PATH_FEASIBILITY are unnecessary for this forward implication; the raw-path theorem is therefore valid for every feasible adapted plan in the profile. The countable datewise hypotheses are intersected before applying that pathwise theorem, so no almost-sure hypothesis is used pointwise outside a common full-measure event.

Assumption profiles: BASIC, PATH_FEASIBILITY, NATURAL_CAP. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: NP01. **Source keys:** A93.

Source locator: A93 Appendix Proposition 1, printed p. 37 / PDF p. 38; A94 p. 666 / PDF p. 9, note 16.

Readable proof. Put `R=1+r` and `q=R^-1`. Since `r>0`, `0<q<1`. Bounded labor and positive wage make the discounted income series summable by comparison with a geometric series. The NP01 finite telescope and `a_(T+1) >= -w*laborMin/r` bound every nonnegative discounted-consumption partial sum above by initial resources, total discounted income, and the natural limit. Hence the discounted-consumption series converges. Rearranging NP01 then shows that exactly `q^T*a_(T+1)` converges to the difference of two finite series and initial resources. Finally, `q^T*a_(T+1) >= -q^T*naturalLimit laborMin w r`, whose right side tends to zero, so the terminal limit is nonnegative. Nothing proves or assumes that it equals zero.

Source, dependency and audit. Dependency is exactly accepted NP01 through its M10A1 telescope helper. A93 Appendix Proposition 1, printed p. 37 / original PDF p. 38, states the forward argument and displays the finite identity; A94 printed p. 666 / original PDF p. 9, note 16, records the present-value/natural-borrowing-limit relationship. The detailed summability, partial-sum bound, limit arithmetic, and countable almost-sure lift are project reconstructions. All three new public declarations have `#check`, `assert_no_sorry`, and `#print axioms` coverage in `Probes/M10A2Signatures.lean` and `Audit.lean`; their axiom inventories are exactly `propext`, `Classical.choice`, and `Quot.sound`. This entry records executor status REVIEW_READY for NP02 only. It does not award GREEN, prove NP03, or authorize stage advancement.

NP03 — No Ponzi implies natural Bound

Status: GREEN. Independent Astra acceptance: `reviews/m10a3_acceptance.md`. **Scope:** core. **Milestone:** 10 / gate M10A3.

Target declaration: `Aiyagari1994.noPonzi_implies_naturalBound`.
Module: `Aiyagari1994/Budget/NoPonzi.lean`.

Exact elaborated signature.

```
Aiyagari1994.noPonzi_implies_naturalBound
    (i : Aiyagari1994.IncomeData) (hsupp : Aiyagari1994.IncomeSupport i)
    (hnd : Aiyagari1994.IncomeNondegenerate i)
    (r w : Real) (hr : 0 < r) (hw : 0 < w)
    (p : Aiyagari1994.M10A3.AdaptedOriginalPlan i)
    (hc : forall t : Nat, almost everywhere omega with respect to
      Aiyagari1994.M10A3.iidPathLaw i, 0 <= p.consumption t omega)
    (hbudget : forall t : Nat, almost everywhere omega with respect to
      Aiyagari1994.M10A3.iidPathLaw i,
      p.consumption t omega + p.asset (t + 1) omega =
        (1 + r) * p.asset t omega + w * (omega t : Real))
    (hnoPonzi : almost everywhere omega with respect to
      Aiyagari1994.M10A3.iidPathLaw i, exists L : Real, 0 <= L and
      Filter.Tendsto (fun T : Nat => (Inv.inv (1 + r)) ^ T *
        p.asset (T + 1) omega) Filter.atTop (nhds L))
    (t : Nat) :
  almost everywhere omega with respect to Aiyagari1994.M10A3.iidPathLaw i,
    -Aiyagari1994.naturalLimit i.lower w r <= p.asset t omega
```

Actual assumptions and interpretation. `IncomeData` supplies the compact labor subtype and its probability law; `IncomeSupport` retains the BASIC positive-support profile; `IncomeNondegenerate` says every lower-endpoint neighborhood has positive mass and permits atoms. The path law is the actual infinite IID product `Measure.infinitePi`. An `AdaptedOriginalPlan` represents date-`t` original assets by a measurable function of exactly the first `t` labor draws and date-`t` consumption by a measurable

function of exactly the first t+1 draws. Thus assets are chosen before the current shock, while consumption and next assets may respond to it. Feasibility and nonnegative consumption hold datewise almost surely. The No-Ponzi premise is a finite real limit L >= 0 for exactly (1+r)^(-T) * a_(T+1) almost surely; neither a zero limit nor NATURAL_CAP is assumed.

Assumption profiles: BASIC, NONDEGENERATE, IID, PATH_FEASIBILITY. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: NP01. **Source keys:** A93.

Source locator: A93 Appendix Proposition 1, printed p. 37 / PDF p. 38; A94 p. 666 / PDF p. 9, note 16.

Readable proof. First intersect the countably many consumption and budget premises with the No-Ponzi event. For any path in that full-measure event, shift NP01 to an arbitrary date t. Nonnegative consumption and the pointwise upper labor support bound imply a_t >= -w*i.upper/r: the locally discounted terminal path converges to (1+r)^t*L >= 0, while the maximal discounted future-income sum is w*i.upper*(1+r)/r. This is the coarse maximum-income debt floor.

Fix a date and suppose the natural floor fails with positive probability. The strict violation set is the countable union of uniform-gap events, so one has positive probability for some eta=1/(n+1)>0. Choose h=eta*r/(8*w), retaining the wage factor and giving w*h/r=eta/8<eta/4. Since 1+r>1, choose a finite N such that w*(i.upper-i.lower)/r < (1+r)^N*(eta-w*h/r). The gap event is measurable with respect to the first t coordinates. Independence of the disjoint finite coordinate sets before t and from t through t+N-1, together with positive mass of the lower neighborhood, proves that this past event and N successive labor draws below i.lower+h have positive joint probability. Intersect with the full-measure feasibility/No-Ponzi event.

On the resulting path, the original budget and nonnegative consumption recursively amplify the debt gap. After n low draws, a_(t+n) <= -w*i.lower/r - ((1+r)^n*(eta-w*h/r)+w*h/r). At n=N this is strictly below -w*i.upper/r, contradicting the already proved coarse floor at date t+N. Therefore the natural floor holds almost surely at the fixed arbitrary date. The public quantifier is exactly forall t, almost everywhere omega; no foresighted array, infinite low-income history, density, finite support, stationarity, optimality, Euler equation, or equilibrium premise enters.

Source, dependency and audit. The sole formal theorem dependency is accepted NP01 through M10A1.discounted_budget_telescope_core; NP02 is not imported by any M10A3 helper and is not used in the proof. A93 Appendix Proposition 1, printed p. 37 / original PDF p. 38, supplies the No-Ponzi/natural-limit argument and finite telescope; A94 printed p. 666 / original PDF p. 9, note 16, records the present-value borrowing-limit relation. Uniform-gap extraction, the infinite product construction, finite-block independence, and detailed debt recursion are project reconstructions. All new declarations have #check, assert_no_sorry, and #print axioms coverage in Probes/M10A3Signatures.lean and Audit.lean; their axiom inventories are exactly propext, Classical.choice, and Quot.sound. This entry requests independent review for NP03 only. It does not award GREEN, authorize E01–E03, discharge the Stage10a human checkpoint, or advance the stage.

E01 — Zero Rate cap change exact

Status: GREEN. Independent Astra acceptance: reviews/m10b1_acceptance.md. **Scope:** extension. **Milestone:** 10.

Target declaration: Aiyagari1994.zeroRate_cap_change_exact.
Module: Aiyagari1994/Extensions/BorrowingCap.lean.

Kernel-checked statement. Fix one HouseholdPrimitives object, its accepted smoothness, curvature and income-nondegeneracy premises, a positive wage, and two finite nonnegative caps.

`M10B1.zeroRateHousehold` constructs the normalized zero-rate household with gross return one, wage `w`, and zero intercept. `M10B1.zeroRateHousehold_prices` proves that this object is exactly the normalization of the P02 finite-cap original prices for each cap. The theorem proves equality of the two entire shifted household primitives, canonical H02 value functions, canonical H04 shifted asset policies, consumption policies, and canonical S05 stationary resource laws. For those laws it concludes exactly

$$S(b_2) - S(b_1) = -(b_2 - b_1).$$

The public declaration is:

```
Aiyagari1994.zeroRate_cap_change_exact
    (m : HouseholdPrimitives) (hsmooth : UtilitySmooth m.utility)
    (hcurvature : UtilityCurvature m.utility)
    (hnd : IncomeNondegenerate m.income)
    (b1 b2 : ℝ) (hb1 : 0 ≤ b1) (hb2 : 0 ≤ b2)
    (w : ℝ) (hw : 0 < w) :
  let m1 := M10B1.zeroRateHousehold m b1 hb1 w hw
  let m2 := M10B1.zeroRateHousehold m b2 hb2 w hw
  let h1 := M10B1.zeroRate_beta_grossReturn_lt_one m b1 hb1 w hw
  let h2 := M10B1.zeroRate_beta_grossReturn_lt_one m b2 hb2 w hw
  let pi1 := M06C.stationaryLaw m1 hsmooth hcurvature hnd h1
  let pi2 := M06C.stationaryLaw m2 hsmooth hcurvature hnd h2
  m1 = m2 ∧ valueFunction m1 = valueFunction m2 ∧
    assetPolicy m1 = assetPolicy m2 ∧
    consumptionPolicy m1 = consumptionPolicy m2 ∧ pi1 = pi2 ∧
    stationaryAssetSupply m2 b2 pi2 -
      stationaryAssetSupply m1 b1 pi1 = -(b2 - b1)
```

Actual assumptions. BASIC is carried by `m`; `hsmooth`, `hcurvature`, and `hnd` retain the accepted SMOOTH, CURVATURE and NONDEGENERATE conditions on the general compact iid income model. The wage is positive and both finite caps are nonnegative. There is no invariant-law premise: gross return is exactly one, so strict impatience follows from `m.beta_lt_one`, and accepted S05 constructs the canonical law. No density, finite labor support, labor atom, or stationary moment assumption is added.

Dependencies: P02, H02, H04, S05, A02. **Source keys:** A93, A94.

Source locator: A94 borrowing-limit discussion, printed pp. 672-673 / PDF pp. 15-16, and equation (2) on printed p. 666 / PDF p. 9.

Readable proof. P02's zero-rate branch gives `effectiveLimit b lo w 0 = b`; normalization has gross return one and intercept `-0*b=0`, hence effective income is exactly `w*l`. Therefore both caps produce the identical shifted primitive object. The H02 value and H04 shifted asset and consumption policies are equal as canonical functions of that object. Since `beta*1=beta<1`, S05 supplies the same canonical stationary resource law in both cases. A02 is invoked separately for each original debt shift and supplies the integrable accounting identities `S(bi)=Integral A dpi-bi`. Equality of the shifted policy and law makes the two integrals identical; subtraction and ring algebra give the displayed result.

Source and reconstruction boundary. A94 equation (2), printed p. 666/PDF p. 9, gives `phi=b` for nonpositive rates and the shifted resource equations. Its borrowing-limit discussion, printed pp. 672–673/PDF pp. 15–16, explicitly states that at zero interest `b` is absent from the shifted asset-demand function and mean net assets decrease one-for-one as `b` rises. The approved A93 full scanned original is supplied under its exact manifest hash because its source ID has no extension-specific locator and its text extraction is unusable; no A93 page attribution is invented. The equality transport through canonical Lean objects and the A02 subtraction are project reconstruction.

Qualification. The policy equality is in shifted assets. It does not say that original/net asset policies are numerically equal: those differ by their respective caps. The result is only the fixed-wage zero-rate identity. It proves no nonzero-rate formula, global cap monotonicity, equilibrium implication, or invariant existence outside the accepted strict-impatience theorem. All predecessor qualifications q1–q956 remain operative without supersession. The implementation and complete audit coverage request independent review only; no GREEN status or later gate is self-awarded. The transitive axiom inventory is exactly `propext`, `Classical.choice`, and `Quot.sound`.

E02 — Caps above natural Limit equivalent

Status: UNFORMALIZED. **Scope:** extension. **Milestone:** 10.

Target declaration: `Aiyagari1994.caps_above_naturalLimit_equivalent`.
Module: `Aiyagari1994/Extensions/BorrowingCap.lean`.

Mathematical contract. At r>0, any two finite caps weakly above w*l_min/r induce identical effective household primitives, policies and, when subcritical, stationary laws and net asset supply.

Assumption profiles: BASIC, FINITE_CAP. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: P02, H02, H04, S05. **Source keys:** A93, A94.

Source locator: A94 borrowing-limit discussion, printed pp. 672-673 / PDF pp. 15-16, and equation (2) on printed p. 666 / PDF p. 9.

Readable proof plan: Architecture §12.2.

Adequacy note. No proof or adequacy certification is asserted by this initial entry.

E03 — Natural Bound government Debt correspondence

Status: UNFORMALIZED. **Scope:** extension. **Milestone:** 10.

Target declaration: `Aiyagari1994.naturalBound_governmentDebt_correspondence`.
Module: `Aiyagari1994/Extensions/GovernmentDebt.lean`.

Mathematical contract. In pure exchange under tax-adjusted natural borrowing, x=a-d maps budget, admissibility, lifetime optimality, invariant laws and bond clearing bijectively between government-debt and debt-free economies. It is not a statement about capital demand or a fixed borrowing cap.

Assumption profiles: BASIC, IID, PURE_EXCHANGE_DEBT. These are branch-sensitive context tags; the completed signature must list the actual premises.

Dependencies: P01, H05. **Source keys:** A93, A94.

Source locator: A94 government-debt reinterpretation, printed pp. 673-674 / PDF pp. 16-17.

Readable proof plan: Architecture §12.2.

Adequacy note. No proof or adequacy certification is asserted by this initial entry.